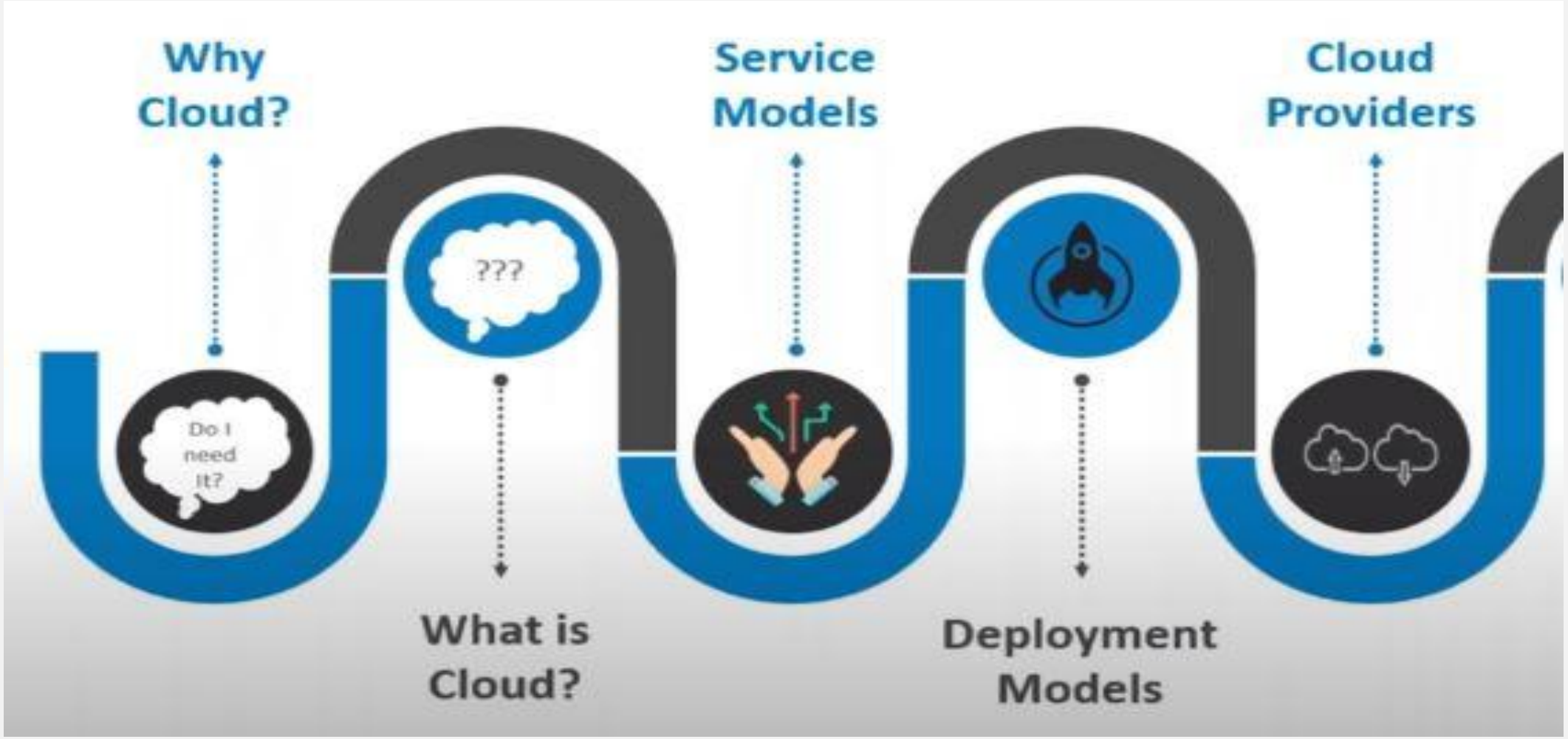
Introduction to Cloud

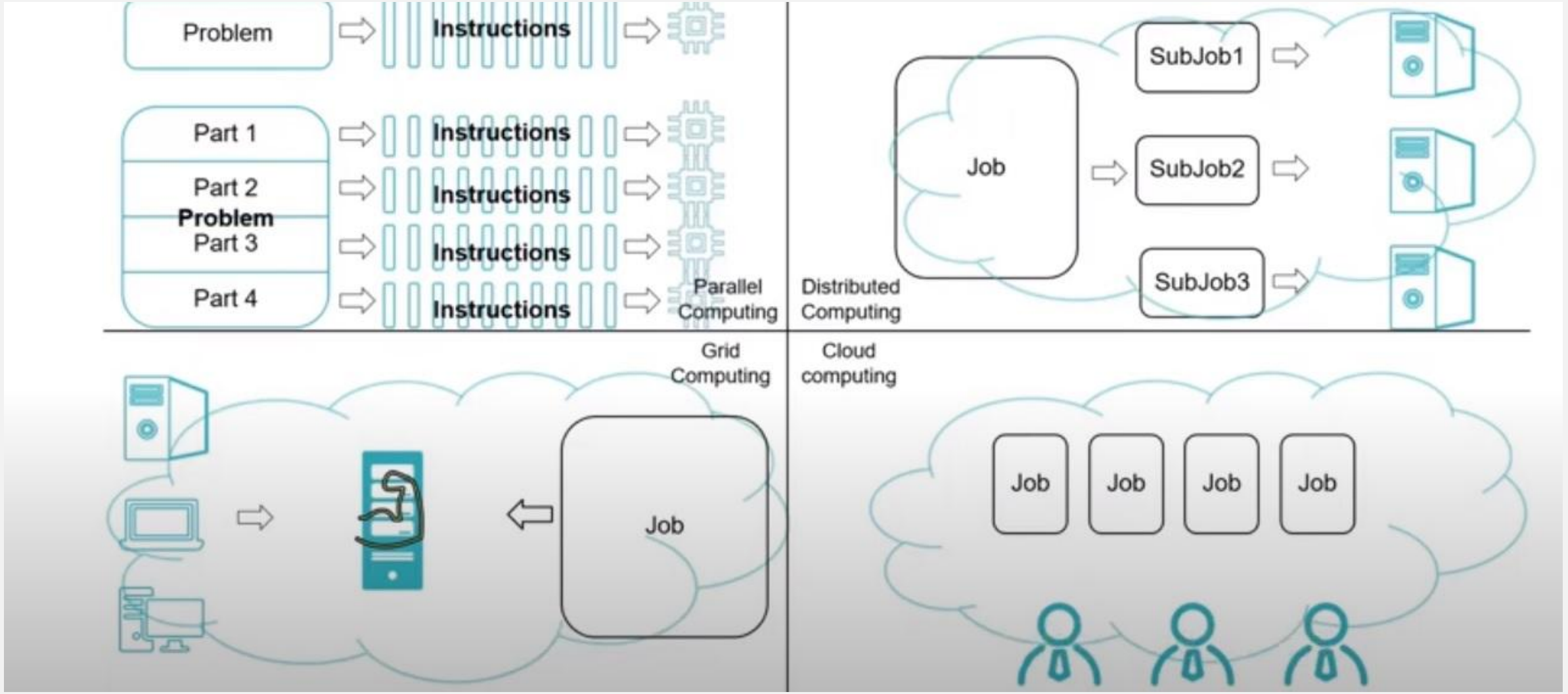
Dr. Mukti Padhya

Assistant Professor, NFSU

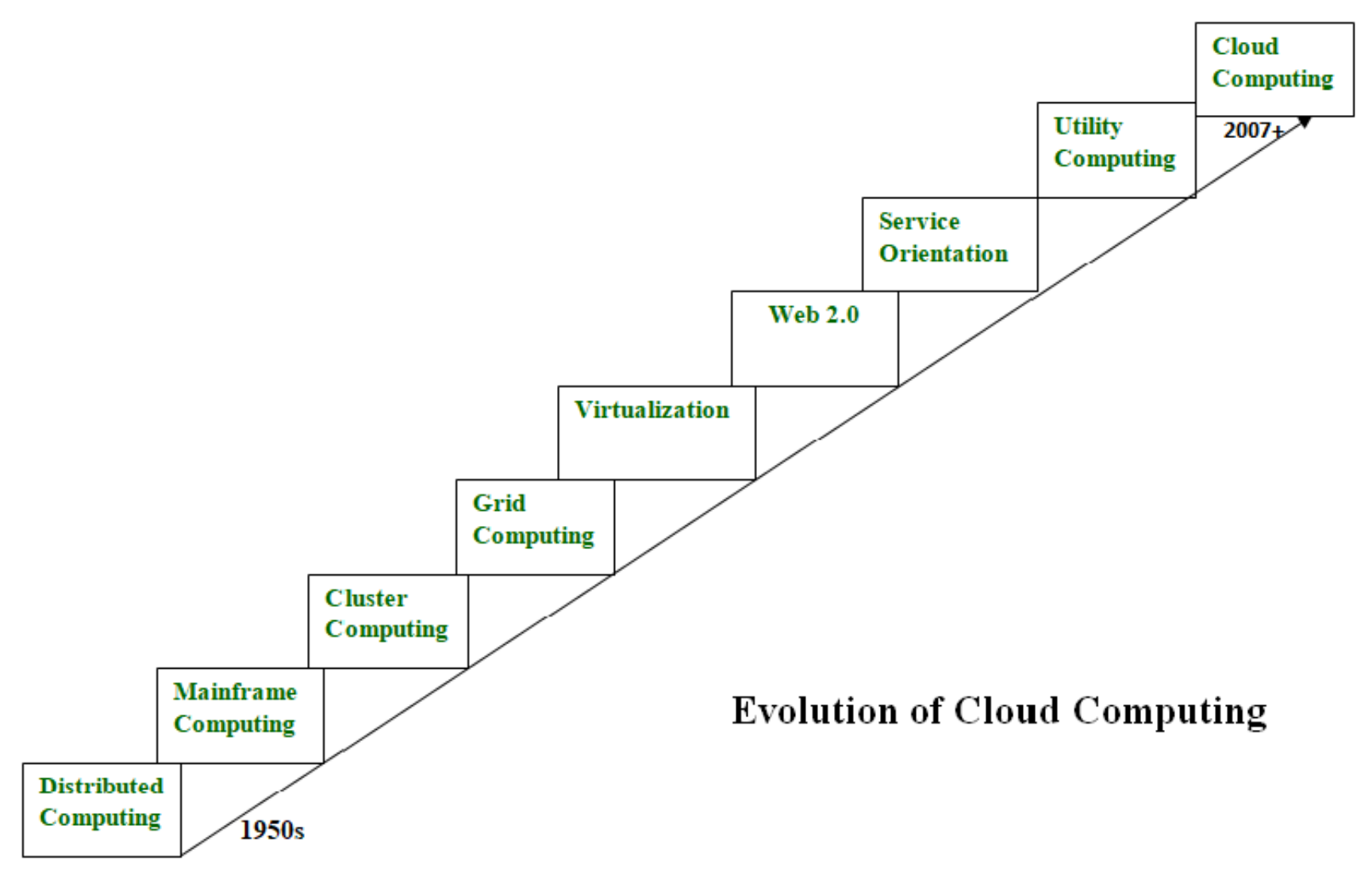
Offerings of This Session : Unit I



Evolution of Cloud Computing

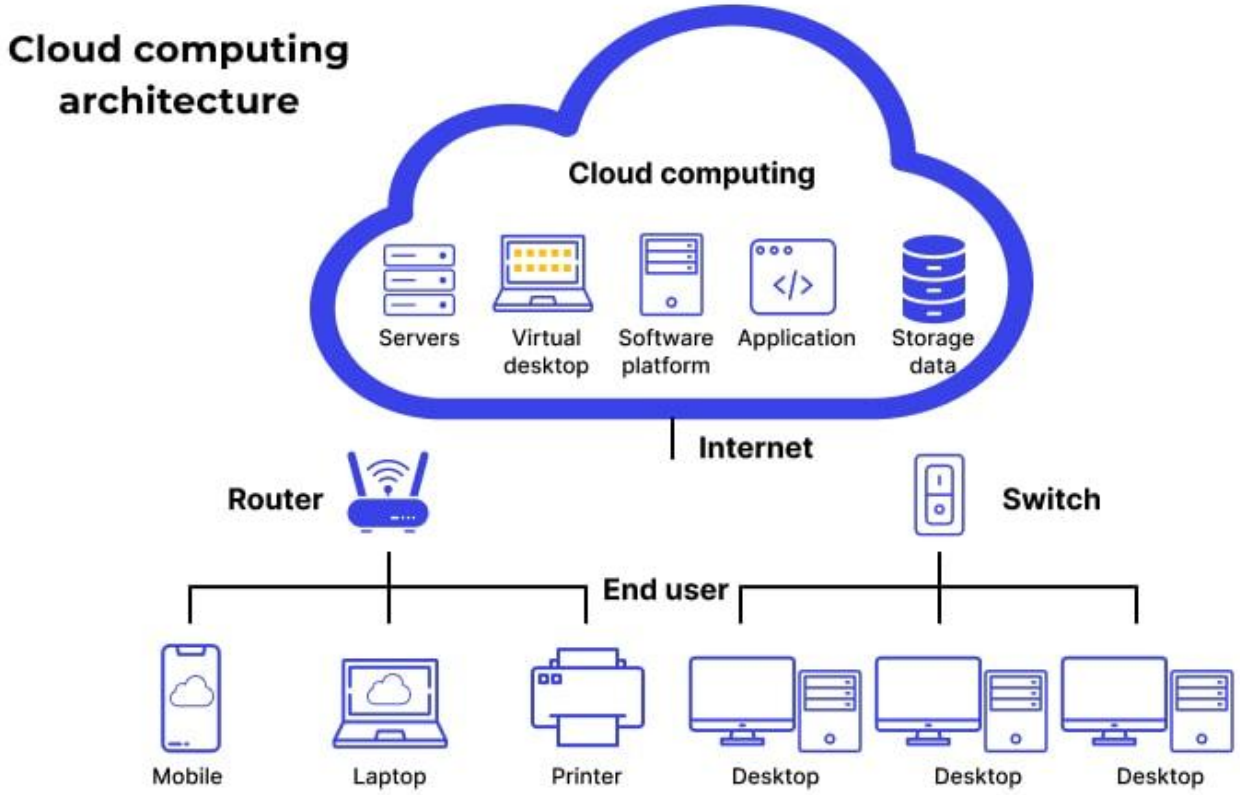


Evolution of Cloud Computing



For Details Refer : https://www.geeksforgeeks.org/evolution-of-cloud-computing/

What is Cloud Computing?



- The term **'Cloud'** in technical terms, was used to refer to distributed computing as early as 1993.
- Cloud computing is basically the on-demand provision of computing resources like **storage, applications, networking capabilities, databases, software and services, development tools, processing capabilities**, and more, by service providers (known as Cloud Service providers or CSP), to its users, via the internet.
- Cloud computing is often referred to as **internet-based computing**.
- In simple terms, users can access data, applications, and services hosted in remote services, instead of accessing it from their computer's hard drive.

You’re Using Cloud Computing Every Day……..

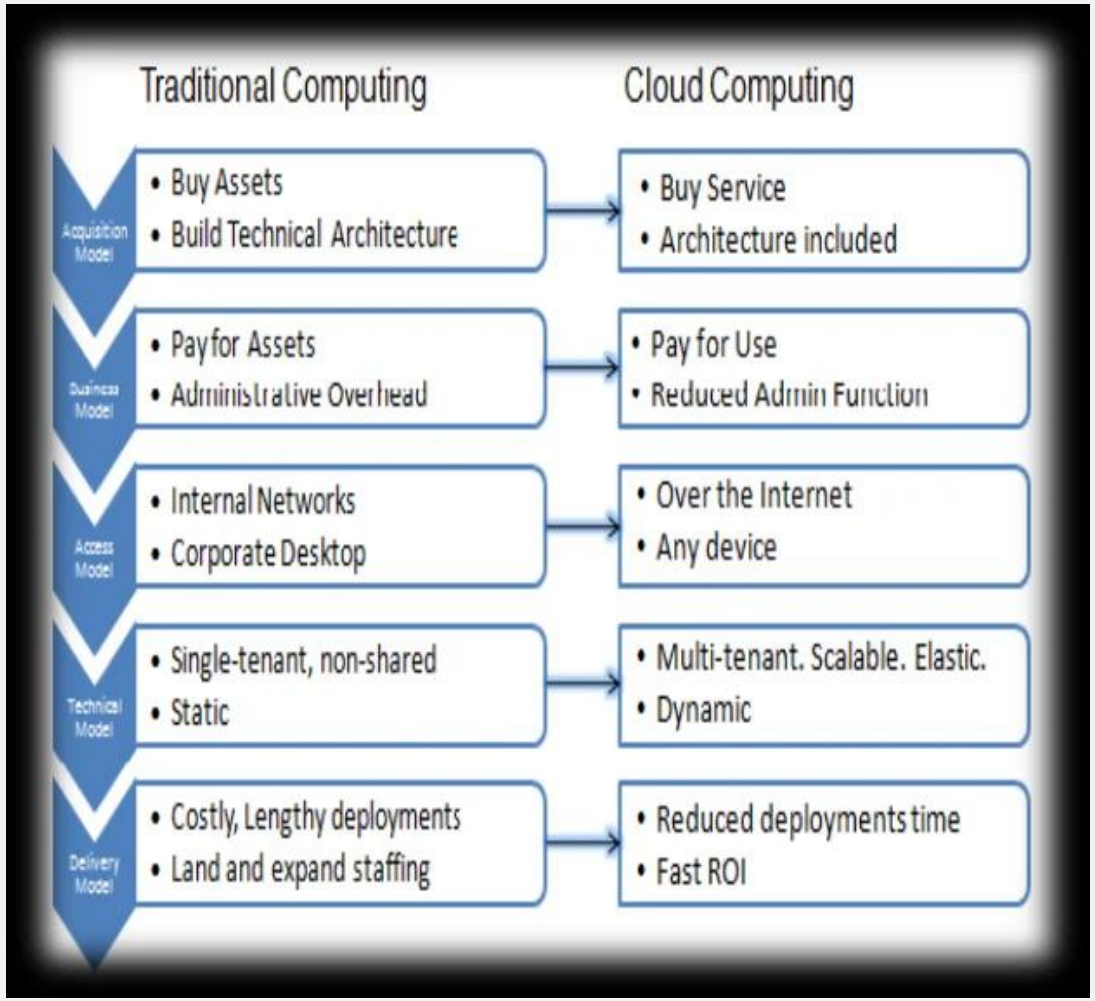


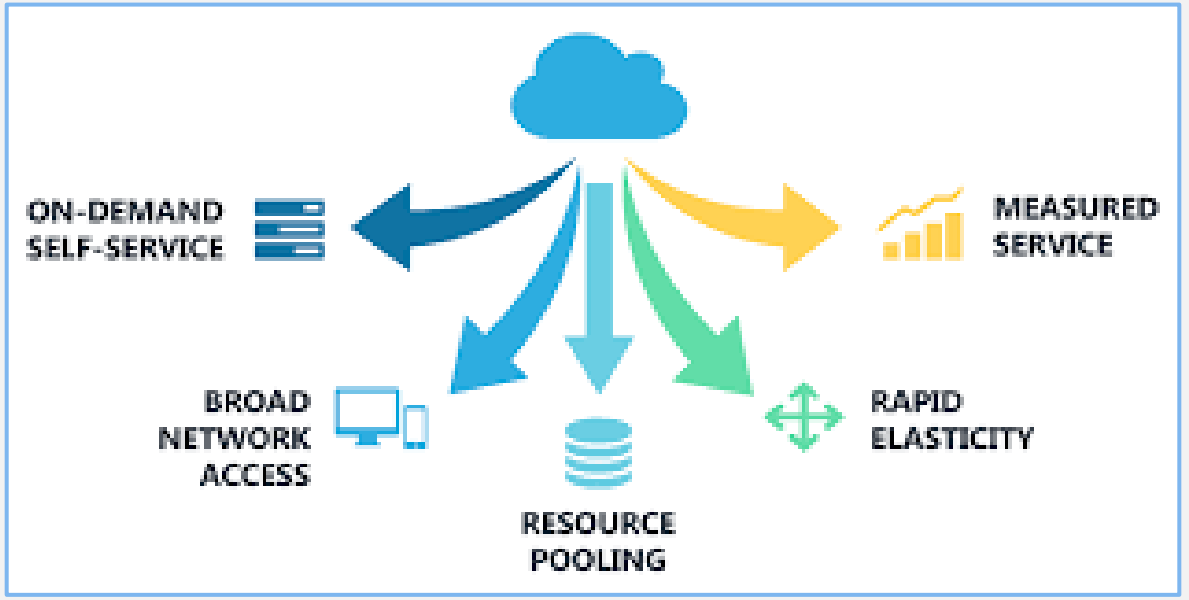




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Cloud Computing Vs. Traditional Computing





What is Cloud?

- The term ‘cloud’ in technical terms, was used to refer to distributed computing as early as 1993.
- Cloud computing is basically the on-demand provision of computing resources like storage, applications, networking capabilities, databases, software and services, development tools, processing capabilities, and more, by service providers (known as Cloud Service providers or CSP), to its users, via the internet.
- These services can be provided with minimum management effort or service provider interaction. Cloud computing is often referred to as internet-based computing.
- In simple terms, users can access data, applications, and services hosted in remote services, instead of accessing it from their computer’s hard drive.

What is Cloud?

- Global network of servers each with a unique function
- Cloud is not a physical entity - Virtualization
- It is a vast network of remote servers around the globe which are hooked together and meant to operate as a single ecosystem
- The information will be available anywhere you go and anytime you need it
- Cloud is nothing but a server that we access over the internet, it contains a large amount of data such as text files, video, audio, images, docs, pdfs, and so on.
- It is just like developing software for millions to use as a service rather than distributing software to run on their PCs.

What is Cloud Computing?

"Cloud computing is a model for enabling ubiquitous convenient, on-demand network access to a shared pool of configurable computing resources (e.g.,networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

- National Institute of Standards and Technology (NIST)

Cloud : DataCenter

- Clouds can be built with physical or virtualized resources over large data centers that are distributed systems. Cloud computing is also considered to be a form of utility computing or service computing.

- A single-site cloud (as known as a “Datacenter”) consists of
  - Compute nodes (grouped into racks).
  - Switches, connecting the racks.
  - A network topology, e.g., hierarchical.
  - Storage (backend) nodes are connected to the network.
  - Front-end for submitting jobs and receiving client requests.
  - Software Services.

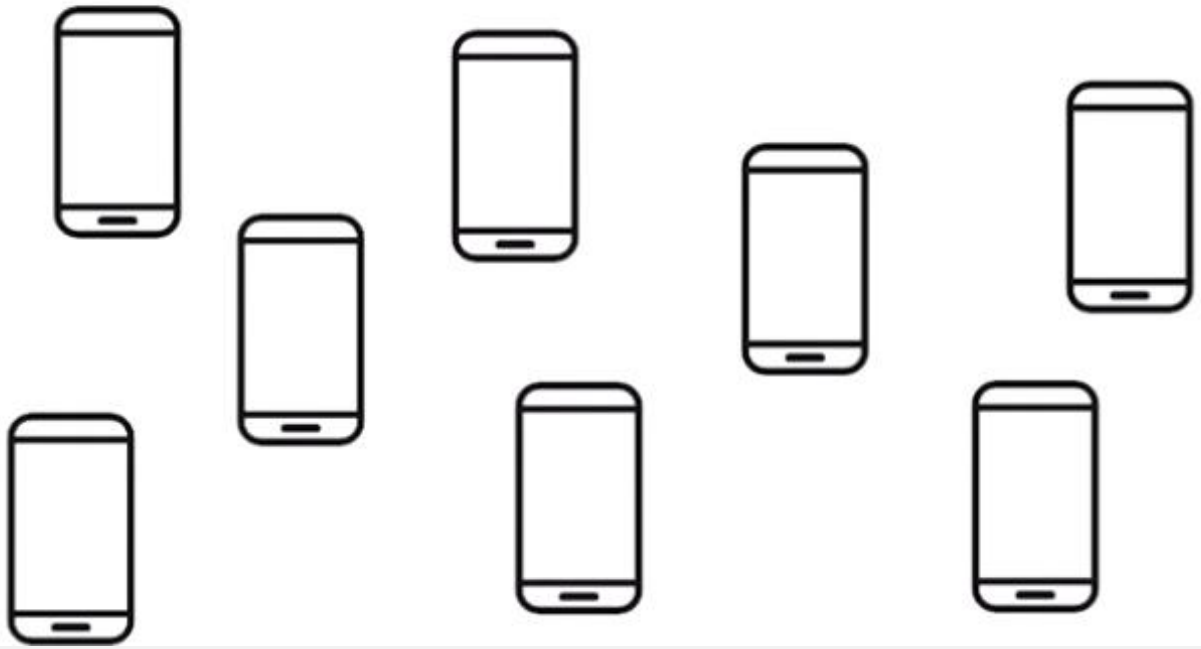
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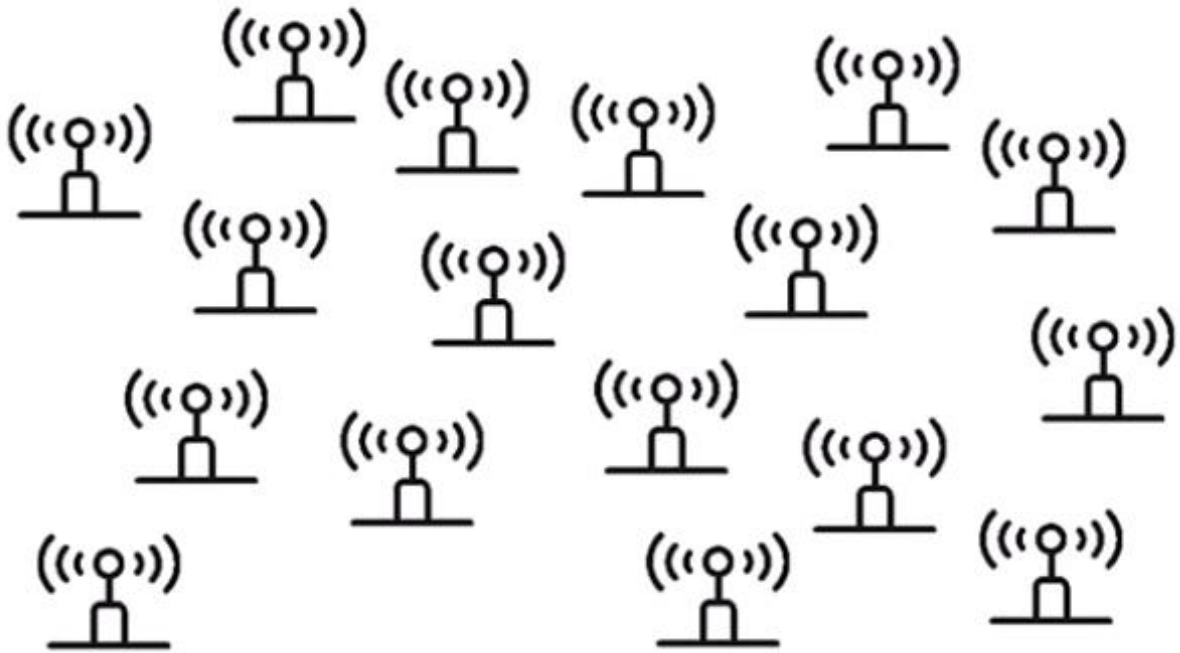
7.3 billion smartphones

Source: Ericsson

41.6 billion IoT devices

Source: International Data Corporation (IDC)





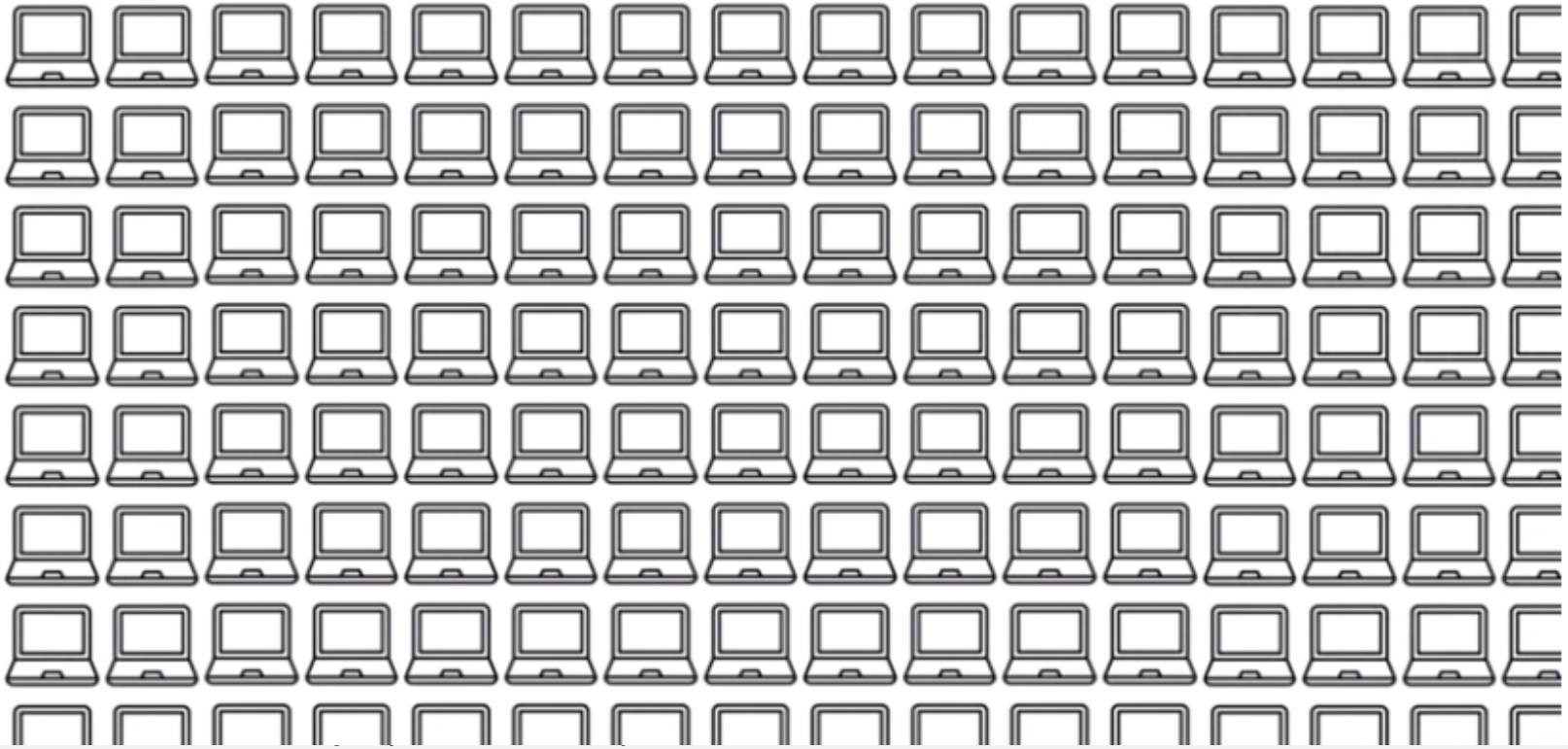
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80 zettabytes

200,000,000 terabytes per day

Source: International Data Corporation (IDC)

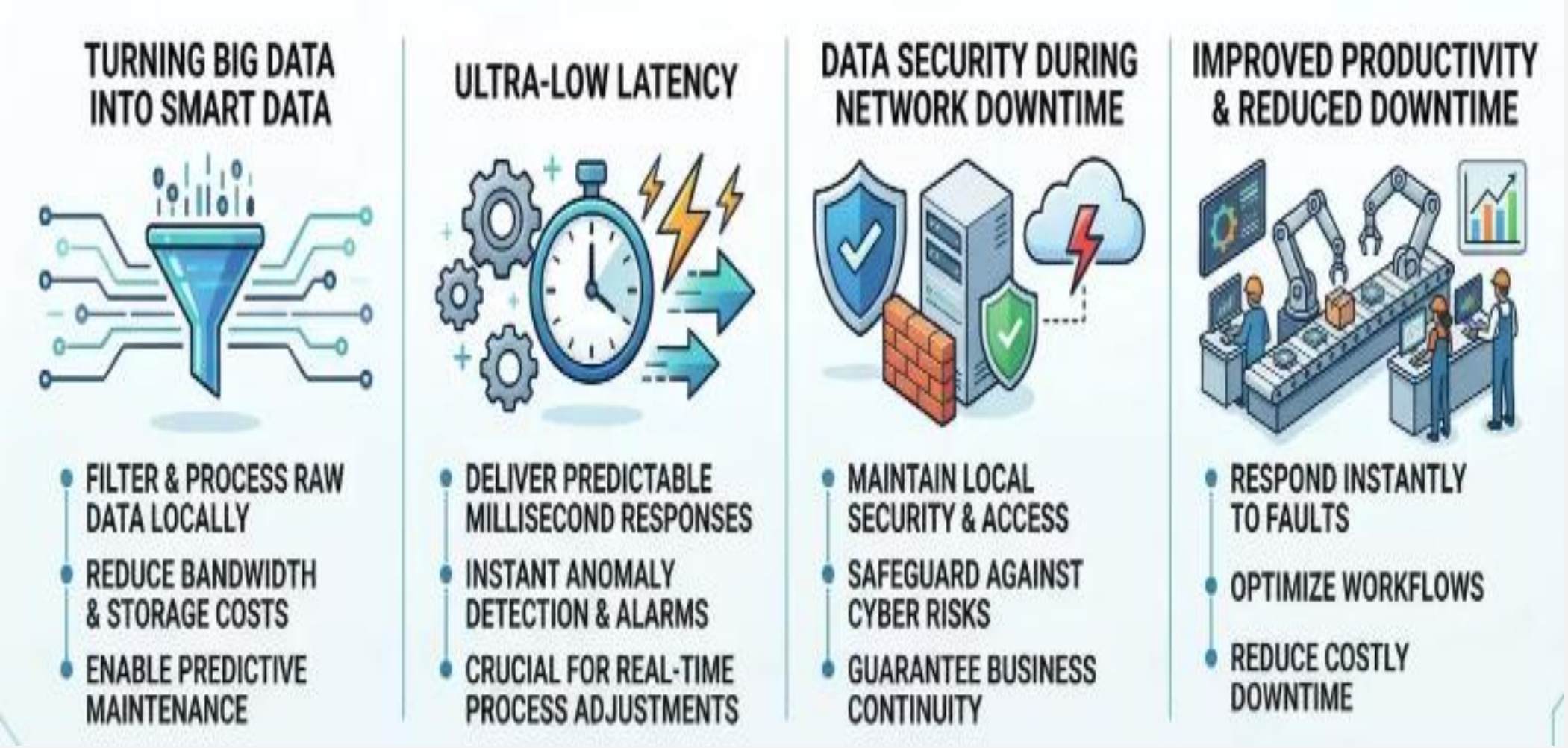




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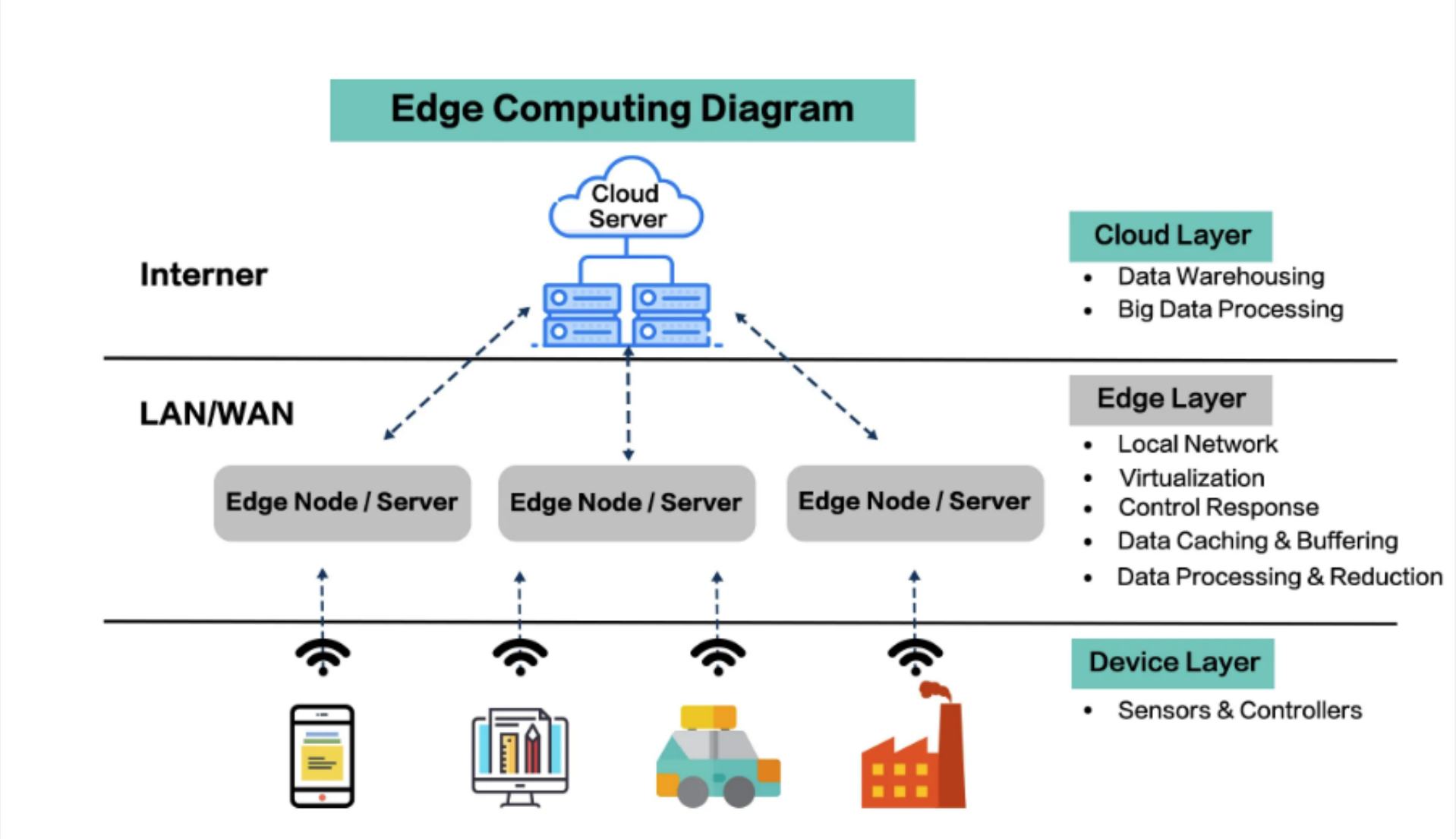

EDGE COMPUTING:
THE ESSENTIAL PILLAR FOR INDUSTRY 4.0
SMART DATA PROCESSING
Bandwidth
optimized
Historical vs. Projected
ULTRA-LOW LATENCY
0.001s
Express lane
DATA SECURITY & RESILIENCE
IMPROVED PRODUCTIVITY
IoT

Why Edge Computing ?



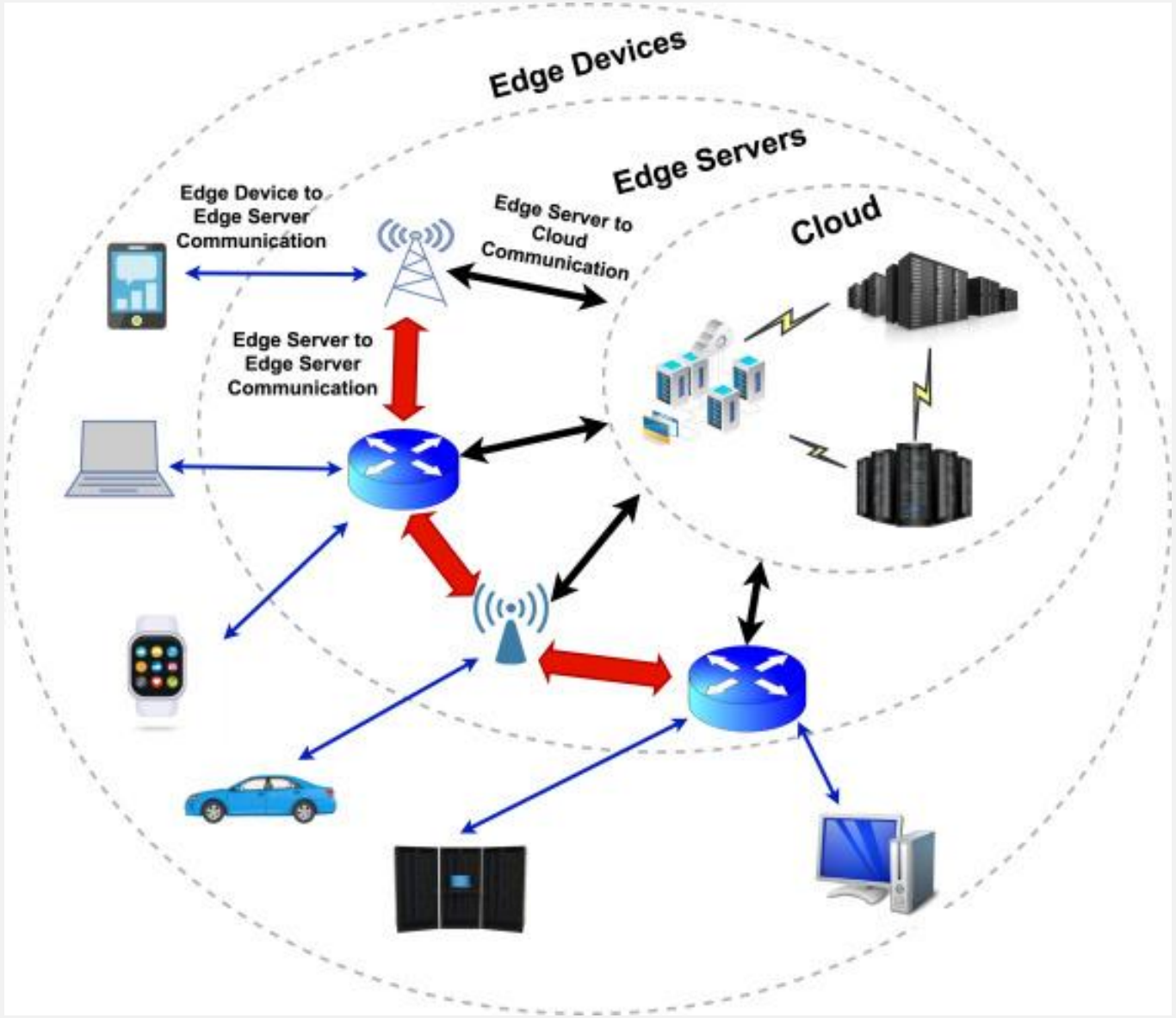
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Edge Computing Working



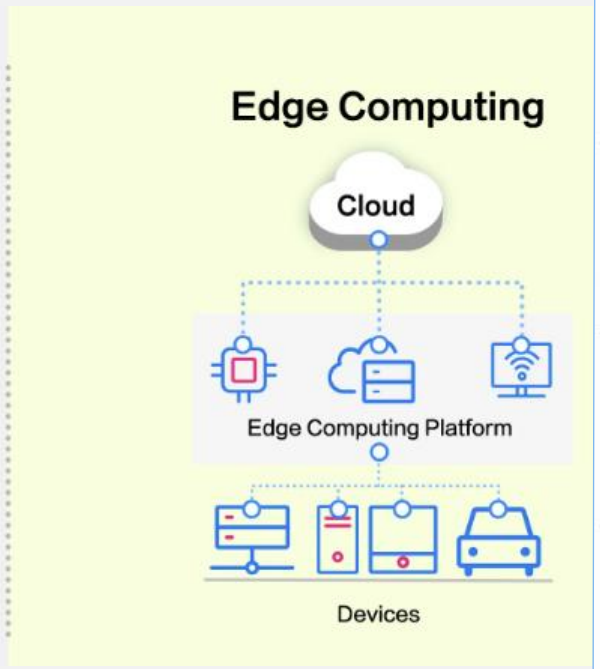
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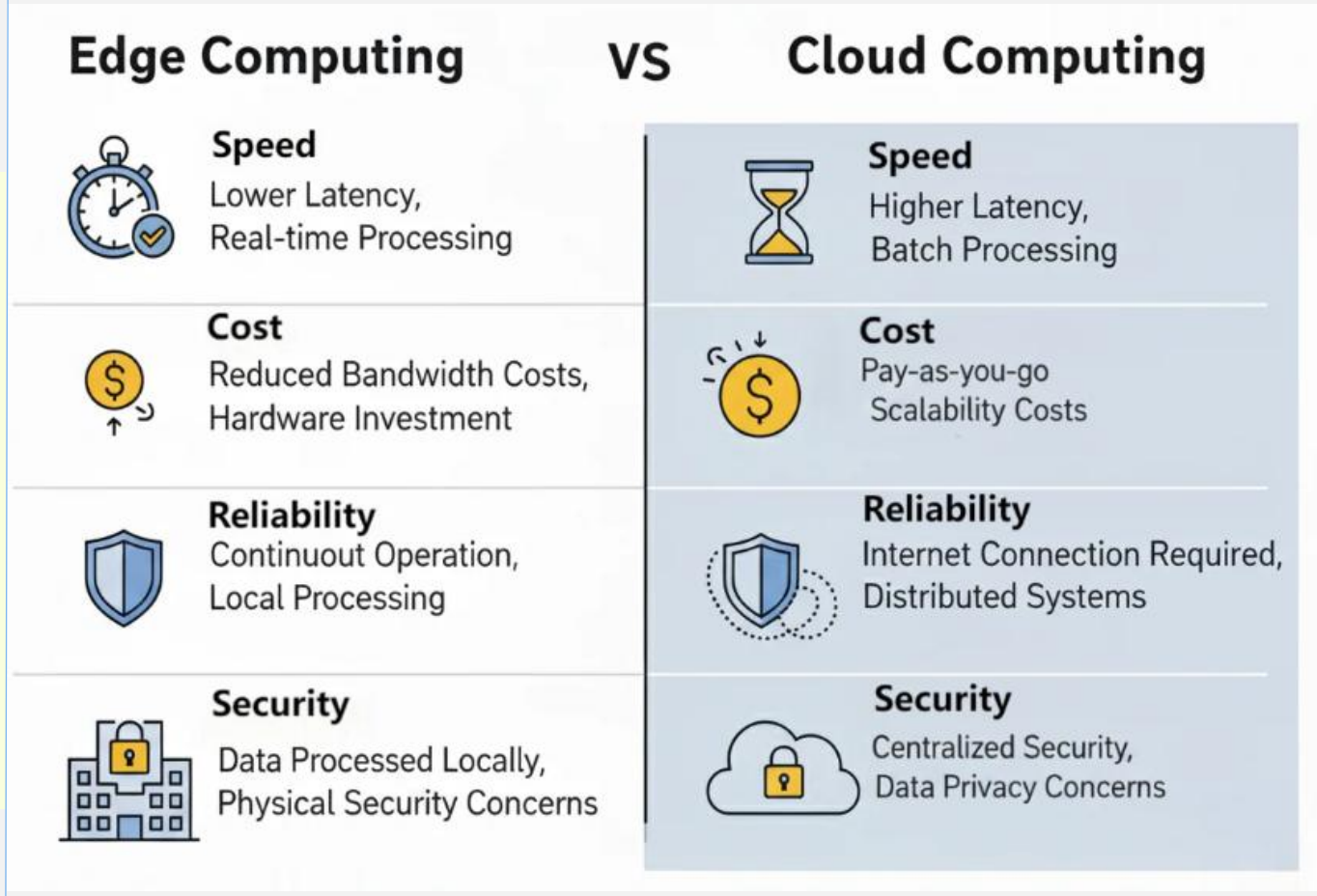
Edge Computing Working



Edge vs Cloud Computing







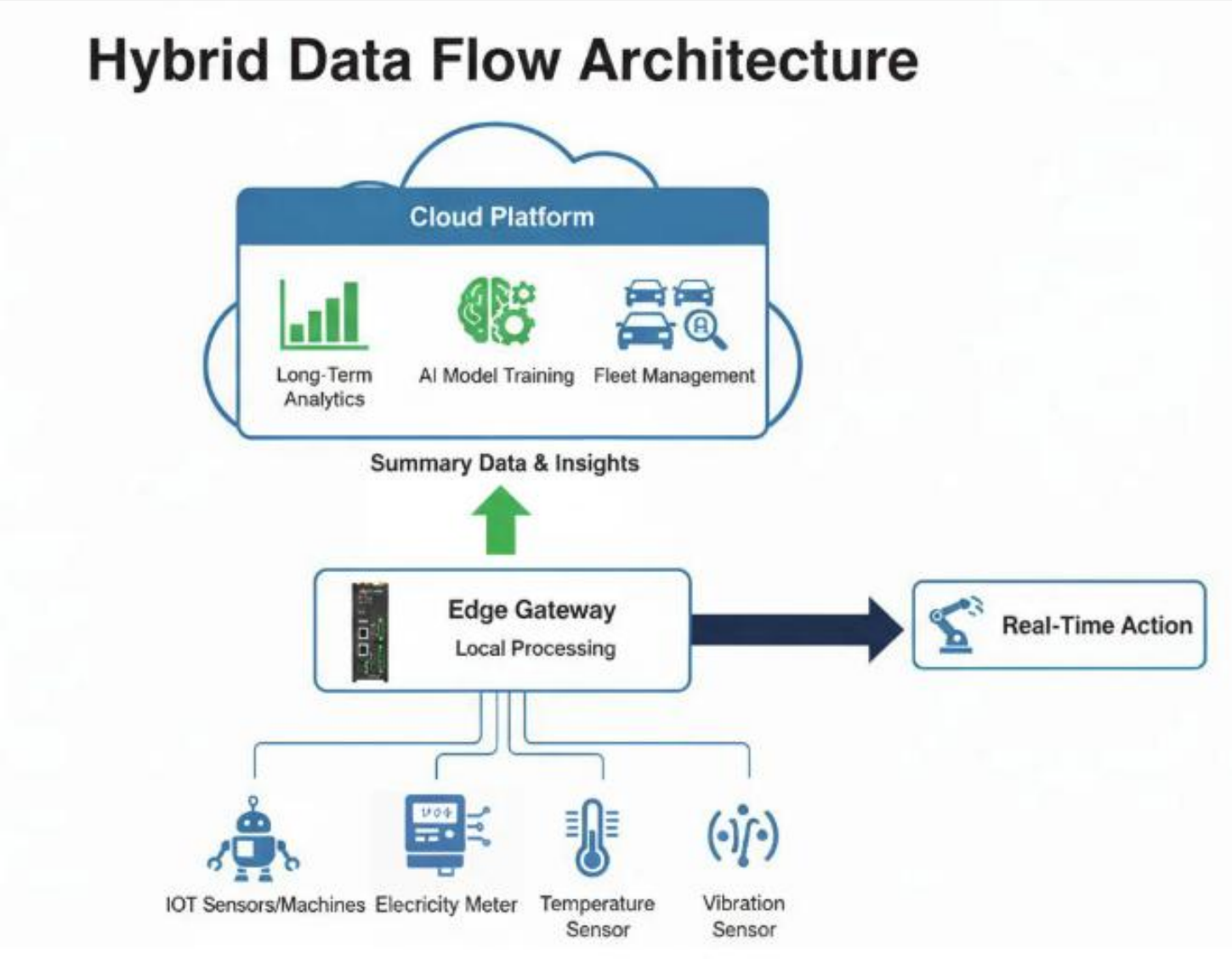
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Edg

Feature	Edge Computing	Cloud Computing
Data Processing Location	Locally, on or near the IoT device	In a remote, centralized data center
Response Time (Latency)	Extremely low (milliseconds), enabling real-time action	High (seconds or more), limited by internet travel time
Bandwidth Requirement	Very low, as only processed insights are transmitted	Very high, as all raw data must be transmitted
Reliability	High; can operate even if the internet connection is lost	Low; completely dependent on a stable internet connection
Security	Enhanced; sensitive data stays on-premise, reducing exposure	Broader attack surface as all data travels over the internet

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Cloud and Edge : Hybrid Architecture



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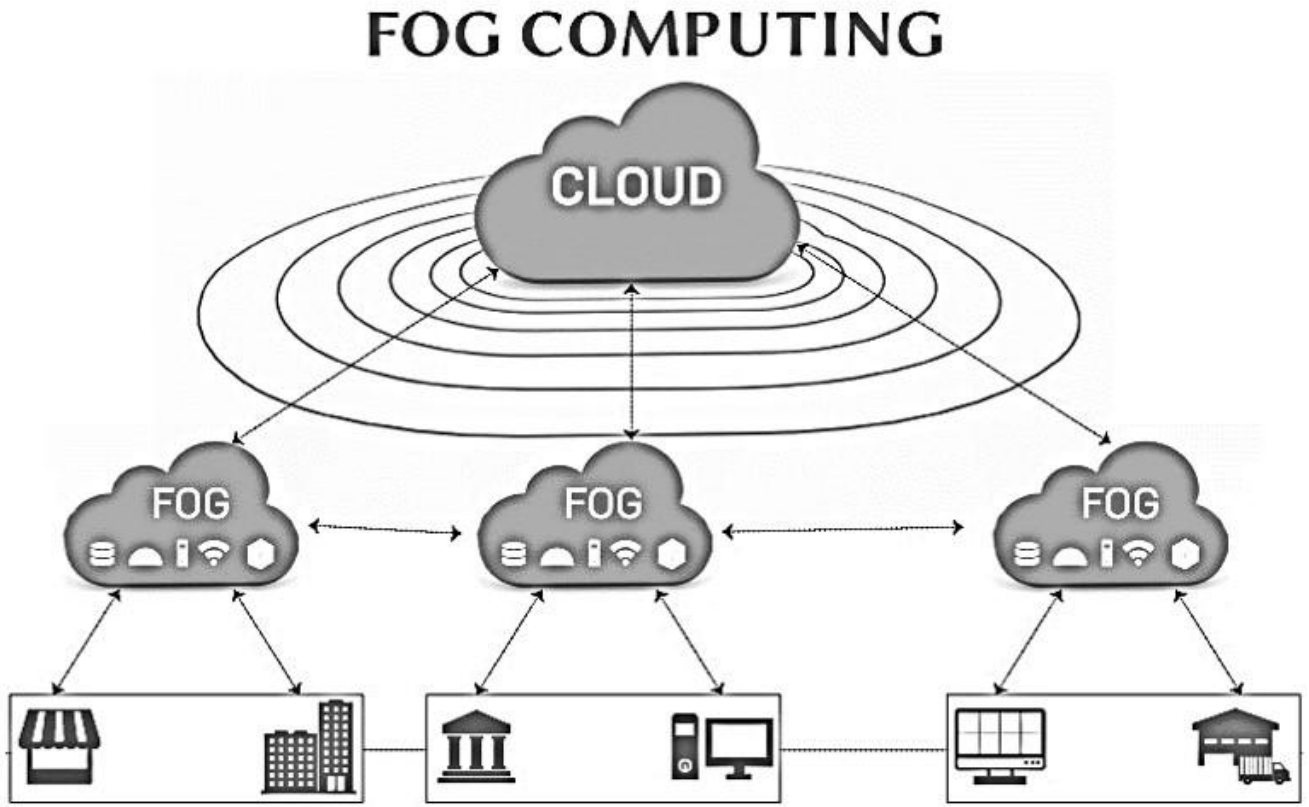
Cloud Computing : Recent Trends

- An edge is a computing location at the edge of a network, along with the hardware and software at those physical locations. Cloud computing is the act of running workloads within clouds, while edge computing is the act of running workloads on edge devices.
  - Clouds are places where data can be stored or applications can run. They are software-defined environments created by datacenters or server farms.
  - Edges are also places where data is collected. They are physical environments made up of hardware outside a datacenter.
  - Cloud computing is an act; the act of running workloads in a cloud.
  - Edge computing is also an act; the act of running workloads on edge devices.

- Fog computing is a compute layer between the cloud and the edge. Where edge computing might send huge streams of data directly to the cloud, fog computing can receive the data from the edge layer before it reaches the cloud and then decide what is relevant and what isn't. The relevant data gets stored in the cloud, while the irrelevant data can be deleted, or analyzed at the fog layer for remote access or to inform localized learning models.

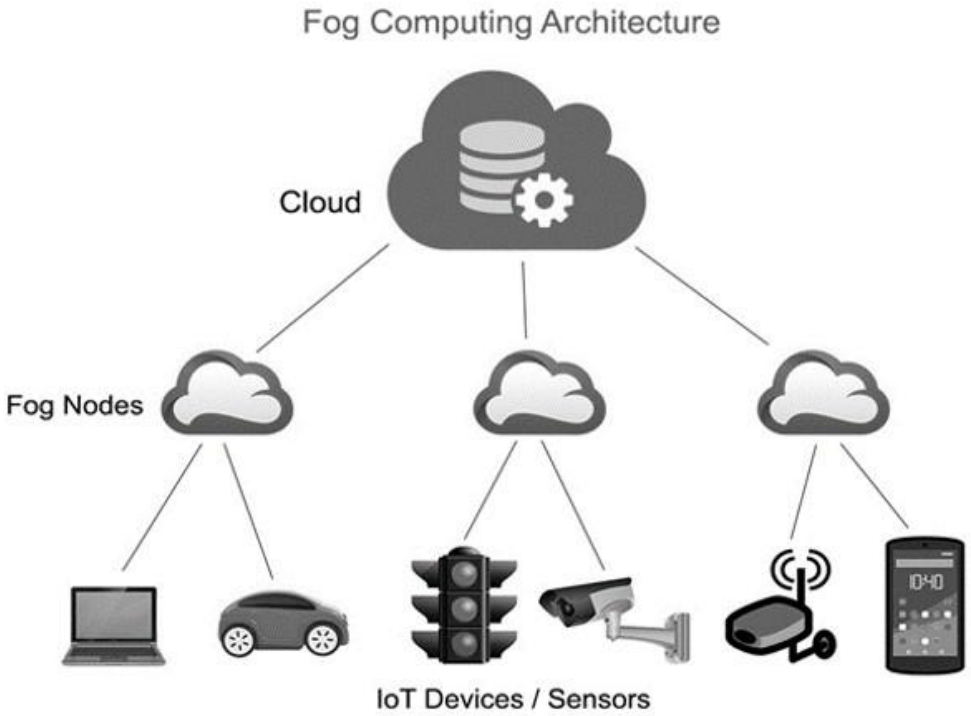
Introduction to Fog Computing

- The most sensitive data should be analyzed in the area closer to the place where it is generated.
- With fog computing, it is possible.
- Using fog computing we can process the data locally and to avoid the trouble by not sending the data to the cloud.
- Respond much faster because of data is moving locally so data travel is reduced considerably.
- It thus process the data in milliseconds.



Working of Fog Computing

- Sensors/devices generate data transmit it to the middle layer, which is very close the data source.
- These nodes in the middle layer are capable of handling the data.
- This requires minimum power and lesser resources.
- All the data need not go to the cloud at the instant.
- Also, sensitive data gets processed very fast, which results in an instant response.
- Fog is not meant for hefty storage. It is still the cloud that does the task of storing big data.
- Fog is just an intermediary layer for faster data processing, and the faster response time.



Summarize the concepts

Concept of fog nodes

- It receives the data feed from the sensors, in real-time.
- Response time is minimal, ideally in milliseconds.
- Fog computing is transit, where data is stored for a limited time only.
- Data is then sent to cloud as a summary.
- It is important to note that not all data goes to the cloud.

Concept of cloud computing platform

- It receives the data summary from the fog.
- Data prediction, data analytics, data storage, etc. takes place here.

Benefits of fog computing model

- Minimal amount of data sent to the cloud.
- Reduced bandwidth consumption.
- Reduced data latency.
- Improved data security. When limited data goes to cloud, it is easier to protect it.
- Immediate processing of data in real time (this is very much needed in industrial applications).

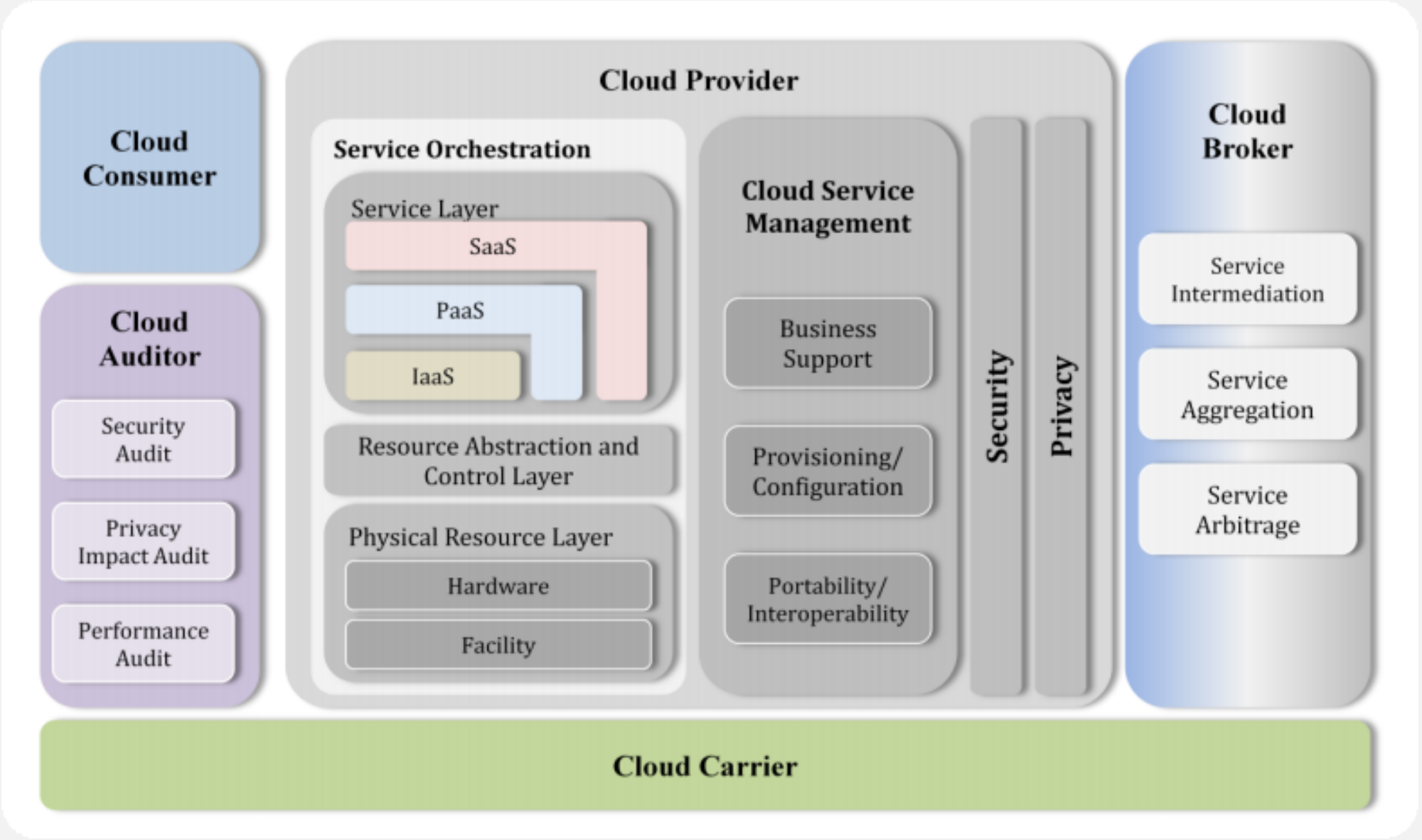
Difference between Edge and Fog Computing

- Both fog and edge are concerned with the computing capabilities to be executed locally, before passing it to the cloud.
- Both aim at reducing complete dependency on the cloud to perform computation.
- Analyzing data and processing it at the cloud is to be avoided.
- Both these reduce the time delay for making faster decision for real-time applications.
- The main difference between edge and fog computing is where data processing takes place.
- Edge computing is the computing carried out at the device itself, where all the sensors are-connected.
- In fog computing, data processing is moved to the processors that are connected to the local area network (LAN), making it a little farther from the sensors and actuators.
- Thus, the main difference between edge and fog computing is the distance.

Cloud Computing?

- NIST defined cloud model is composed of five essential characteristics, three service models, and four deployment models
- If a cloud does not have any of these essential characteristics and deployment models it is likely not a cloud
- It is a convenient, on-demand way to access a shared pool of configurable resources (networks, servers, storage, applications and services)
- Enables users to develop, host and run services and applications on demand in a flexible manner in any devices, anytime, and anywhere

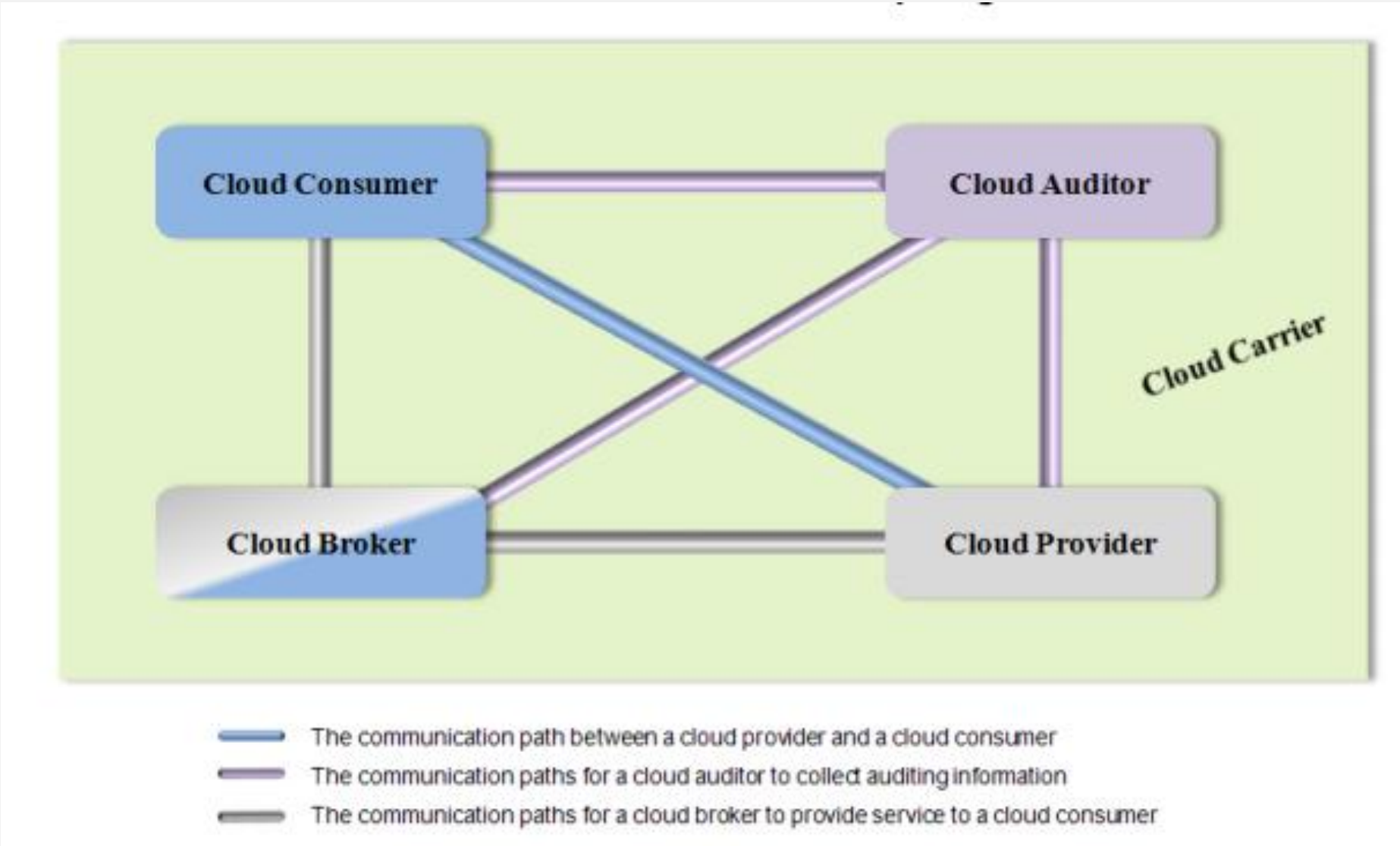
Cloud Architechture : NIST



Cloud Stakeholders : NIST

Actor	Definition
Cloud Consumer	A person or organization that maintains a business relationship with, and uses service from, *Cloud Providers*.
Cloud Provider	A person, organization, or entity responsible for making a service available to interested parties.
Cloud Auditor	A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.
Cloud Broker	An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between *Cloud Providers* and *Cloud Consumers*.
Cloud Carrier	An intermediary that provides connectivity and transport of cloud services from *Cloud Providers* to *Cloud Consumers*.

Interaction between Stakeholders



Example Usage Scenario : 1

- **Example Usage Scenario 1**: A cloud consumer may request service from a cloud broker instead of contacting a cloud provider directly. The cloud broker may create a new service by combining multiple services or by enhancing an existing service. In this example, the actual cloud providers are invisible to the cloud consumer and the cloud consumer interacts directly with the cloud broker.



Cloud Broker

- A cloud broker can provide services in three categories :

- **Service Intermediation:** A cloud broker enhances a given service by improving some specific capability and providing value-added services to cloud consumers. The improvement can be managing access to cloud services, identity management, performance reporting, enhanced security, etc.

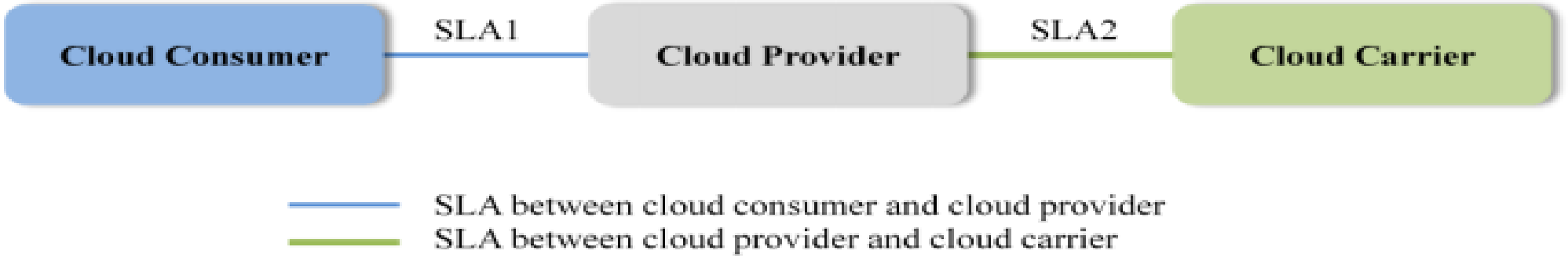
- **Service Aggregation:** A cloud broker combines and integrates multiple services into one or more new services. The broker provides data integration and ensures the secure data movement between the cloud consumer and multiple cloud providers.

- **Service Arbitrage:** Service arbitrage is similar to service aggregation except that the services being aggregated are not fixed. Service arbitrage means a broker has the flexibility to choose services from multiple agencies. The cloud broker, for example, can use a credit-scoring service to measure and select an agency with the best score.

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Example Usage Scenario : 2

Example Usage Scenario 2: Cloud carriers provide the connectivity and transport of cloud services from cloud providers to cloud consumers. As illustrated in Figure 4, a cloud provider participates in and arranges for two unique service level agreements (SLAs), one with a cloud carrier (e.g. SLA2) and one with a cloud consumer (e.g. SLA1). A cloud provider arranges service level agreements (SLAs) with a cloud carrier and may request dedicated and encrypted connections to ensure the cloud services are consumed at a consistent level according to the contractual obligations with the cloud consumers. In this case, the provider may specify its requirements on capability, flexibility and functionality in SLA2 in order to provide essential requirements in SLA1.



- **SLAs can cover terms regarding the quality of service, security, remedies for performance failures. A cloud provider may also list in the SLAs a set of promises explicitly not made to consumers, i.e. limitations, and obligations that cloud consumers must accept.**

Cloud Carrier

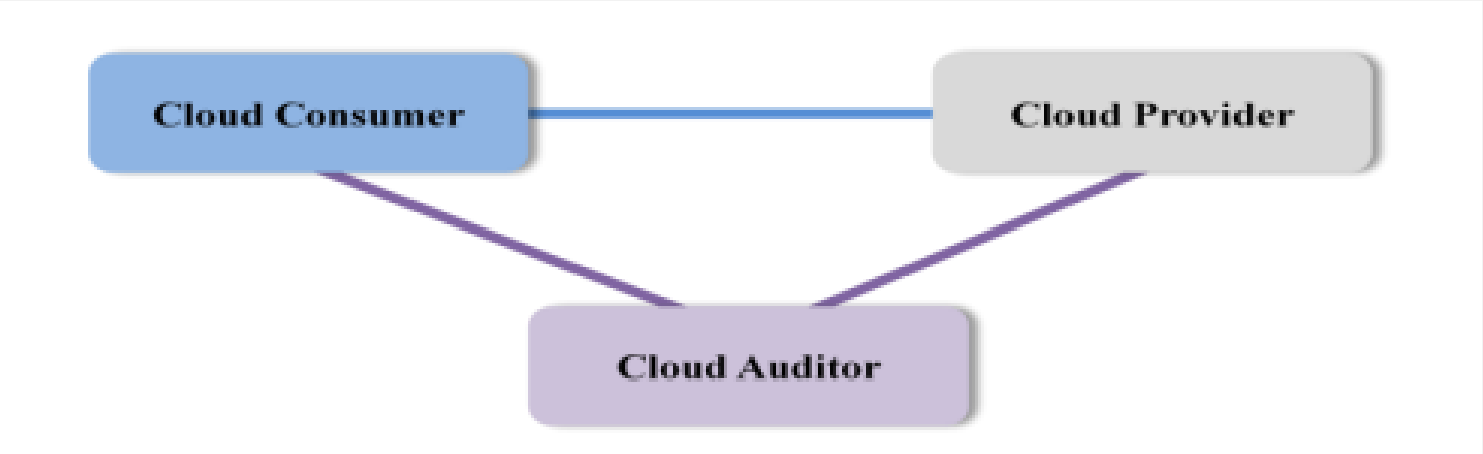
- A cloud carrier acts as an intermediary that provides connectivity and transport of cloud services between cloud consumers and cloud providers.

- Cloud carriers provide access to consumers through network, telecommunication and other access devices.

- For example, cloud consumers can obtain cloud services through network access devices, such as computers, laptops, mobile phones, mobile Internet devices (MIDs), etc .

Example Usage Scenario : 3

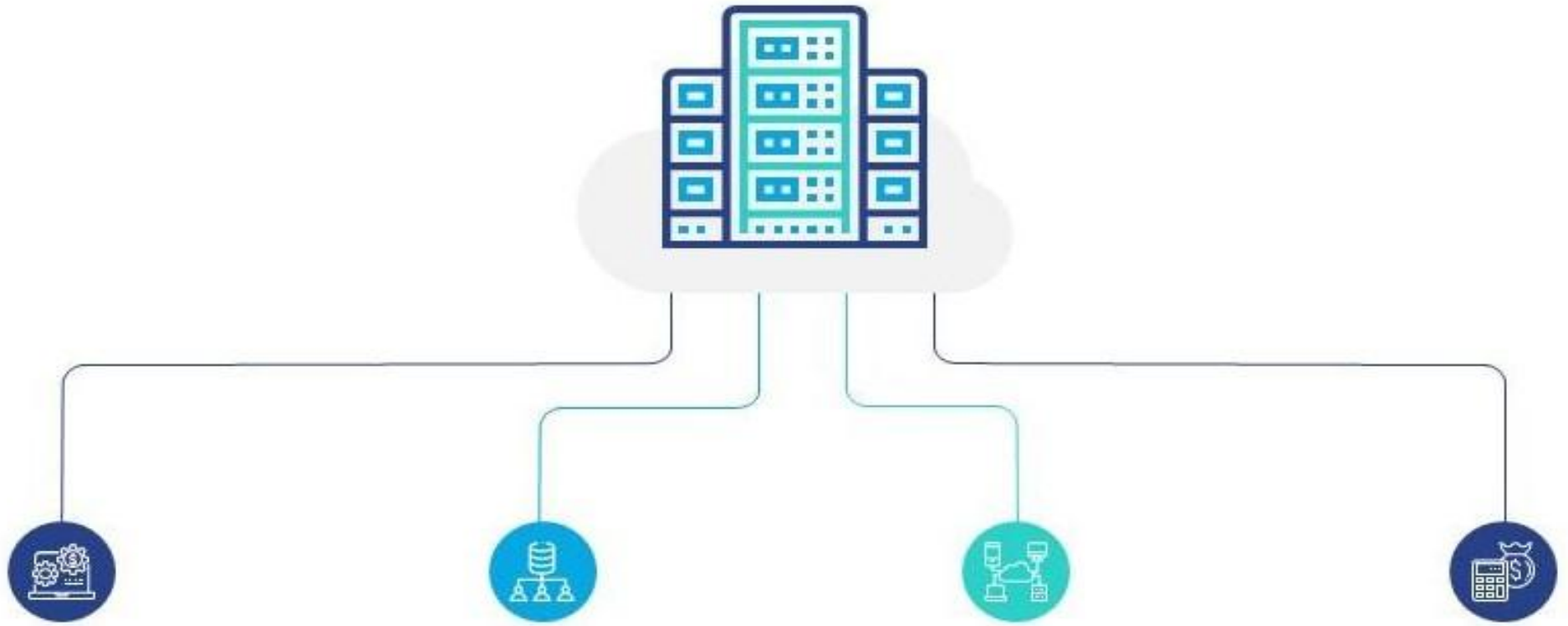
- For a cloud service, a cloud auditor conducts independent assessments of the operation and security of the cloud service implementation. The audit may involve interactions with both the Cloud Consumer and the Cloud Provider.



Cloud Auditor

- Audits are performed to verify conformance to standards through review of objective evidence.
- A cloud auditor can evaluate the services provided by a cloud provider in terms of security controls, privacy impact, performance, etc.
- For security auditing, a cloud auditor can make an assessment of the security controls in the information system to determine the extent to which the controls are implemented correctly, operating as intended, and producing the desired outcome with respect to the security requirements for the system.
- The security auditing should also include the verification of the compliance with regulation and security policy.
- For example, an auditor can be tasked with ensuring that the correct policies are applied to data retention according to relevant rules for the jurisdiction.
- The auditor may ensure that fixed content has not been modified and that the legal and business data archival requirements have been satisfied.

Characteristics of Cloud Computing



On-demand Self-service

Additional computing resources can be added without going through the cloud service provider

Broad Network Access

Cloud computing resources are available over the network and can be accessed by diverse customer platforms

Multi-tenant model

Allows multiple customers to share the same applications while retaining the security over the information

Elasticity and Scalability

Resources can scale up or down rapidly and therefore cost, can be scaled up or down with no additional contract or penalties

On-demand Self-Service

- Cloud computing resources can be provisioned without human interaction from the service provider
- In other words, a manufacturing organization can provision additional computing resources as needed without going through the cloud service provider
- This can be a storage space, virtual machine instances, database instances, and so on

Broad Network Access

- No geographical boundaries.
- Cloud computing has a vast access area and is accessible via the internet.
  - You can access your files and documents or upload your files from anywhere in the world
- Cloud computing resources are available over the network and can be accessed by diverse customer platforms

Multi-tenancy & Resource Pooling

- Cloud computing resources are designed to support a multi-tenant model

- Multi-tenancy allows multiple customers to share the same applications or the same physical infrastructure while retaining privacy and security over their information
  - E.x. : people living in an apartment building, sharing the same building infrastructure but they still have their own apartments and privacy within that infrastructure.

- The IT resource (e.g., networks, servers, storage, applications, and services) present are shared across multiple applications and occupant in an uncommitted manner. Multiple clients are provided service from a same physical resource.

Rapid Elasticity

- Dynamically adjust to capacity requirement
- Cloud computing resources can scale up or down rapidly
- Elasticity is a landmark of cloud computing and it implies that manufacturing organizations can rapidly provision and de-provision any of the cloud computing resources
- Rapid provisioning and de-provisioning might apply to storage or virtual machines or customer applications.

Measured Service

- Cloud computing resources usage is metered and manufacturing organizations pay accordingly for what they have used
- Resource utilization can be optimized by leveraging charge-per-use capabilities
- This means that cloud resource usage - whether virtual server instances that are running or storage in the cloud—gets monitored, measured and reported by the cloud service provider
- The cost model is based on pay for what you use

Other Characteristics of Cloud

- Simplicity
- Flexibility
- Automation
- Scalability (Rapidly adjust to accommodate growth)
- High availability and reliability
- Agility
- Device and Location Independence
- Maintenance
- Low Cost
- Services in the pay-per-use mode

How does Cloud computing work?

- Cloud computing is possible because of a technology called virtualization
- Virtualization allows for the creation of a simulated, digital-only "virtual" computer that behaves as if it were a physical computer with its own hardware
  - Such Computer sustem is called virtual machine
- VMs don't interact with each other at all
- The files and applications from one virtual machine aren't visible to the other virtual machines even though they're on the same physical machine

Virtualization

- Virtualization uses software to create an abstraction layer over computer hardware that allows the hardware elements of a single computer—processors, memory, storage and more—to be divided into multiple virtual computers, commonly called virtual machines (VMs).

- Each VM runs its own operating system (OS) and behaves like an independent computer - running on just a portion of the actual underlying computer hardware.

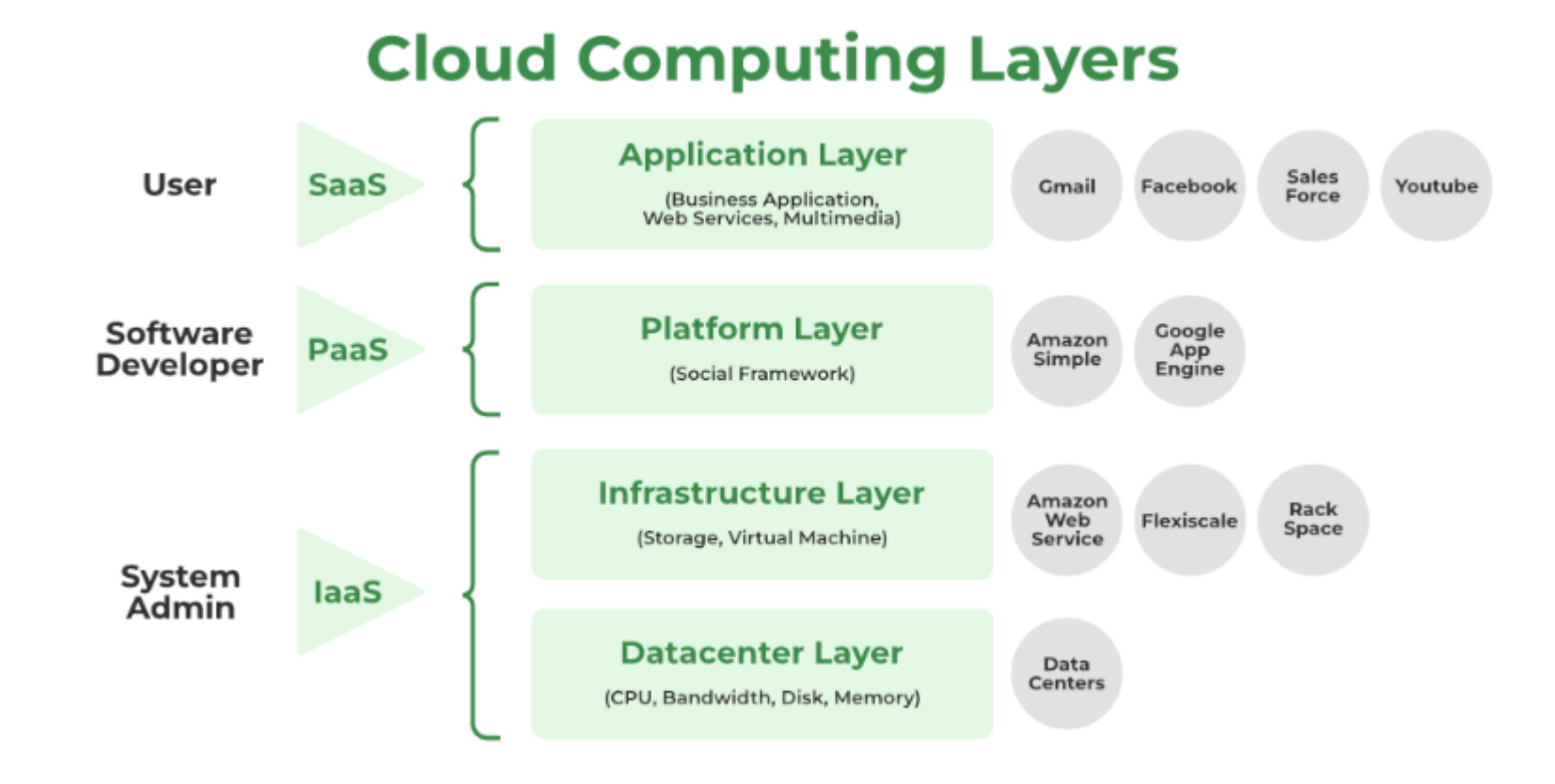
Cloud Computing : Advantage

- Backup and Restoration of Data – Data stored on cloud is easy to backup and restore
- Improved Collaboration – Applications on cloud allow quick and easy sharing of info via shared storage
- Easy Accessibility – Data stored on cloud is easily accessible without any geographic boundaries 24 X 7
- Low maintenance cost – Hardware and Software maintenance cost is reduced in cloud computing
- Pay-per-use Model – Allows user to pay for service being used thereby saving huge costs
- Unlimited storage – Nearly unlimited storage is available to user for storing photos, movies, etc

Cloud Computing : Disadvantage

- Internet Connection – A reliable and high speed internet connection is required for accessing cloud resources
- Vendor Lock-in – Cross compatibility of platforms provided by different vendors is an issue
- Limited Control – Cloud resources are managed by service providers hence user has limited control
- Security – Sensitive information of an organisation stored in the cloud can be misused by hackers
- Loss of data – Due to natural calamity/ incident

Layers of Cloud Computing



Layers of Cloud Computing

- **Application Layer**

1. The application layer, which is at the top of the stack, is where the actual cloud apps are located. Cloud applications, as opposed to traditional applications, can take advantage of the **automatic-scaling** functionality to gain greater performance, availability, and lower operational costs.
2. This layer consists of different Cloud Services which are used by cloud users. Users can access these applications according to their needs. Applications are divided into **Execution layers** and **Application layers**.
3. In order for an application to transfer data, the application layer determines whether communication partners are available. Whether enough cloud resources are accessible for the required communication is decided at the application layer. Applications must cooperate in order to communicate, and an application layer is in charge of this.
4. The application layer, in particular, is responsible for processing IP traffic handling protocols like Telnet and FTP. Other examples of application layer systems include web browsers, SNMP protocols, HTTP protocols, or HTTPS, which is HTTP's successor protocol.

Layers of Cloud Computing

- **Platform Layer**

1. The operating system and application software make up this layer.
2. Users should be able to rely on the platform to provide them with **Scalability, Dependability, and Security Protection** which gives users a space to create their apps, test operational processes, and keep track of execution outcomes and performance. SaaS application implementation's application layer foundation.
3. The objective of this layer is to deploy applications directly on virtual machines.
4. Operating systems and application frameworks make up the platform layer, which is built on top of the infrastructure layer. The platform layer's goal is to lessen the difficulty of deploying programmers directly into VM containers.
5. By way of illustration, Google App Engine functions at the platform layer to provide API support for implementing storage, databases, and business logic of ordinary web apps.

Layers of Cloud Computing

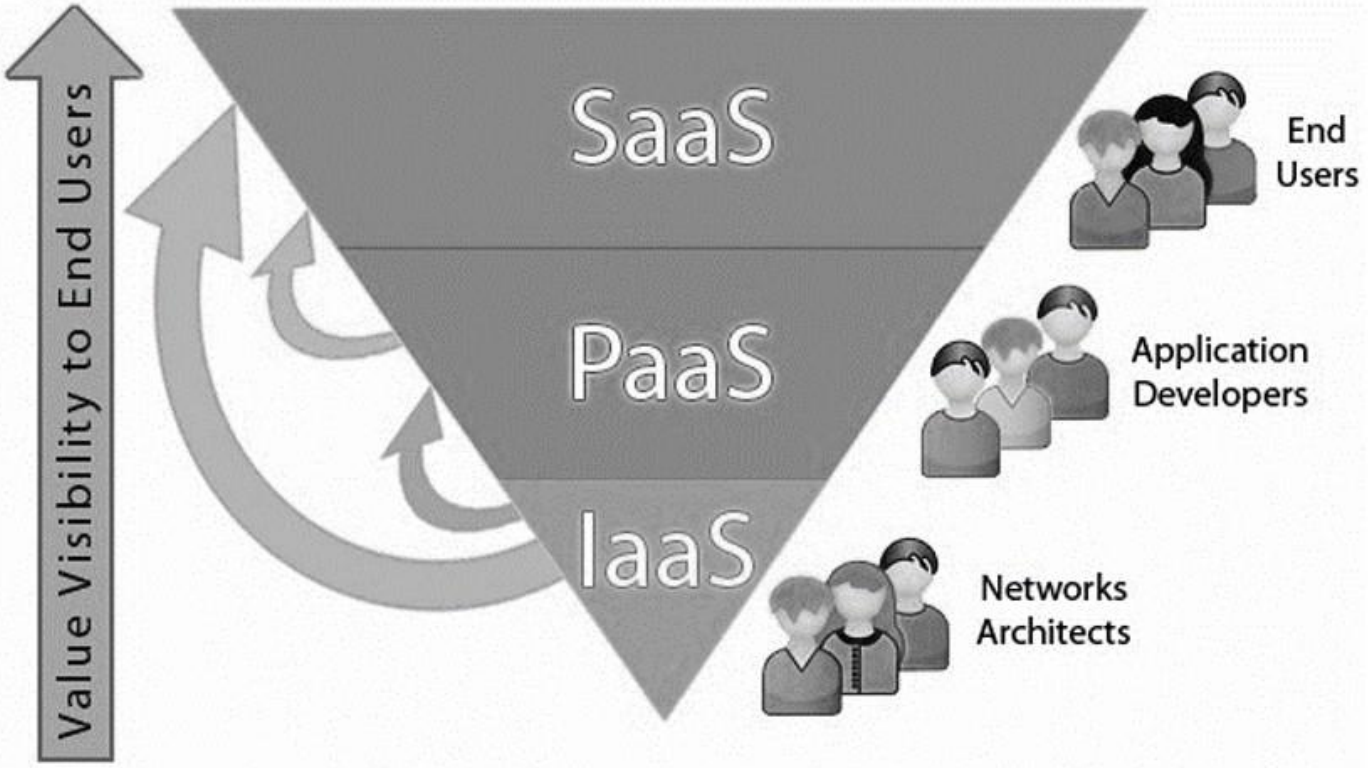
- **Infrastructure Layer**

1. It is a layer of virtualization where physical resources are divided into a collection of virtual resources using virtualization technologies like Xen, KVM, and VMware.
2. **This layer serves as the Central Hub of the Cloud Environment**, where resources are constantly added utilizing a variety of virtualization techniques.
3. A base upon which to create the platform layer. constructed using the virtualized network, storage, and computing resources. Give users the flexibility they want.
4. Automated resource provisioning is made possible by virtualization**,** which also improves infrastructure management.
5. The infrastructure layer sometimes referred to as the virtualization layer, partitions the physical resources using virtualization technologies like **Xen, KVM, Hyper-V, and VMware** to create a pool of compute and storage resources.
6. The infrastructure layer is crucial to cloud computing since virtualization technologies are the only ones that can provide many vital capabilities, like dynamic resource assignment.

Layers of Cloud Computing

- **Datacenter Layer**
- In a cloud environment, this layer is responsible for **Managing Physical Resources** such as servers, switches, routers, power supplies, and cooling systems.
- Providing end users with services requires all resources to be available and managed in data centers.
- Physical servers connect through high-speed devices such as routers and switches to the data center.

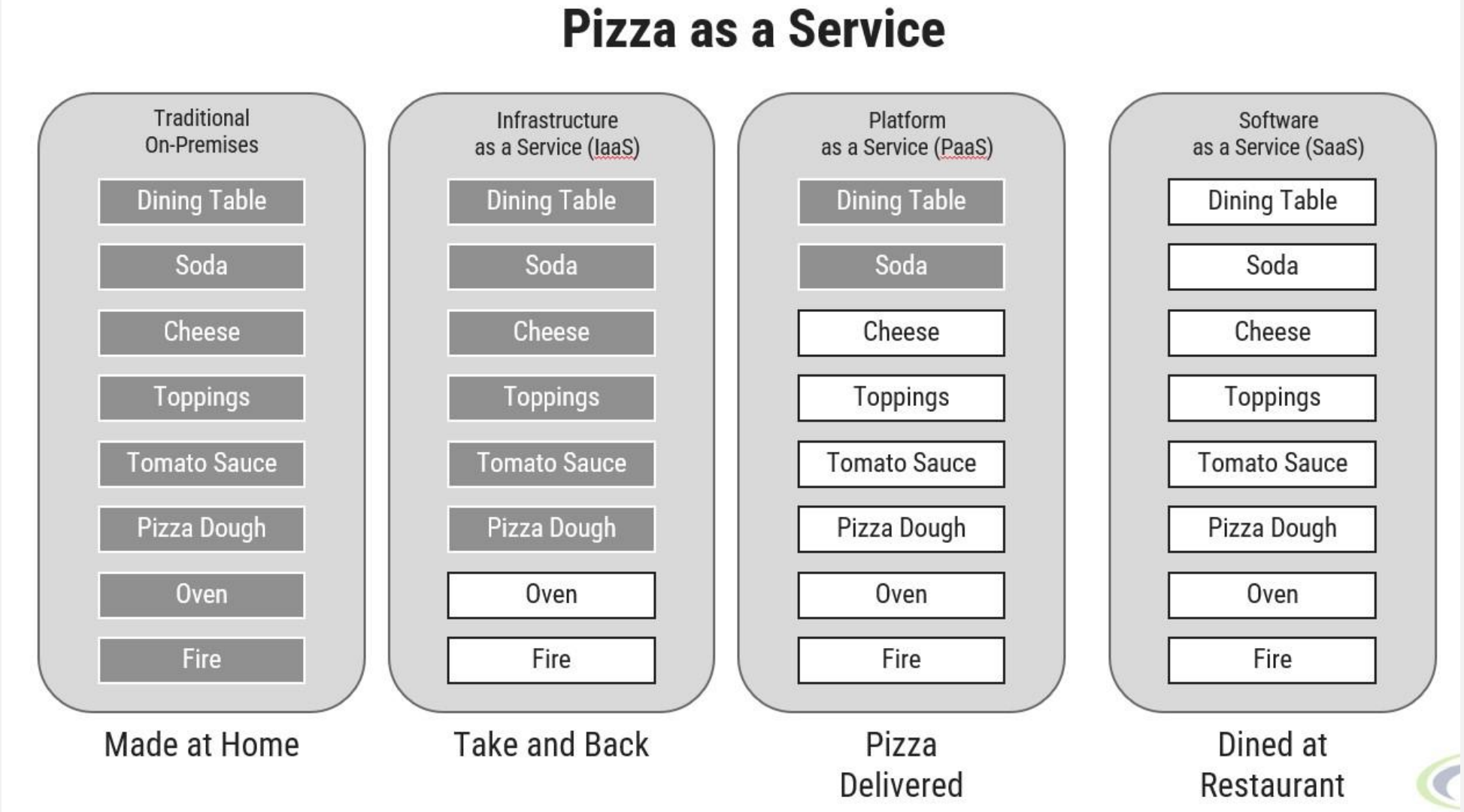
Cloud Service Models



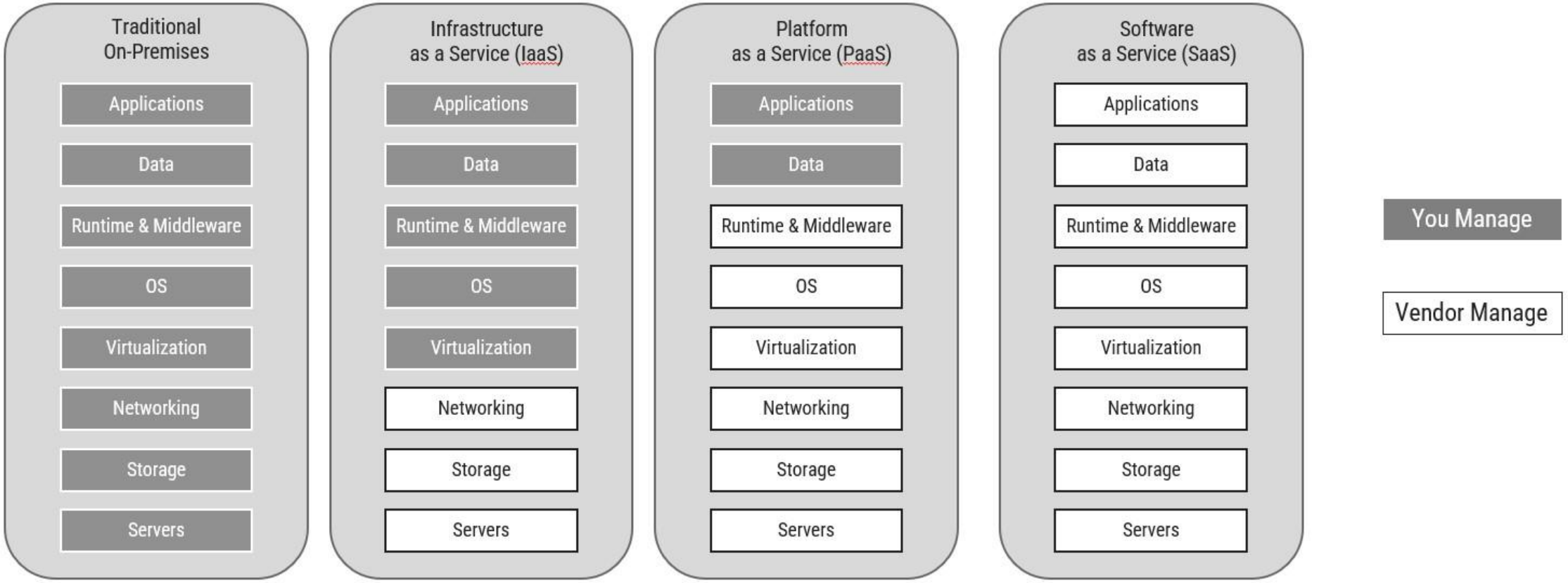
- There are **three** main service models of **cloud computing** – Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS).
- Each model represents a different part of the cloud computing stack.
- Each type of cloud service provides you with different levels of control, flexibility, and management.



Example of “as a Service”



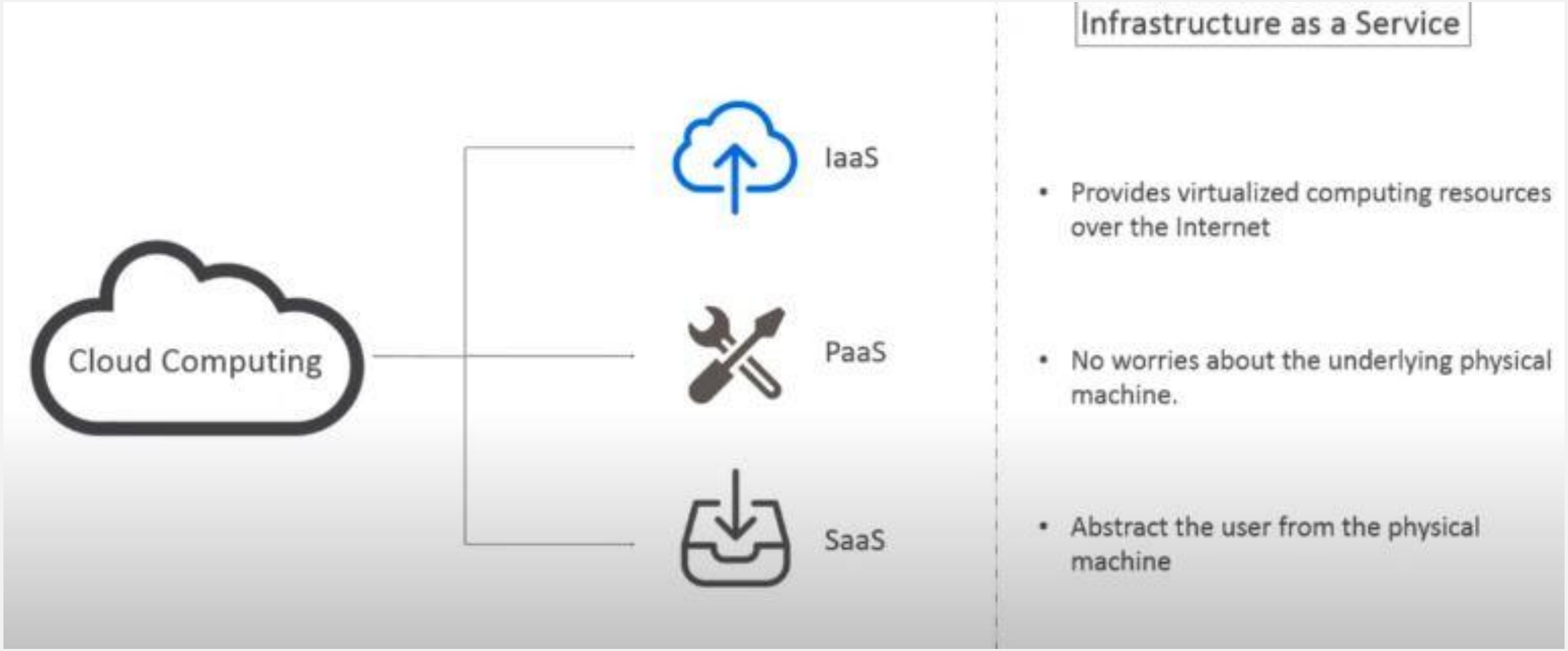
Cloud Services



Service Models

Type of Cloud Service	What It Does	Examples
Infrastructure as a service (IaaS)	Compute power, networking, and storage provided over the internet	Amazon Elastic Compute Cloud (Amazon EC2), Rackspace, Google Compute Engine
Platform as a service (PaaS)	Tools provided over the internet for making programs and applications	AWS Elastic Beanstalk, Microsoft Azure, Google App Engine
Software as a service (SaaS)	Applications and programs that are accessed and provided over the internet	Dropbox, Slack, Spotify, YouTube, Microsoft Office 365, Gmail

Service Models : IaaS



Service Models : IaaS

- A standardized way of acquiring computing capabilities on demand and over the web
- Resources include storage facilities, networks, processing power, and virtual private servers.
- This is a Pay as you go model

Service Models : Why IaaS?

- IaaS is the most flexible of cloud models
- It gives the best option when it comes to IT hardware infrastructure
- If control is required over the hardware infrastructure such as in managing and customizing according to requirements
- Ideal for Startups as it gives access to computing resources without the need to invest in them separately.
- However, the only downside with IaaS is that it is much costlier than SaaS or PaaS cloud models

Service Models : PaaS





IaaS



PaaS



SaaS

Platform as a Service

- No control over the underlying architecture including OS, storage, servers etc.
- The Cloud Provider gives the ability to the customer to deploy customer created apps using programming languages, tools etc that are provided by the Cloud Provider.

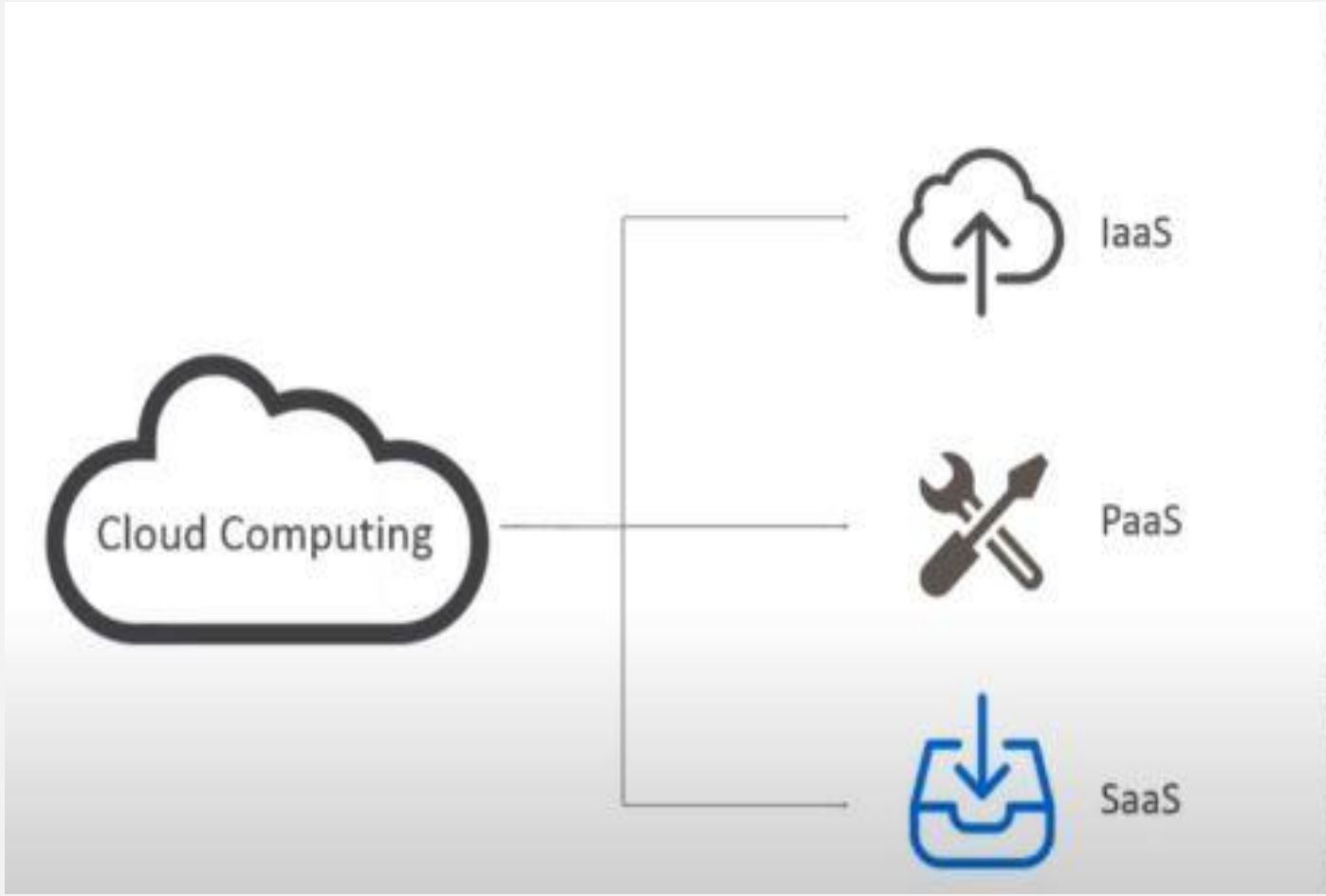
Service Models : PaaS

- In this model, companies don't pay for hosted applications; instead they pay for the things they need to build their own applications
- PaaS vendors offer everything necessary for building an application, including development tools, infrastructure, and operating systems, over the Internet
- PaaS can be compared to renting all the tools and equipment necessary for building a house, instead of renting the house itself
- PaaS examples include Heroku and Microsoft Azure

Service Models : Why PaaS?

- Preferred option for projects involving multiple developers and vendors
- Easy to create customized applications as it leases all the essential computing and networking resources
- PaaS simplifies the app development process that minimizes organizational costs
- It is flexible and delivers the necessary speed in the process, which rapidly improves development times
- Less flexible compared to the IaaS cloud model

Service Models : SaaS



Software as a Service

- Cloud Provider leases applications or softwares which are owned by them to its client
- **Example:** salesforce.com provides the CRM(Customer Relation Manager) on a cloud infrastructure to its client and charges them for it, but the software is owned by the salesforce company only

Service Models : SaaS

- Instead of users installing an application on their device, SaaS applications are hosted on cloud servers, and users access them over the Internet
- SaaS is like renting a house: the landlord maintains the house, but the tenant mostly gets to use it as if they owned it
- Examples of SaaS applications include Salesforce, MailChimp, and Slack

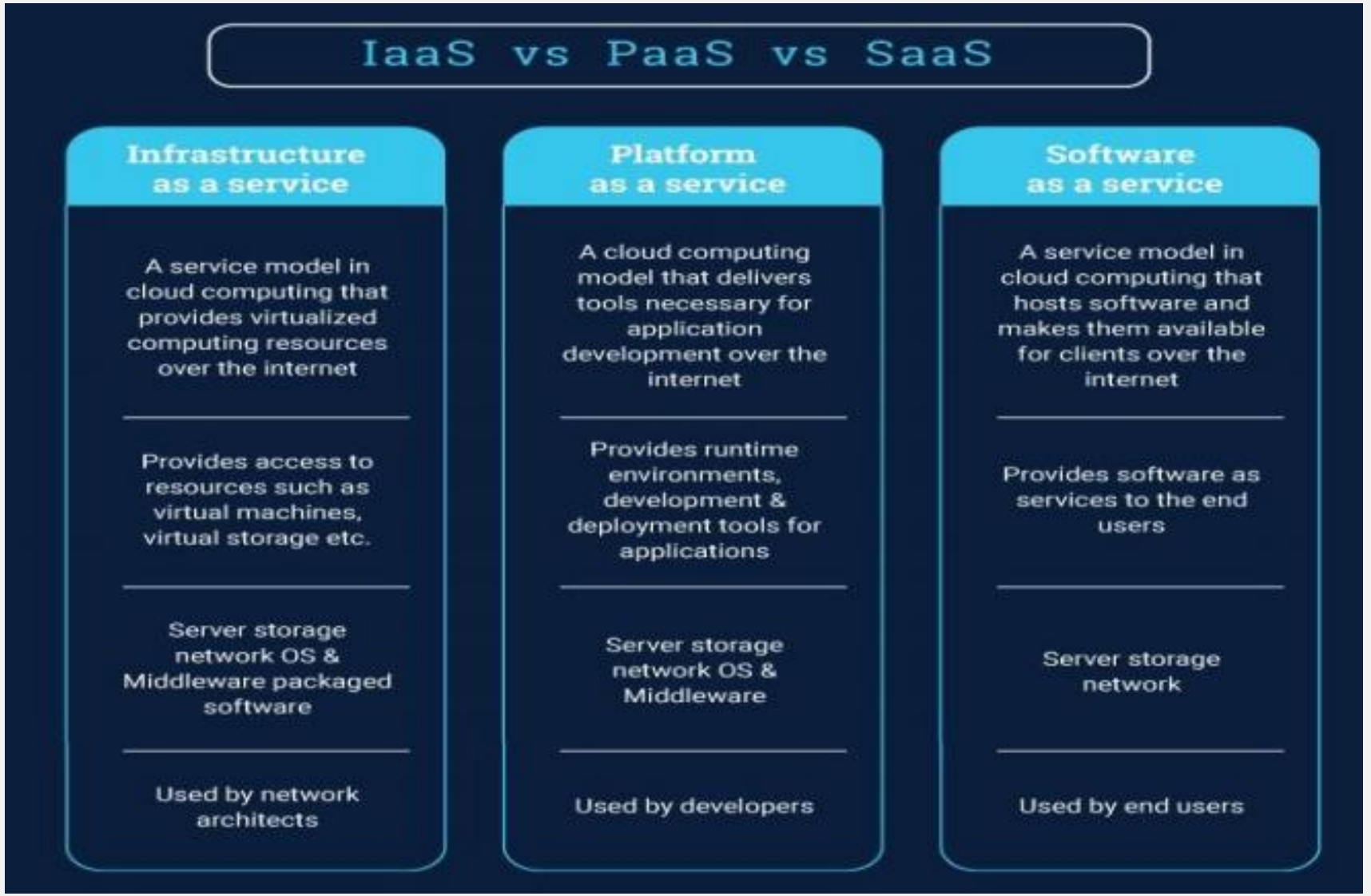
Service Models : Why SaaS?

- With SaaS, communication, transferring of content, and scheduling meetings are made easy
- Ideal choice for small-scale businesses
- Companies that do not have the necessary budget and resources to deploy on-premise hardware
- Organisations requiring frequent collaboration on their projects will find SaaS platforms useful
- Supply Chain Management, Business Intelligence, Enterprise Resource Planning (ERP), and Project and Portfolio Management are some high growth SaaS applications

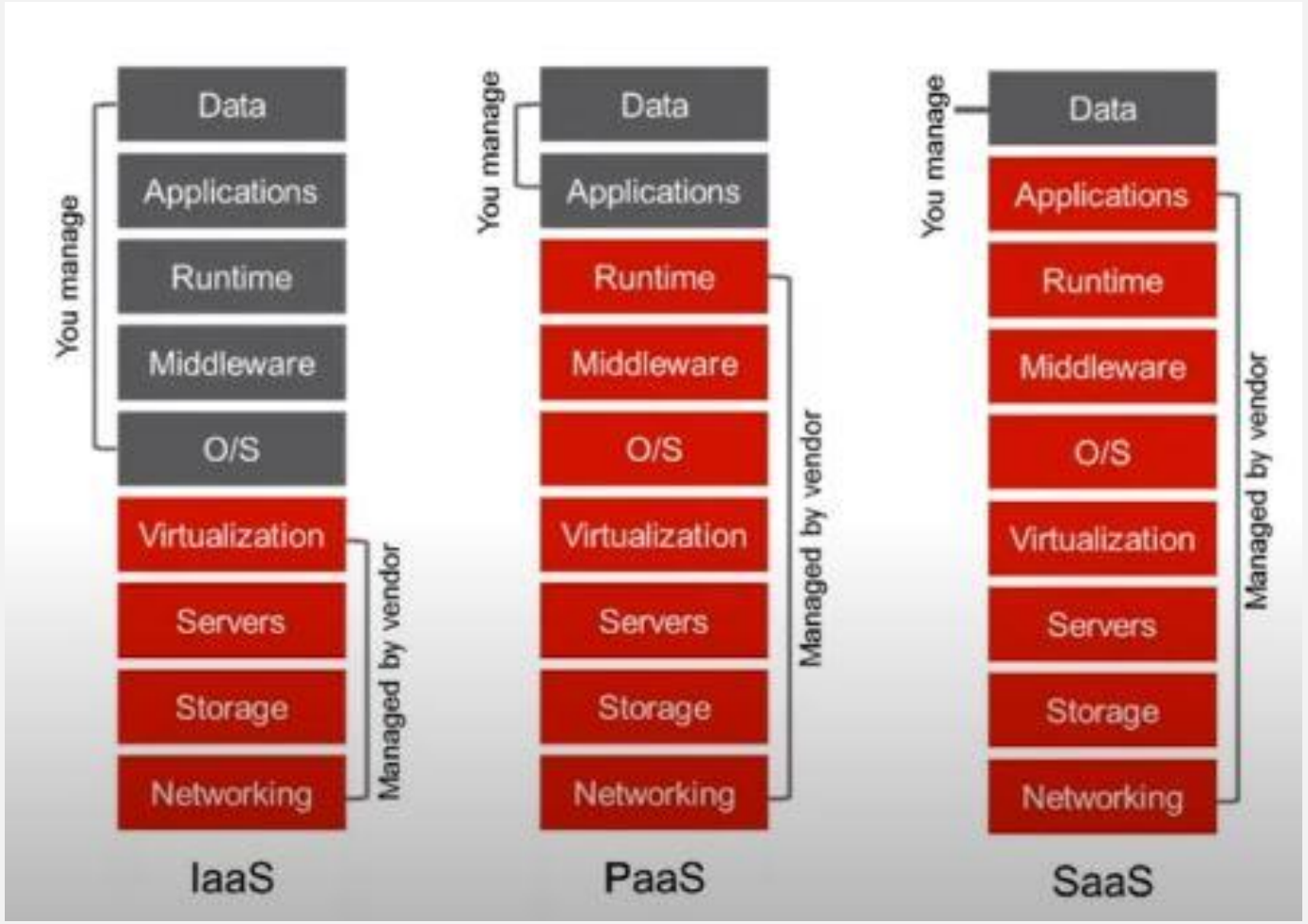
Service Models : Comparison

- IaaS - Best option where a complete virtual computing platform with powerful resources is required
- PaaS – Perfect choice if need is there to develop and test software and applications
- SaaS – Ideal for cloud-based software like email, CRM, and productivity tools

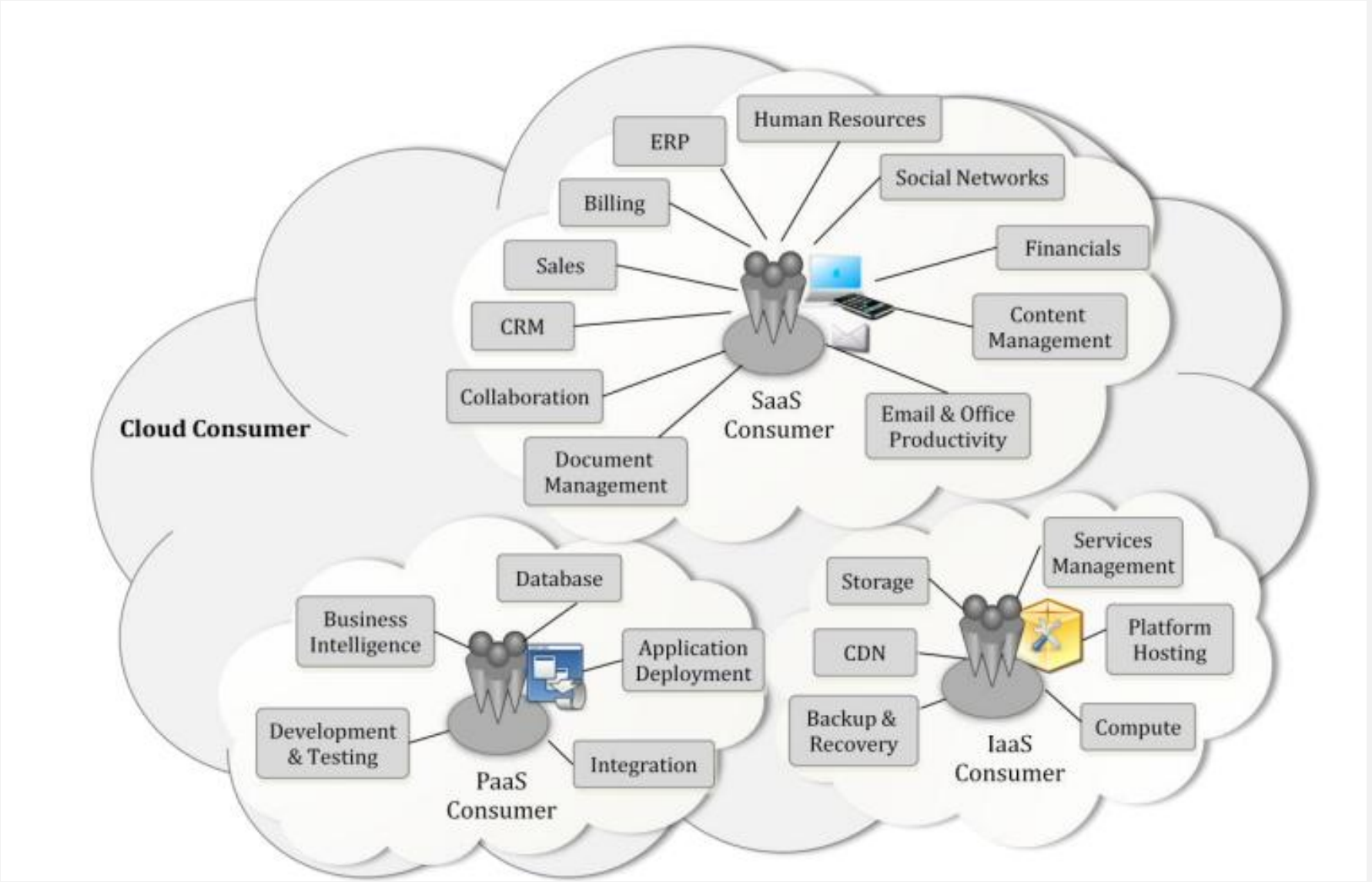
Service Models : Summary



Service Models



Cloud Consumer



Cloud Deployment Models

- Cloud deployment describes the way a cloud platform is implemented, how it's hosted, and who has access to it

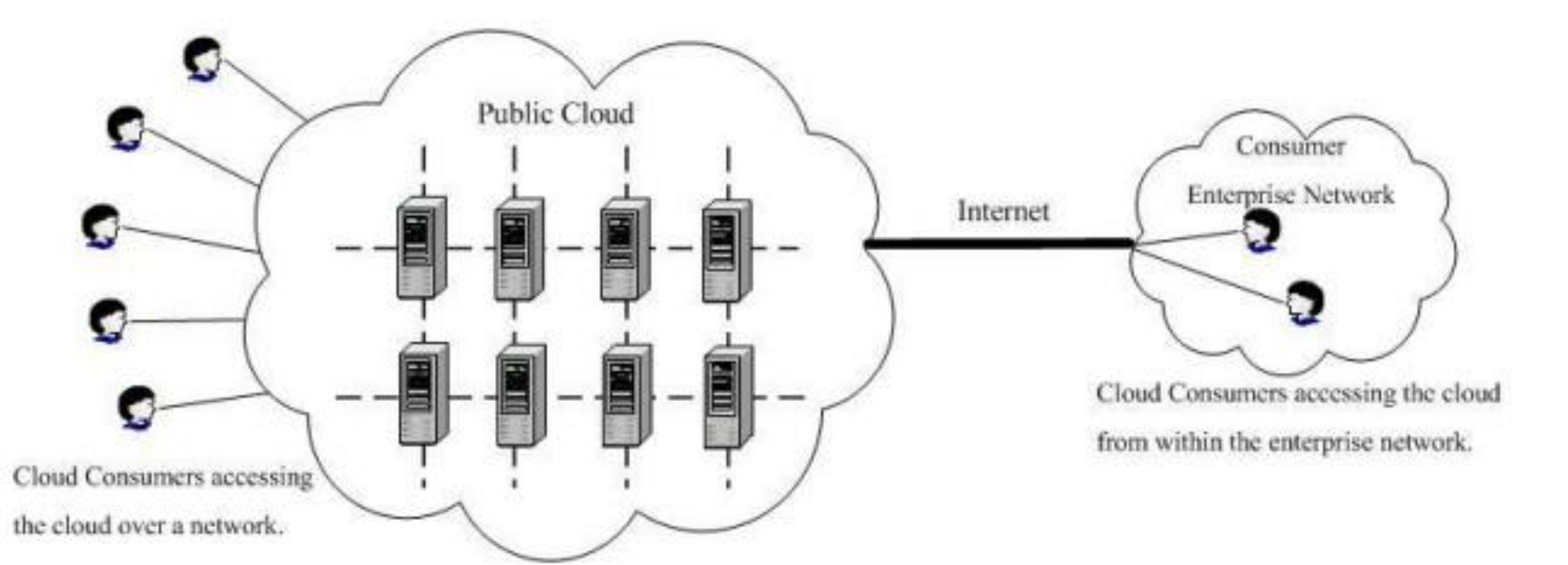
- All cloud computing deployments operate on the same principle by virtualizing the computing power of servers into segmented, software-driven applications that provide processing and storage capabilities

- Types are
  - Public
  - Private
  - Hybrid
  - Community

Public Cloud



Public Cloud



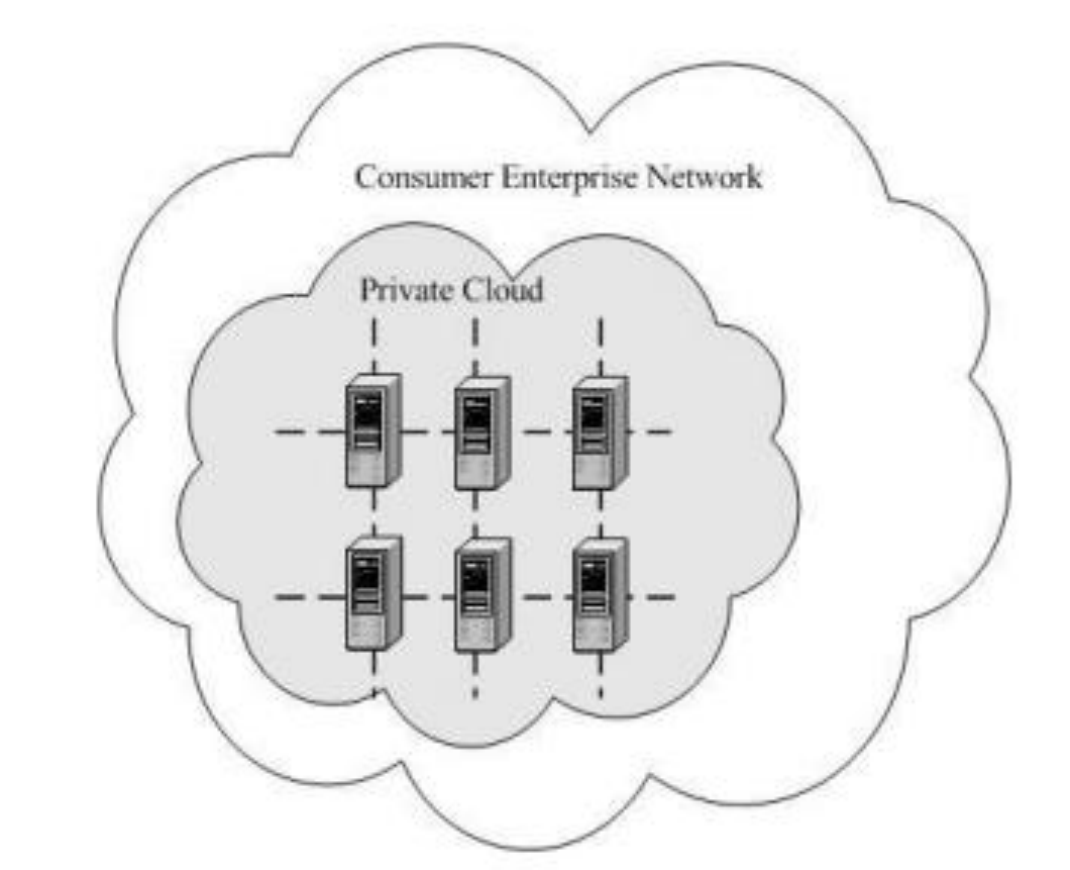
Public Cloud

- Cloud services provided for public use : Data is stored on third party servers
- Resources such as virtual machines, applications or storage are available to users remotely
- Services may be free/through subscription
- Versatility and “pay as you go” structure that allows customers to provision more capacity on demand
- It is also popular among businesses of all sizes for their web applications, webmail, and storage of nonsensitive data
- Public Cloud examples include those offered by Amazon, Microsoft, or Google

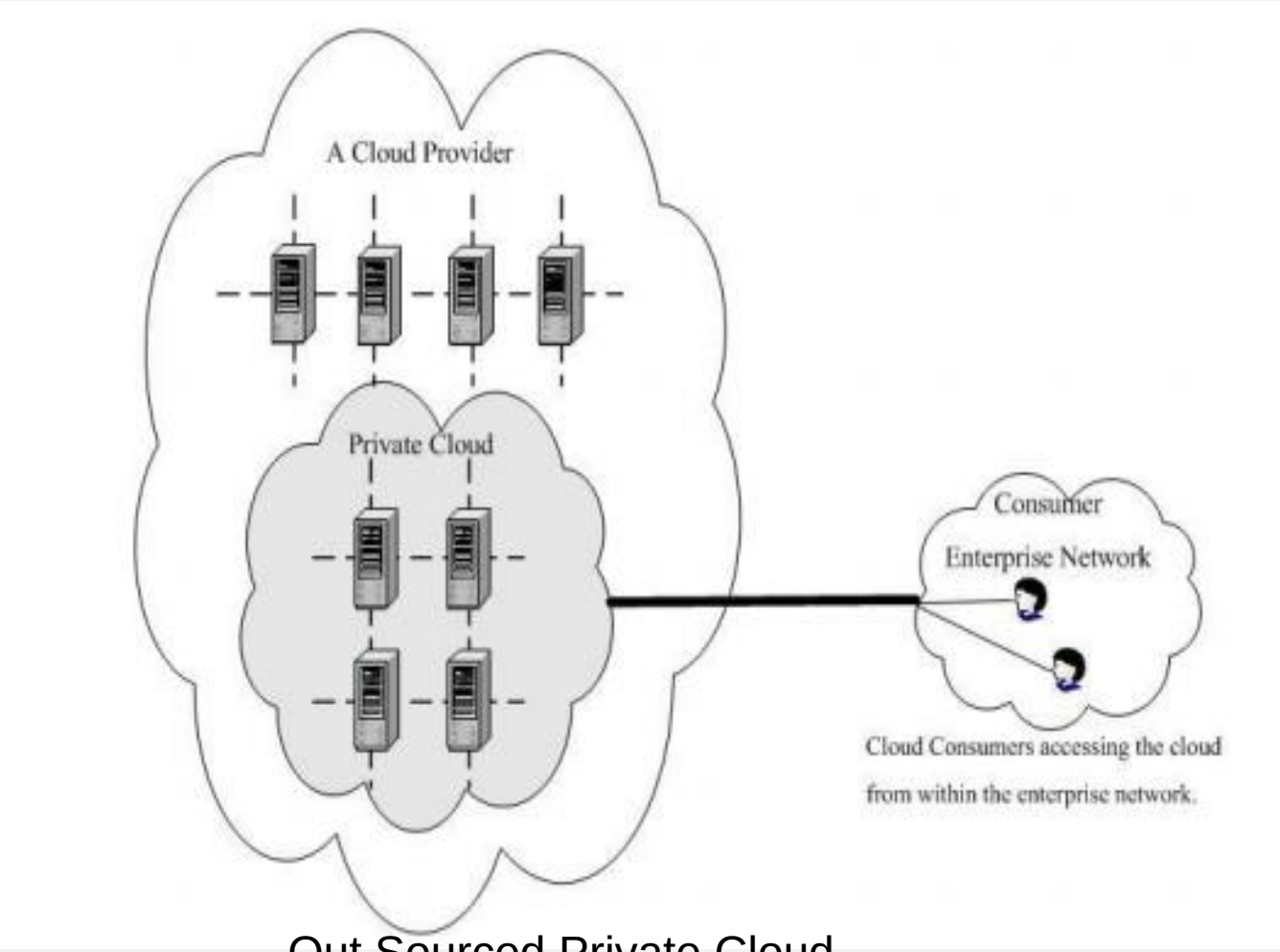
Private Cloud



Private Cloud



On Site Private Cloud

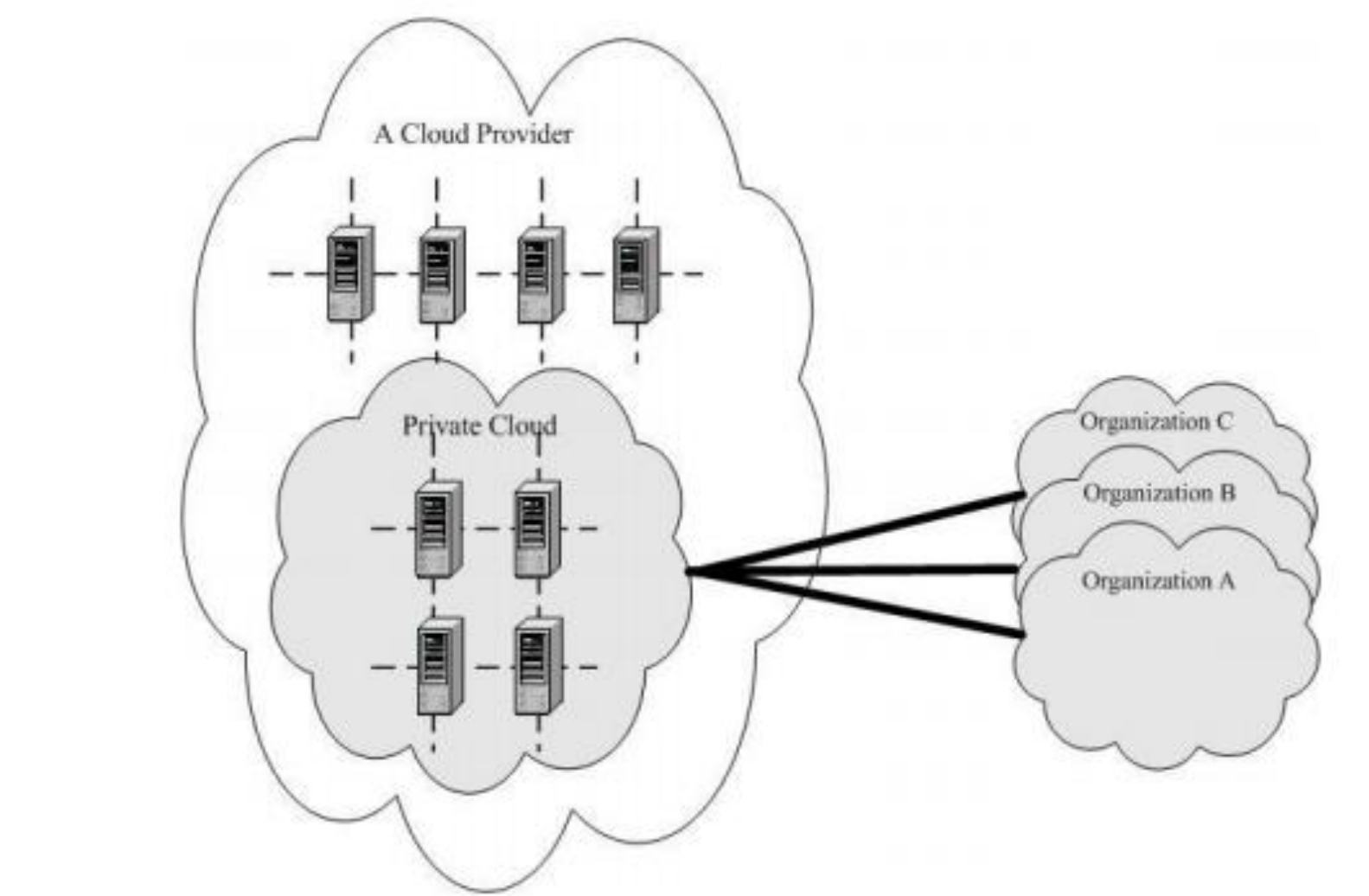


Out Sourced Private Cloud

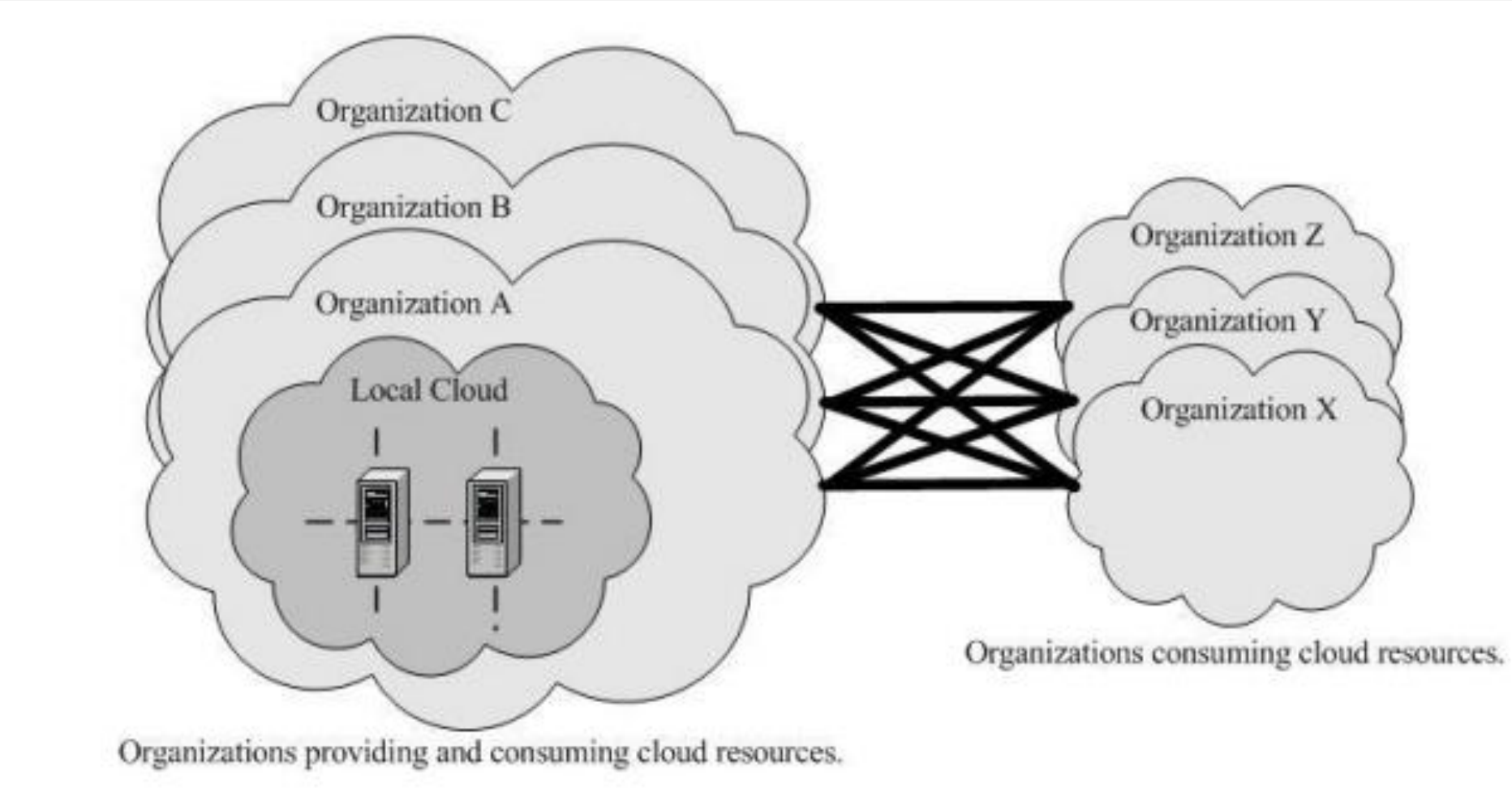
Private Cloud

- A private cloud is a cloud service that is not shared with any other organization
- The private cloud user has the cloud to themselves
- It offers greater control over security
- Private clouds are perfect for organizations that have high security requirements, high management demands and availability requirements
- The server can be hosted externally or on the premises of the owner company
- Multiple public cloud service providers, including Amazon, IBM, Cisco, Dell and Red Hat, also provide private solutions
- Highly scalable, secure, reliable and with privacy but at a high cost

Community Cloud



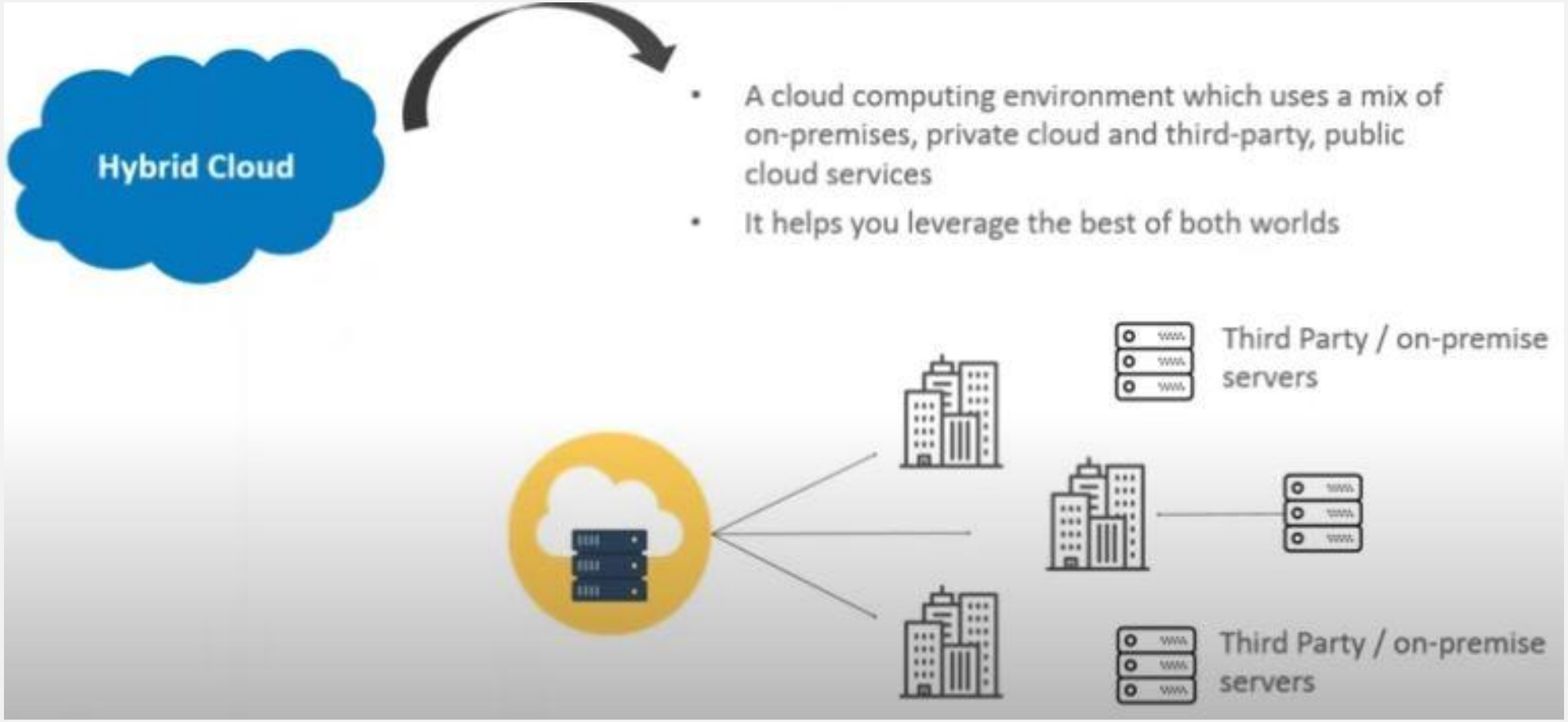
Community Cloud



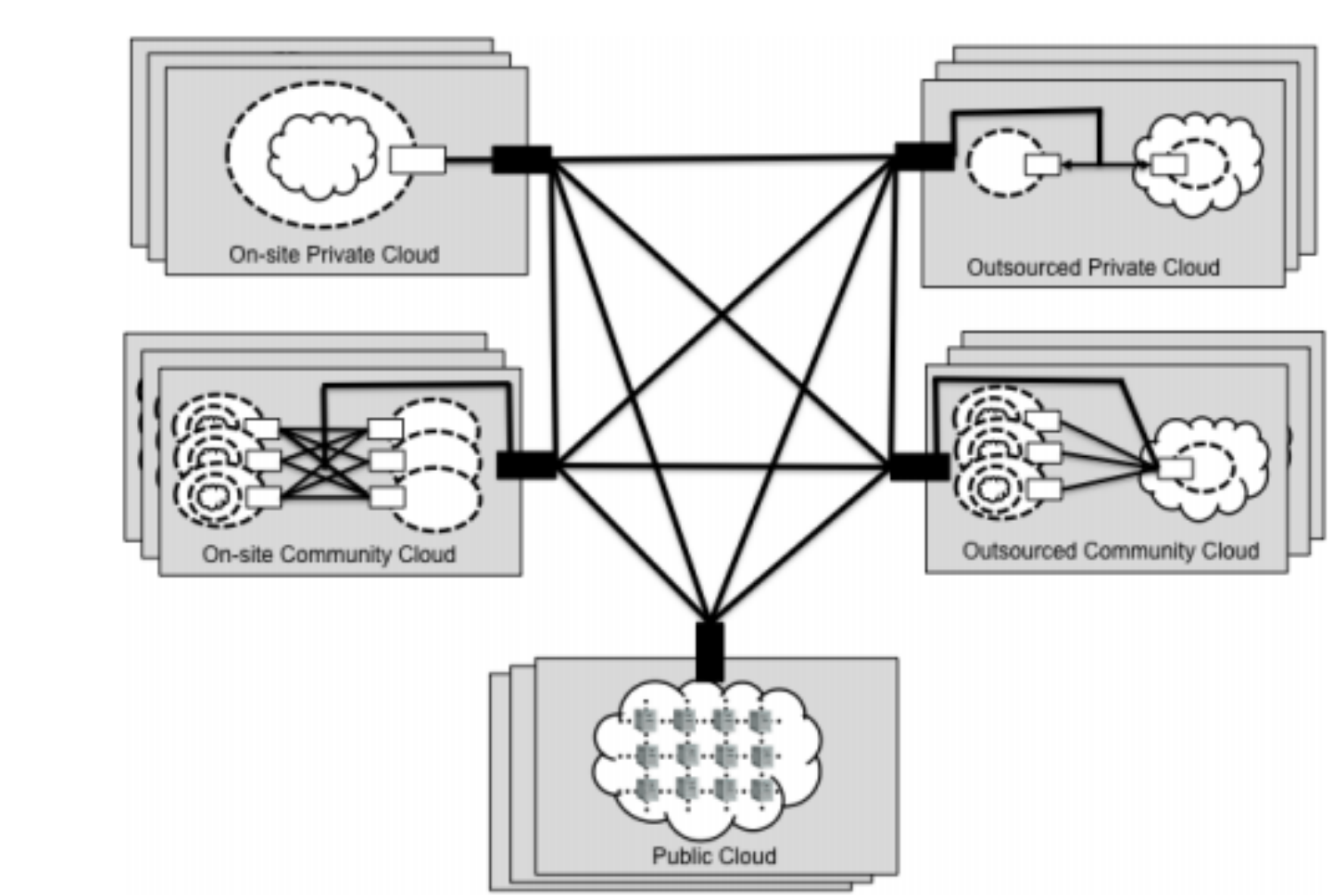
Community Cloud

- A community cloud is a private cloud that functions much like a public cloud
- Collaborative, multi-tenant platform used by several distinct organizations to share the same applications
- Users are typically operating within the same industry or field and share common concerns in terms of security, compliance, and performance
- Cost reduction, Improved security, privacy and reliability, Ease of data sharing and collaboration
- E.g. Government agencies, healthcare organizations, financial services firms, and other professional communities

Hybrid Cloud



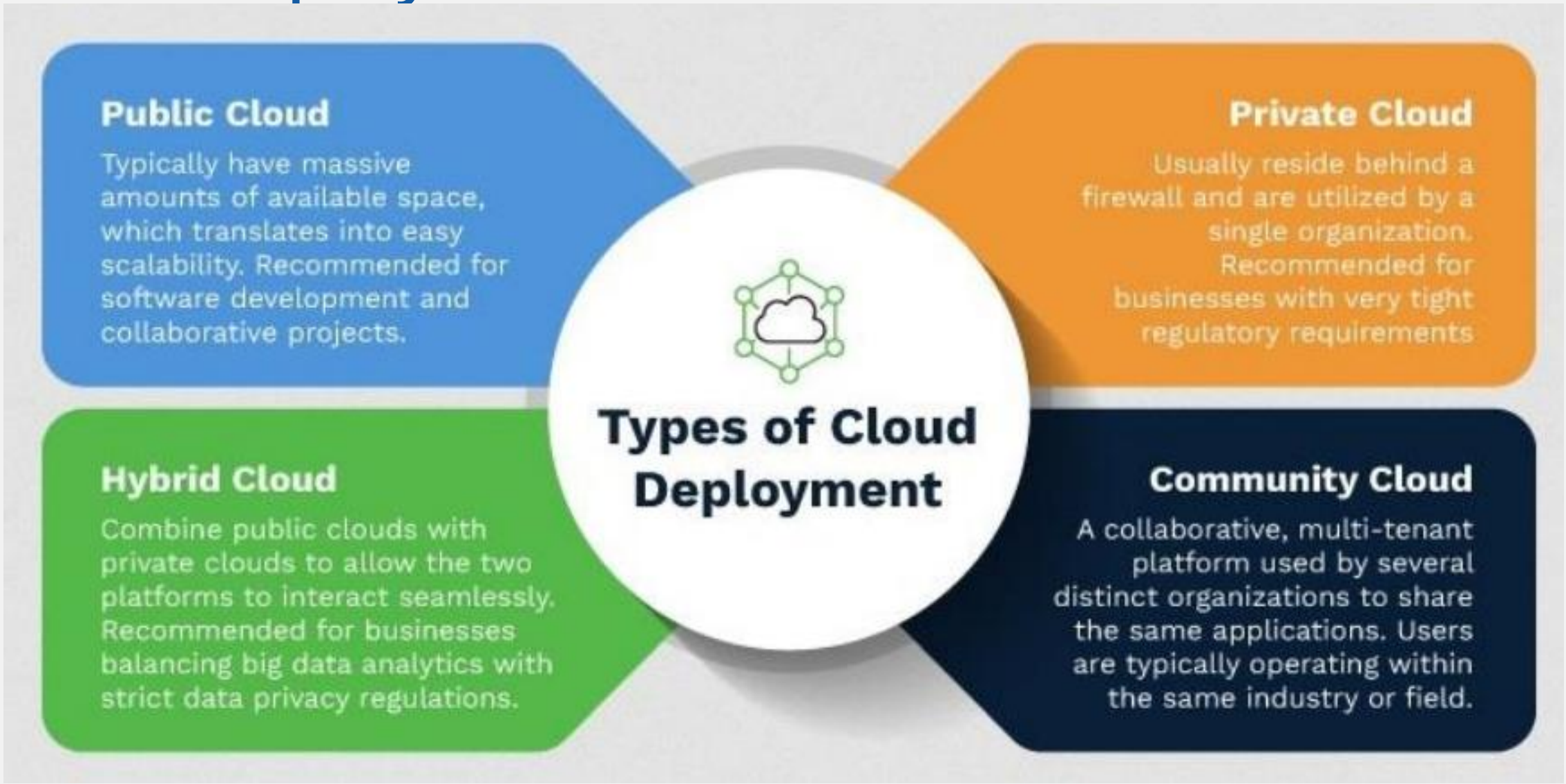
Hybrid Cloud



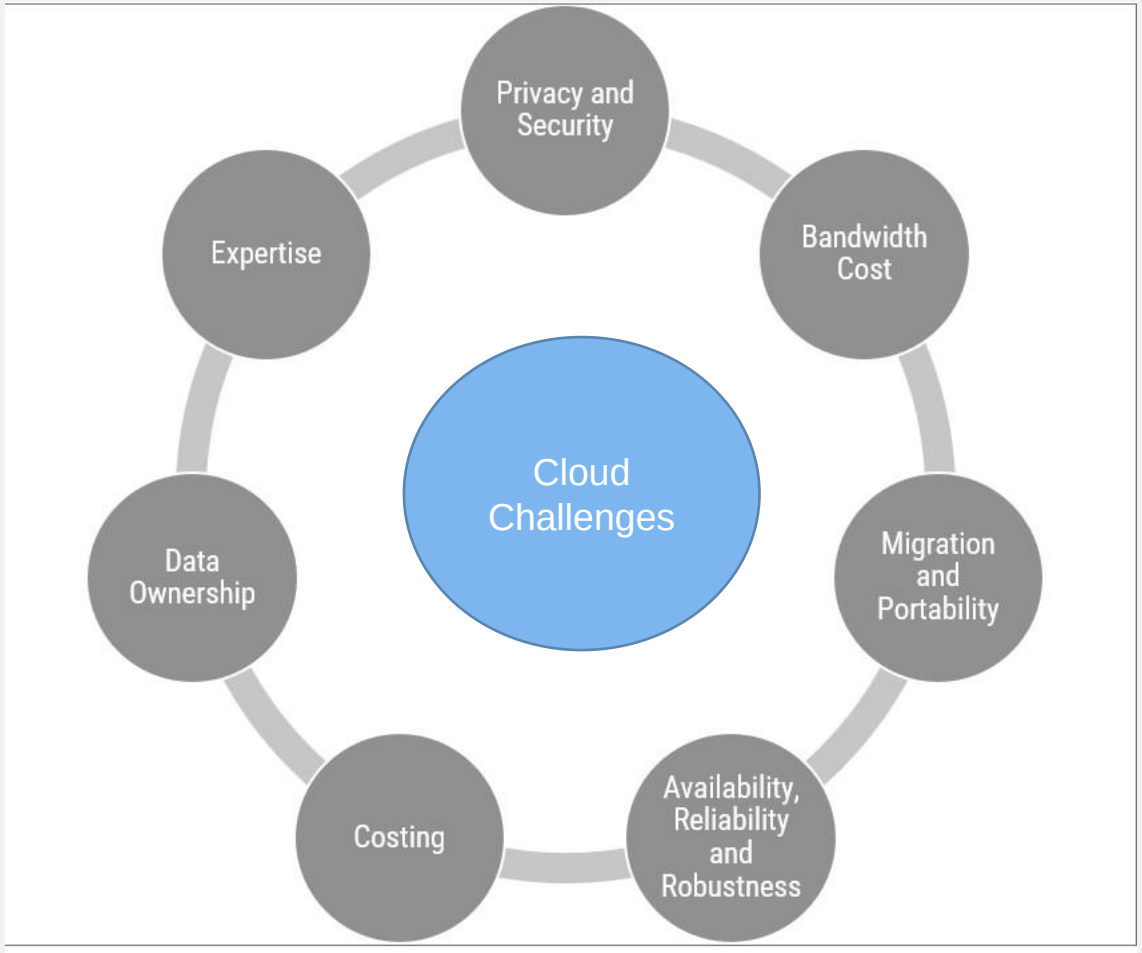
Hybrid Cloud

- A mixture of public and private cloud
- The critical activities are performed using private cloud while the non-critical activities are performed using public cloud
- The best features of the three deployment models is taken
- Mission-critical workloads on a secure private cloud and deploying less sensitive ones to a public one
- Benefits are Improved security and privacy,
- Enhanced scalability and flexibility and Reasonable price

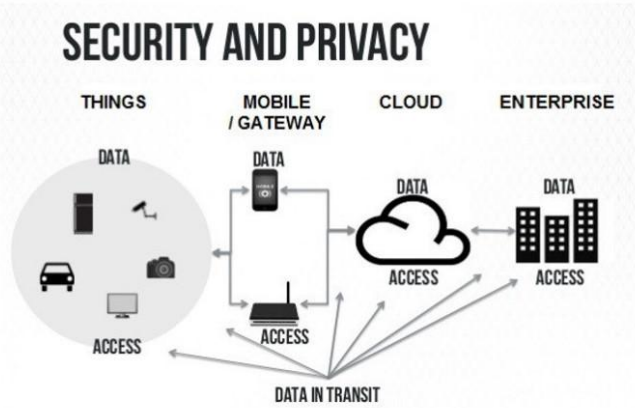
Cloud Deployment Models



Cloud Challenges



Cloud Challenges



The following are solutions to this privacy and security challenge.

- Periodic monitoring of the network activities
- Select private cloud if the data is confidential
- To reduce the risk of being exposed, use recognized antivirus solutions.
- Before signing the contract with a cloud service provider, it is necessary to read and understand the regulations involved in the service being provided.

Security & Privacy

Availability, Reliability and Robustness

- In downtime, it would miss critical data so reliability of the process has to be monitored.
- The process should be robust towards handling data at different rates.
- Data could be flooded or slowed at anytime, in the both situations it should be handled effortlessly.

Cloud Challenges

Legal and regulatory issues also need attention

- Providers may choose to locate data anywhere
  - The physical location of data centers determines the set of laws applied to the management of data
- Specific cryptography techniques can not be used
  - They are not allowed in some countries
- Country laws can impose that sensitive data are to be stored within national borders
  - e.g., patient health records

Privacy & Compliance Law

Vendor Lock-In & Standarization

- A major concern of cloud computing users
  - Having their data locked-in by a certain provider
    - Users may want to move data and applications out from a provider that does not meet their requirements
    - Cloud computing infrastructures and platforms do not employ standard methods of storing user data and applications.
    - User data are not portable
  - The answer to this concern is standardization
- The Cloud Computing Interoperability Forum (CCIF)
  - Formed by organizations like Intel, Sun, and Cisco

Cloud Challenges

- Data centers consume large amounts of electricity
  - 100 server racks can consume 1.3 MW of power
  - Another 1.3 MW are required by the cooling system
    - Costing USD 2.6 million per year
  - Data centers significantly impact the environment
    - In terms of CO2 emissions from the cooling systems

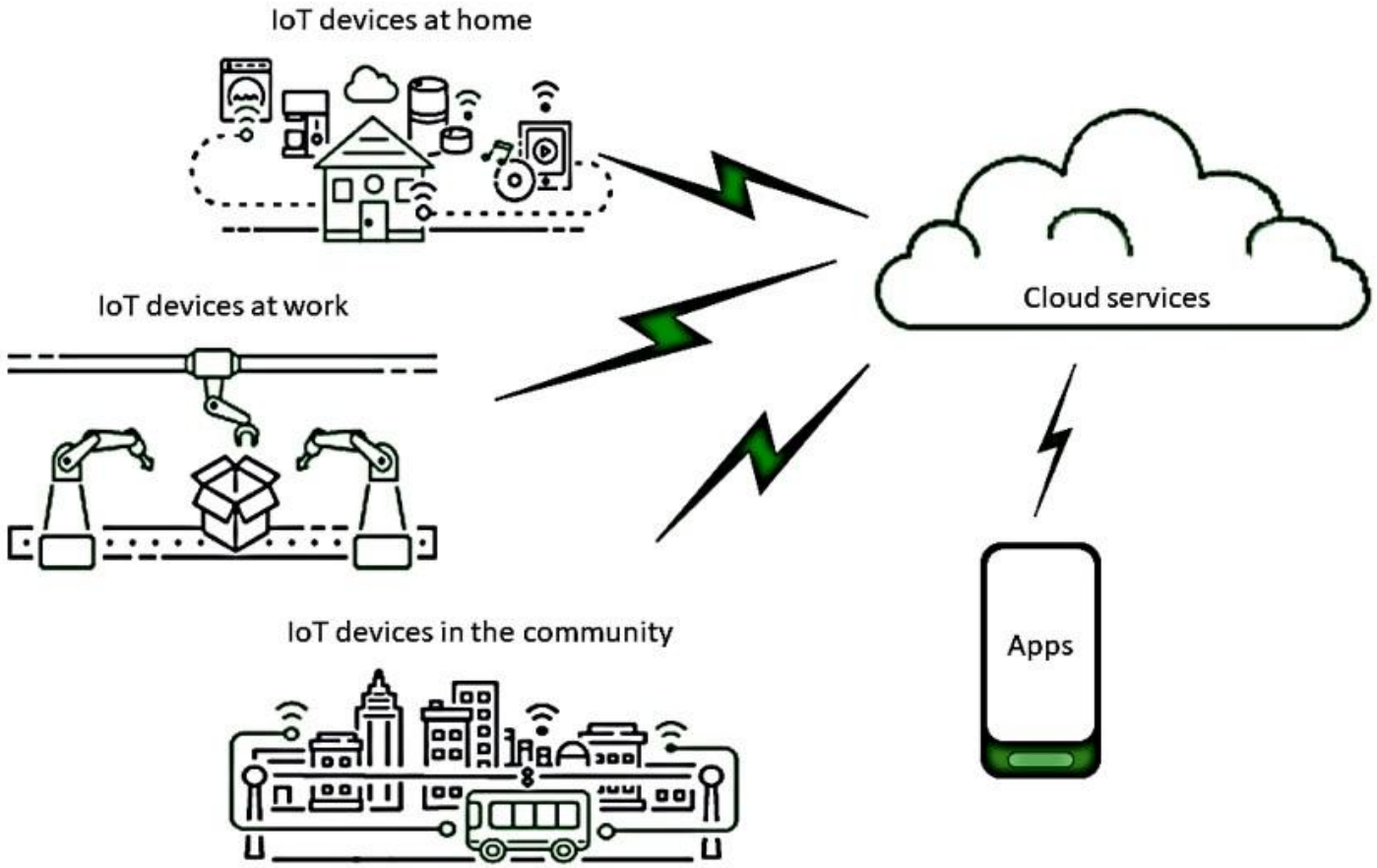
Energy Consumption

Resource Management & Dynamic Migration

- Dynamic VM mapping policies may leverage the ability to suspend, migrate, and resume VMs
  - As an easy way of preempting low-priority allocations in favor of higher-priority ones
- Migration of VMs also brings additional challenges
  - e.g., detecting when to initiate a migration, which VM to migrate, and where to migrate
- Policies may take advantage of live migration of virtual machines to relocate data center load

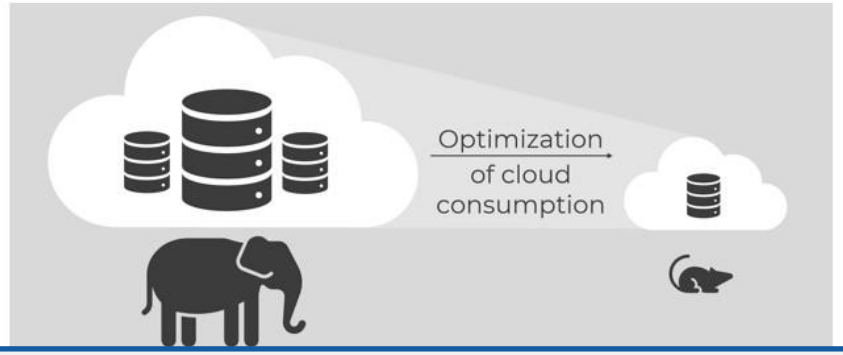
Cloud Challenges : Bandwidth Cost

- Bandwidth is one of the challenges because of continues data transferring from the IoT devices.
- IoT is all about data, and in most cases, this would be big data. Hence, huge investment in storage is needed.
- Cloud computing is preferred for storage and processing in IoT.
- Small-scale IoT application demanding lesser resources.
- But if the application is data concentrated, then the investment in bandwidth would be considerable.



Cloud Challenges

- One of the main advantages of cloud is that it can scale up with rising demand.
- While it is scalable and flexible, an organization should plan its budget carefully.
- Wrong selection for subscription without having clear vision and planning, it may lead to unnecessary cost.



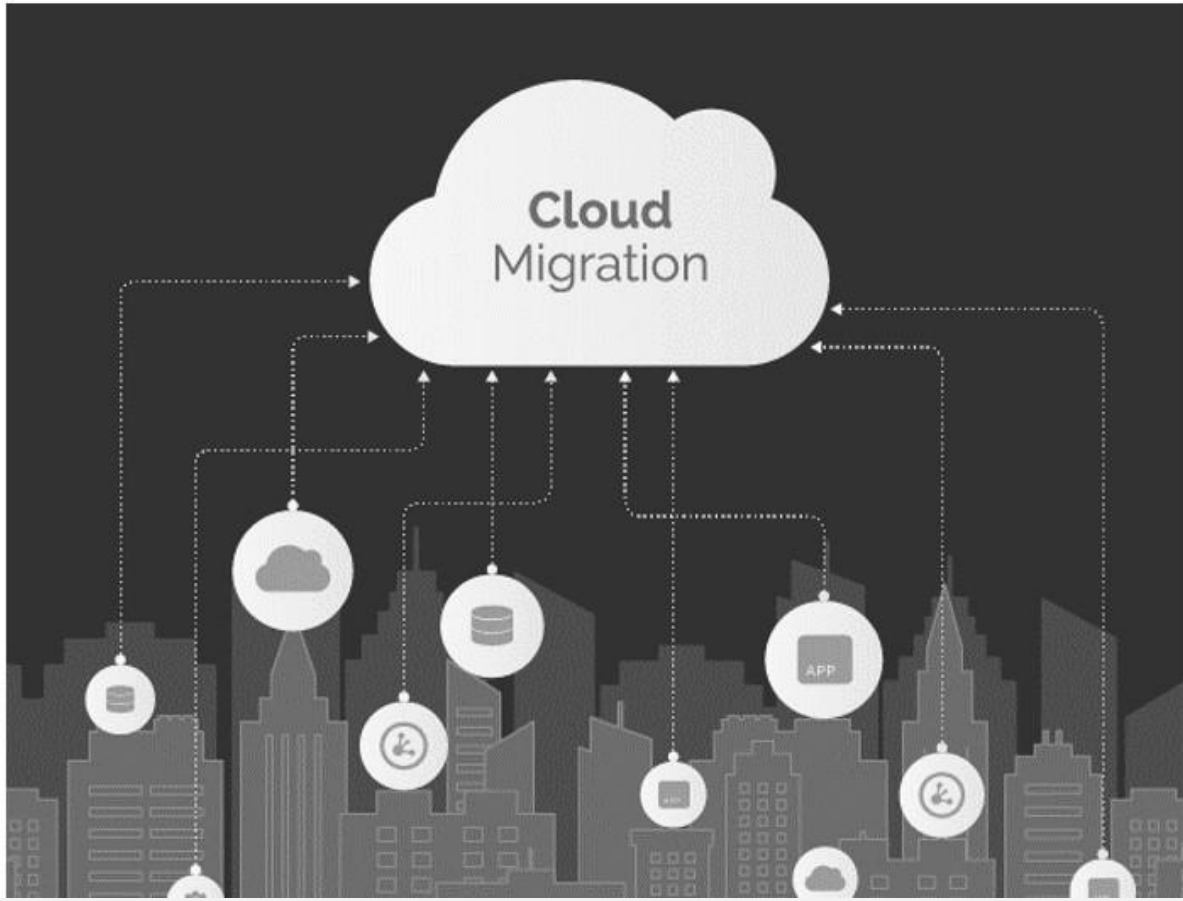
Cost

Data Ownership



Cloud Challenges : Cloud Migration

- When data is to be moved to or migrate from the cloud, we have to take care of the followings.
  - How easy and safe is it to move the data?
  - How much downtime would this process require?
  - What is the strategy to migrate data to the cloud?
  - Will it be easy to select out of the cloud and take data back to the infrastructure?
  - How much would it cost?
  - Would there be support offered to migrate smoothly to another cloud service provider?



Cloud Migration

- A cloud migration strategy is the plan an organization makes to move its data and applications from an on-premises architecture to the cloud.
  - To get advantages Scalability, Cost, Performance, Digital experience

- Cloud Migration Challenges
  - Data Security and Compliance
  - Downtime
  - Interoperability
  - Cost Management

For Details Refer: https://www.cisco.com/c/en_in/solutions/cloud/what-is-a-cloud-migration-strategy.html#~process

The Cloud migration process

- Planning your migration:
  - List out reasons for the move and which strategy can best support them.
  - calculate your cloud server requirements based on current application resource requirements

- Choosing your cloud environment:
  - Decide what kind of cloud model you want to adopt.
  - Whether you choose public cloud, hybrid cloud, private cloud, or multicloud

- Migrating your apps and data:
  - complying with security policies and planning for data backup and recovery

- Validating post-move success:
  - comparing pre- and post-move application performance, from both a technical and business perspective, in a low-risk test environment

The Cloud migration strategies

- **Rehosting ("lift and shift"):** lifting your stack and shifting it from on-premises hosting to the cloud. You transport an exact copy of your current environment without making extensive changes

- **Replatforming:** making a few further adjustments to optimize your landscape for the cloud.

- **Repurchasing:** moving your applications to a new, cloud-native product, most commonly a SaaS platform (for example, moving a CRM to Salesforce). The challenge is losing the familiarity of existing code and training your team on the new platform.

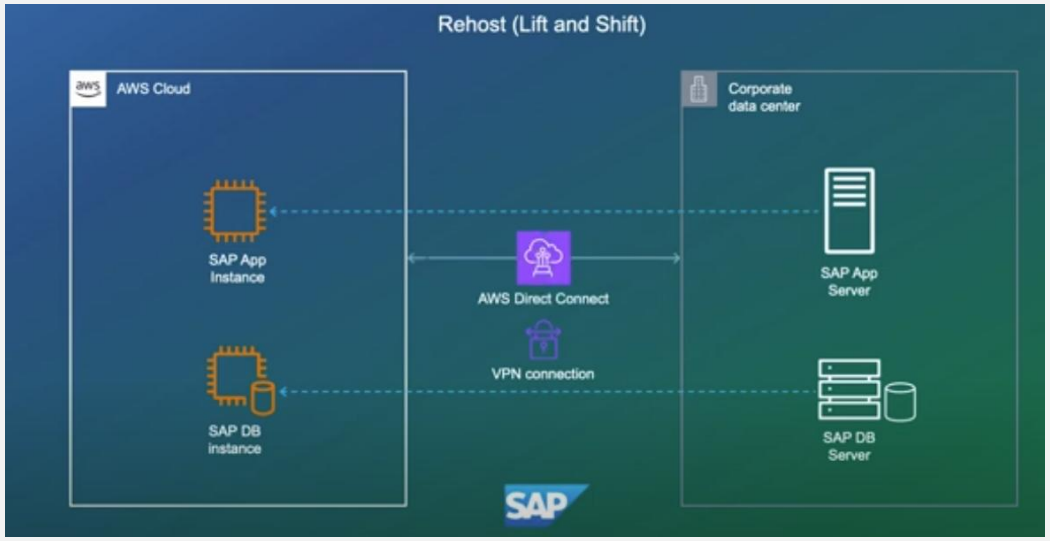
- **Refactoring:** rebuilding your applications from scratch. This is usually driven by a business need to leverage cloud capabilities that are not available in your existing environment, such as cloud auto-scaling or serverless computing

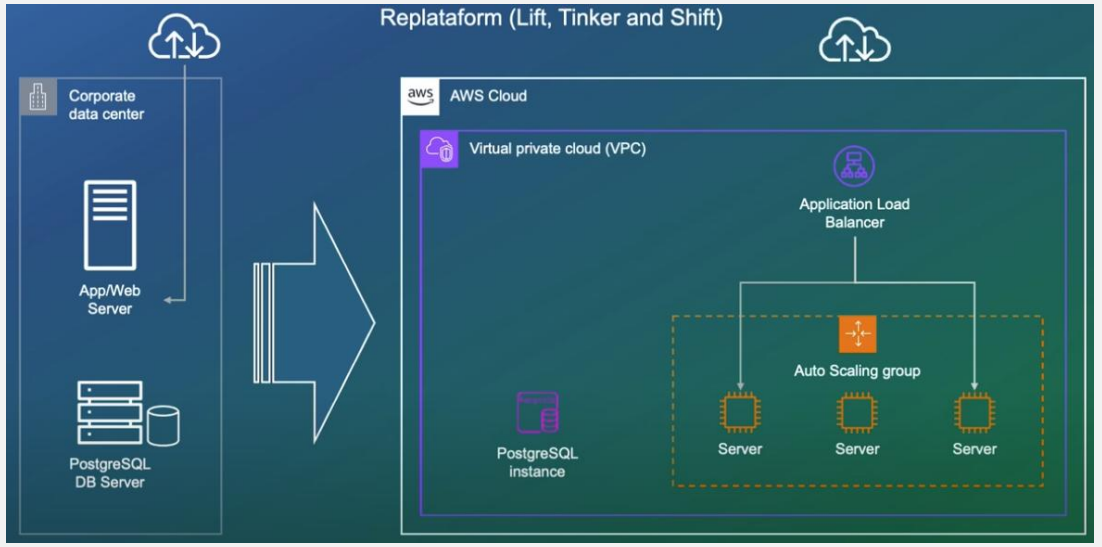
The Cloud migration strategies

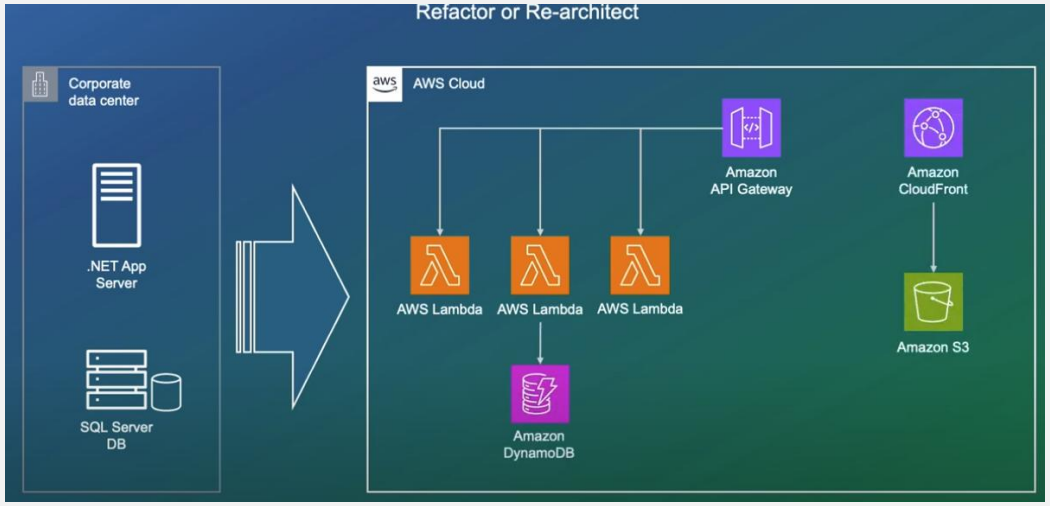
- **Retiring:** Once migration is somplete, find some applications which are not longer useful and simply turn them off. The resulting savings might even boost your business case for applications

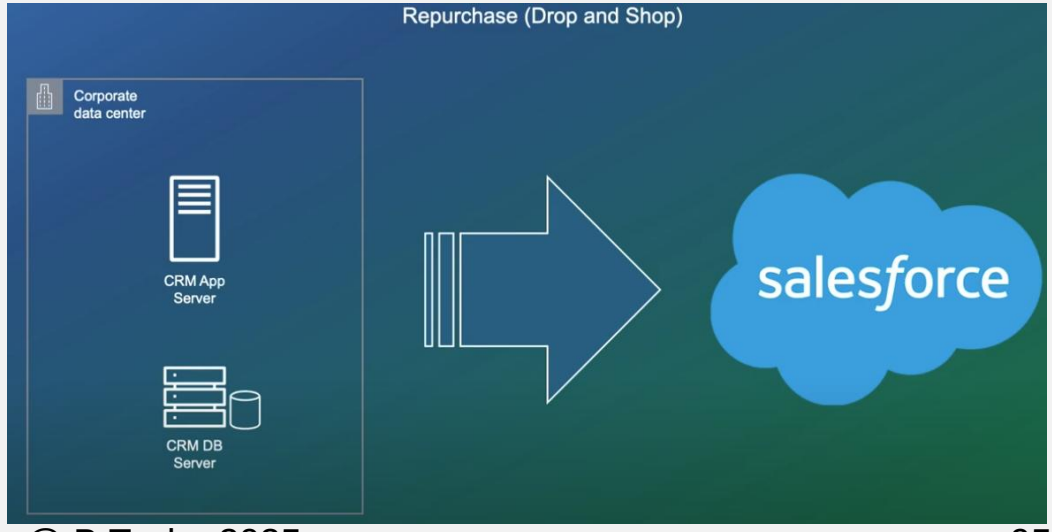
- **Retaining:** For some organizations, cloud adoption does not yet make sense. In this case, plan to revisit cloud computing at a later date

The Cloud migration strategies

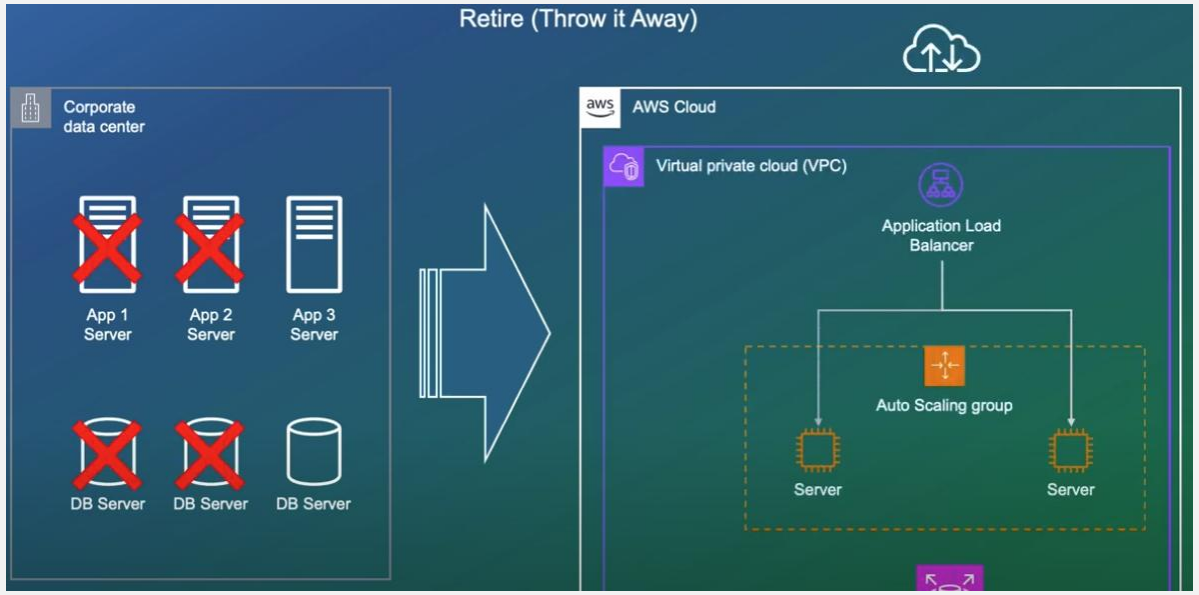


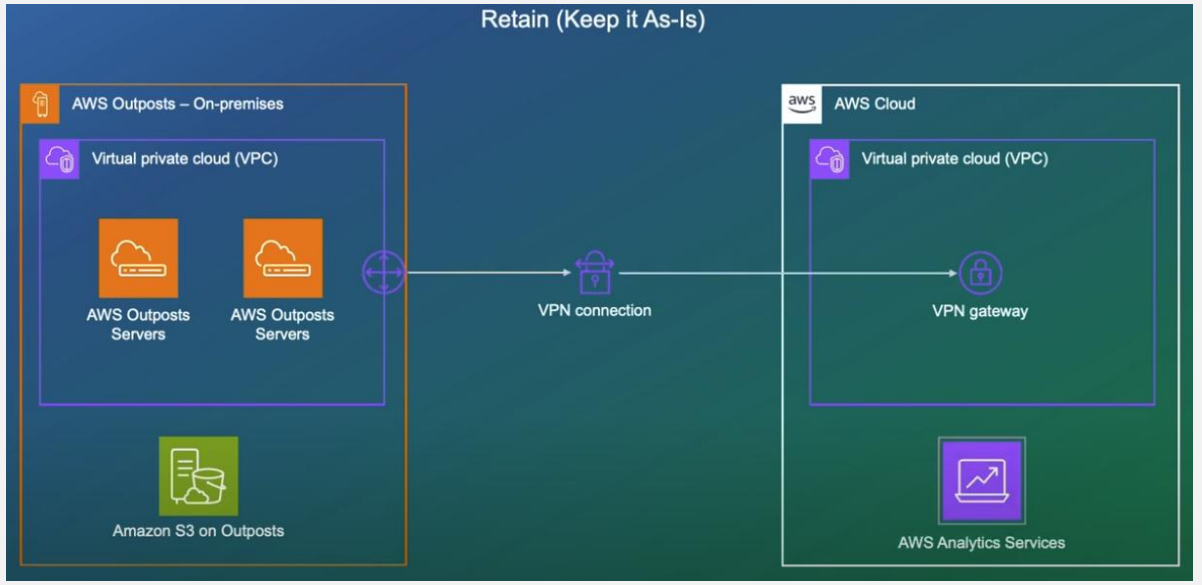






The Cloud migration strategies





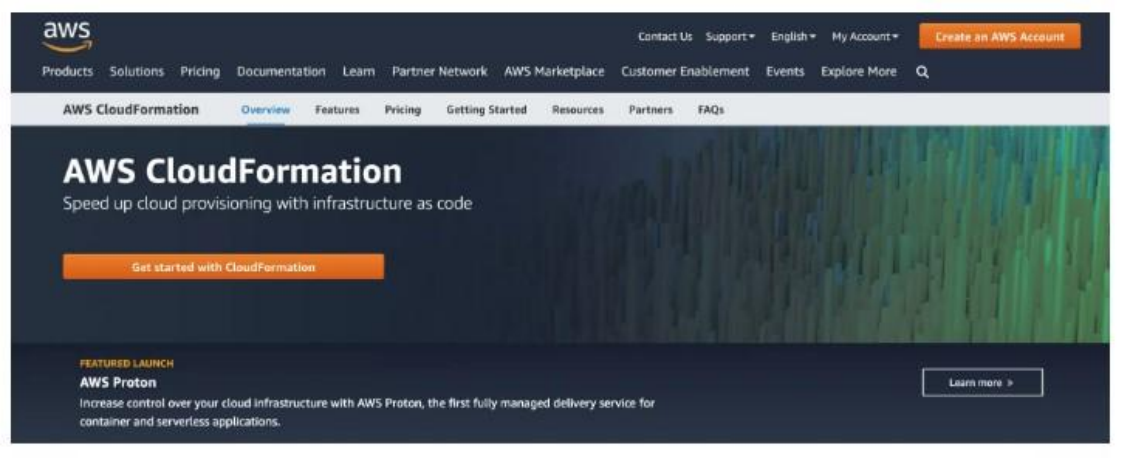
Cloud Orchestration

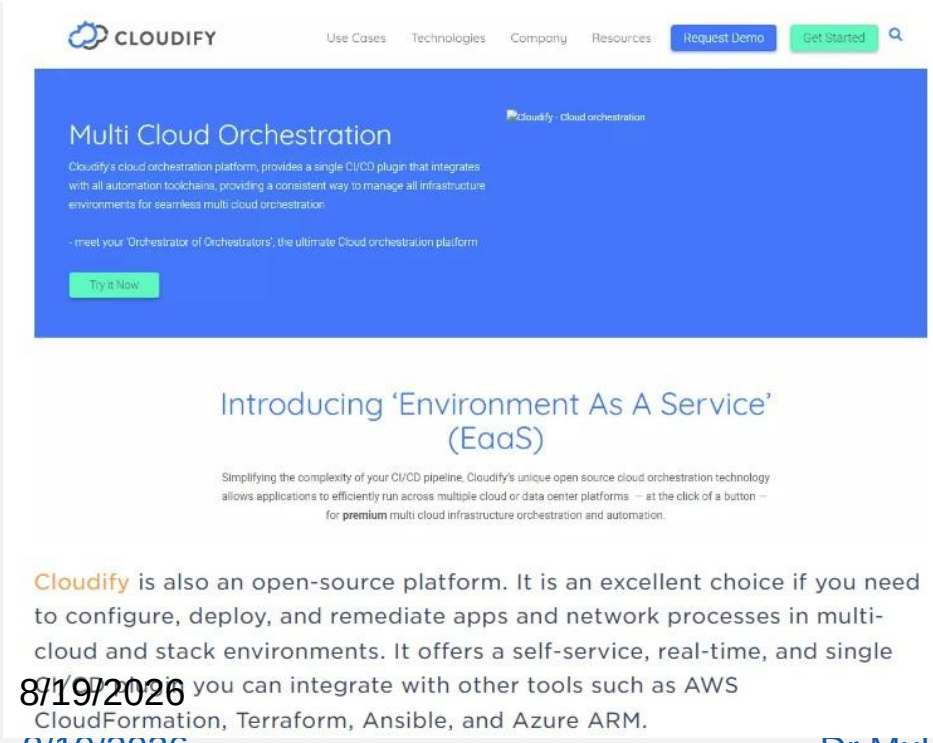
- Cloud orchestration is the coordination of multiple tools, processes, and APIs an organization uses in a cloud environment, from start to finish. This involves coordinating many automated tasks to occur within a single, synchronized IT process.
- Automated tasks are orchestrated by aligning appropriate resources, timing, security protocols, and other deployment mechanisms to ensure they start and run correctly.
- Cloud orchestration is often a complex, automated cloud infrastructure management technique. It brings together complex computing systems, including instances in multi-cloud and hybrid cloud environments, server instances in different physical locations, and automation at scale.
- That coordination takes place in what's referred to as an orchestration layer. The orchestration layer governs, controls, and coordinates service delivery among servers, networks, security protocols, virtual machines, and storage options.
- Cloud platform users generally have no access to the orchestration layer, which is only accessible by cloud provider engineers who maintain and update it.

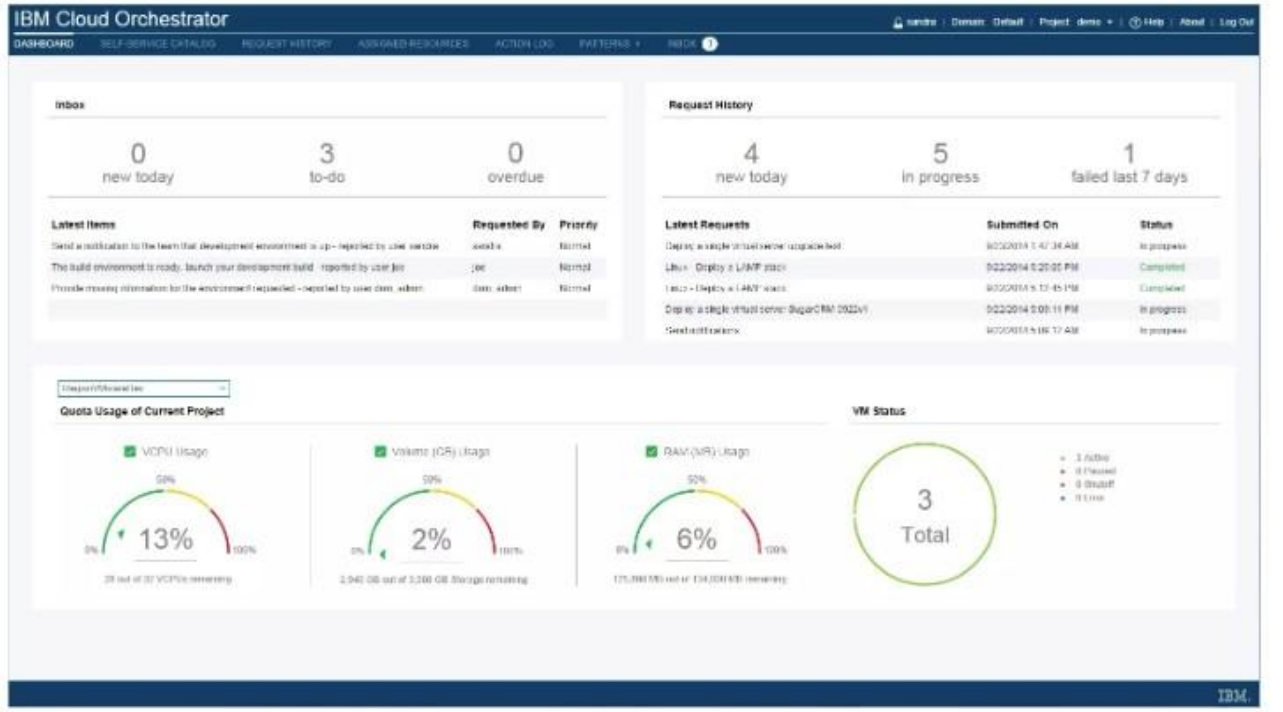
Cloud Orchestration

- Cloud orchestration includes two types of models:
  - Single Cloud model
  - Multi-cloud model
- In Single cloud model, all the applications designed for a system run on the same IaaS platform (same cloud service provider).
- Applications, interconnected to create a single workflow, running on various cloud platforms for the same organization define the concept of multi-cloud model.
- IaaS requirement for some applications, though designed for same system, might vary. This results in availing services of multiple cloud service providers.
- For example, application with patient's sensitive medical data might reside in some IaaS, whereas the application for online OPD appointment booking might reside in another IaaS, but they are interconnected to form one system. This is called multi-cloud orchestration.
- Multi-cloud models provide high redundancy as compared to single IaaS deployments.
- This reduces the risk of down time.

Cloud Orchestration : Provider







Cloud Security

Cloud Computing: Security Aspects

- The security of any computing platform including cloud computing depends on
  - Software security,
  - Infrastructure security,
  - Storage security, and
  - Network security,
- If any of these is compromised, it would result in security violation and could cause damages.
- Let us discuss these security aspects briefly.

Cloud Security

Cloud Computing: Security Aspects

1. Software Security:

- Software is the core component and plays a vital role in presenting and ensuring a secure environment.
- If there are defects created/generated during the development phase, it is a software security threat.
- Defects such as simple software implementation defects, memory allocation, design issues, and exception handling all contribute to security issues.
- This can be ensured by complete and comprehensive testing carried out at all-stages.



Cloud Security

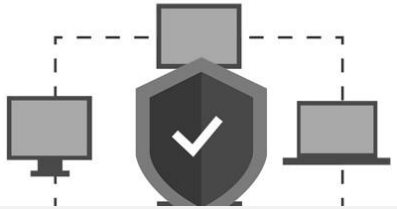
Cloud Computing: Security Aspects

2. Infrastructure Security:

- Making sure that the infrastructure provided by the CSP is safe is a must.
- Third party could also contribute to the infrastructure.
- It is extremely important to check the security vulnerabilities with the infrastructure.
- All infrastructure related guidelines should be mentioned clearly in the agreements and should be made transparent to the customer.
- If data is damaged, everything is damaged and lost.
- Hence, care should be taken to protect the infrastructure.

3. Storage Security:

- It is important to be informed of who owns the data and the location where it is stored.
- Data leak, snooping, malware attacks, etc. are all threats to the stored data and can be listed under storage security.
- Appropriate antivirus software and periodic monitoring, should help protect the data.





4. Network Security:

- Data is stored in the cloud via the Internet, and hence all network threats become a possibility.

Security Terminology

- Security Vulnerability
- Threat/ Attack
- Attack Types:
  - Passive attack
  - Active AttackThreat

Security Terminology

- Security Vulnerability : It is a weakness, flaw, or error found within a security system that has the potential to be leveraged by a threat agent in order to compromise a secure network.
  - Broken Authentication:
  - SQL Injection:
  - Cross-Site Scripting:
  - Security Misconfiguration:

- Threat/ Attack: A threat refers to the hypothetical event wherein an attacker uses the vulnerability. An attempt to gain unauthorized access to information resource or services, or to cause harm or damage to information systems.

- Attack Types:
  - Passive attack
  - Active AttackThreat

Cloud Security??

- Cloud security is the set of strategies and practices for protecting data and applications that are hosted in the cloud
- Like cyber security, cloud security is a very broad area, and it is never possible to prevent every variety of attack.
- However, a well-designed cloud security strategy vastly reduces the risk of cyber attacks
- The goal of a cloud security strategy is to reduce the threat posed by these risks as much as possible by protecting data, managing user authentication and access, and staying operational in the face of an attack

Cloud cyber attack

"Any cyber attack that targets off-site service platforms that offer storage, computing, or hosting services via their cloud infrastructure can be classified as a cloud cyber attack. This can include attacks on service platforms that utilise service delivery models like SaaS, IaaS, and PaaS."

Cloud cyber attack :Example

1. Accenture. In August of 2021, Accenture fell prey to a LockBit ransomware attack. The culprits claimed to have stolen 6TB worth of data, for which they requested a ransom of $50 million.

The largest exposed server appeared to contain credentials linked to Accenture customer accounts. One backup database contained nearly 40,000 passwords – the majority of which were in plain text.

"This cloud leak shows that even the most advanced and secure enterprises can expose crucial data and risk serious consequences," wrote security researcher Chris Vickery.

Lesson learned: Ensure that IT departments and/or cyber security personnel check to ensure correct configuration of AWS cloud servers. Attacks on misconfigured servers can cause extreme reputational, client and financial damage.

3. Cognyte. In May of 2021, the cyber analytics firm Cognyte left a database unsecured without authentication protocols. In turn, hackers managed to expose 5 billion records. Information such as names, email addresses, passwords, and vulnerability data points within their system were leaked. Information was even indexed by search engines.

Lessons learned: The company managed to secure the data within four days, but the incident highlighted how persistent cyber attackers can effectively exploit the smallest of flaws. In this instance, the importance of cyber attack prevention cannot be overstated. Prevent as many attacks as possible through a combination of policies, tools, education and vigilance.

4. Facebook. In April of 2021, Facebook reported a breach affecting hundreds of millions of user records, which were publicly exposed on Amazon's cloud computing service. Although Facebook confirmed that it identified and resolved the issue immediately, the attack managed to impact founder Mark Zuckerberg.

In precipitating the incident, two third-party Facebook app development companies posted the records in plain sight. The database exposed contained private information that social engineers could use in targeted attacks or within hacking attempts.

Lessons learned: In resolving this issue, Facebook reached out to Amazon, which took down the exposed servers. "...If you're still opening AWS buckets [to the public], you're not paying attention," says business advisor Corey Quinn.

Why Cloud Security?

Hackers accessed **Uber's** cloud-based storage and obtained personal data of 57 million users and drivers. Uber initially failed to report the breach.

•CAM4—2020

CAM4 is an adult live streaming website that fell victim to a cloud cyber attack in March 2020 that exposed 10.8 billion sensitive entries amounting to 7 TB of data. The leaked database included location details, email addresses, IP addresses, payment logs, usernames and more.

Abuse of Cloud Services - use cheap cloud services to arrange DoS and brute force attacks. Bryan and Anderson arranged a DoS using Amazon's EC2 cloud infrastructure in 2010, by spending only $6 to rent virtual services.

Facebook exposed over 540 million records through unsecured cloud storage, including data such as user comments, likes, and reactions, making it vulnerable to external access.

•Advanced Info Service (AIS)—2020

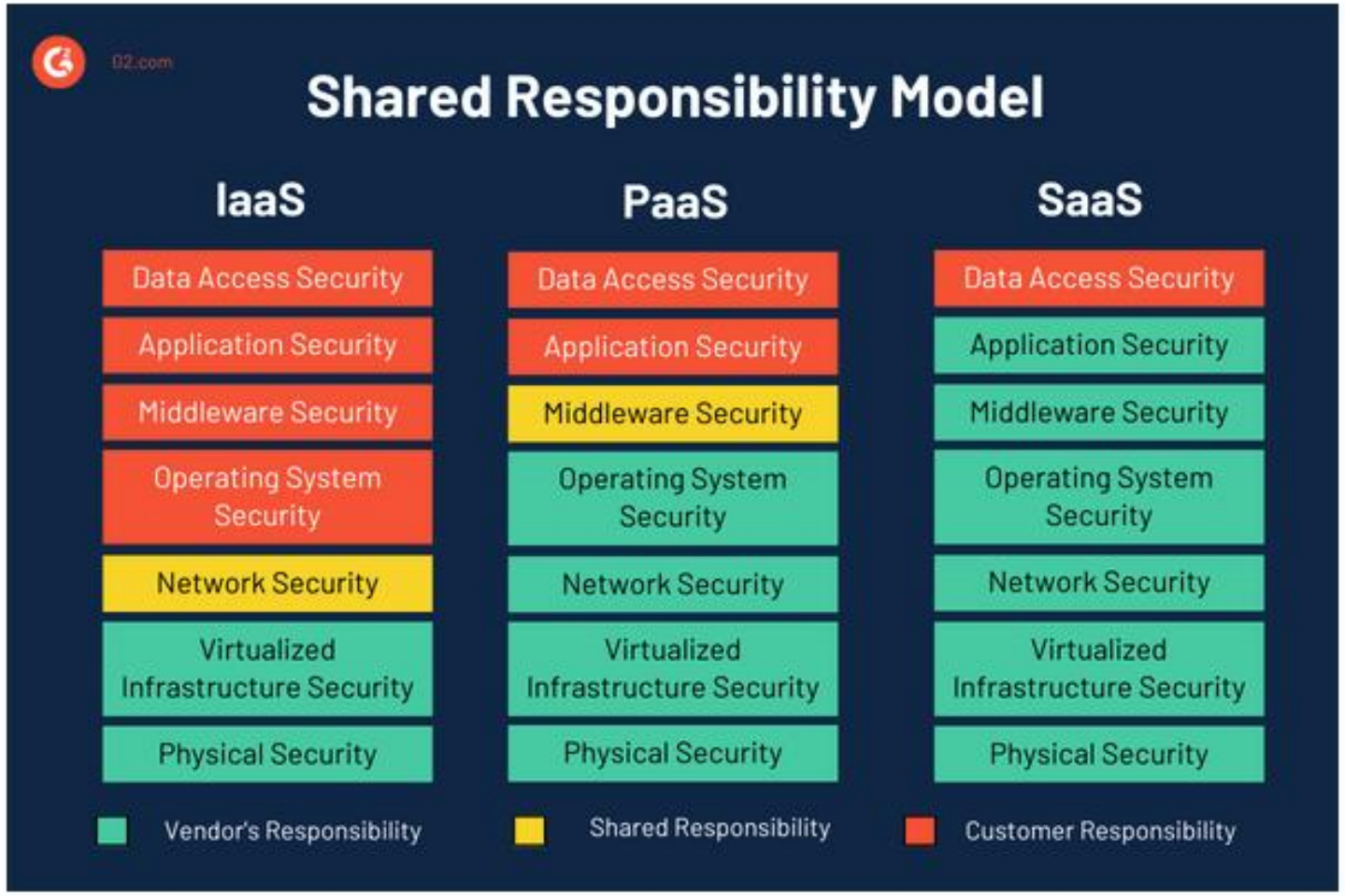
The AIS data breach was discovered by cybersecurity researcher Justin Paine when browsing BinaryEdge and Shodan. According to Paine, the leaked database included 8.3 billion network flow logs and DNS query logs of AWN customers of the Thailand-based telecommunications company.

Why Cloud Security?

Year	Organization	Type of Incident	Primary Reason for Attack/Breach	Key Security Lesson
2012	**Dropbox**	Account compromise	Stolen credentials from a third-party breach; **lack of MFA**	Implement MFA and monitor suspicious logins
2014	**Snapchat**	Photo data leak	**Insecure third-party applications** handling cloud-stored data	Control third-party access and secure cloud data storage
2016	**Uber**	Data breach affecting 57 million users	**Weak cloud storage security** and inadequate access controls	Secure cloud storage and enforce strict access management
2017	**Amazon Web Services (S3 incidents)**	Large-scale data exposure	Publicly accessible S3 buckets due to **cloud misconfiguration**	Regularly audit cloud permissions and storage settings
2017	**Accenture**	Exposure of internal databases	Weak cloud security configurations and **lack of encryption**	Encrypt sensitive data and limit access privileges
2018	**GitHub**	Massive DDoS attack	**Exploitation of unsecured internet** services (Memcached amplification)	Deploy cloud-based DDoS protection mechanisms
2019	**Capital One**	Exposure of 100+ million customer records	**Misconfigured AWS** environment and inadequate access controls	Continuous configuration management and security auditing
2019	**Microsoft**	Exposure of support records	**Misconfigured Azure cloud storage** by third-party vendor	Continuous monitoring and cloud security audits
2019	**Facebook**	Exposure of 540 million records	**Unsecured cloud storage** managed by third-party developers	Secure APIs, encrypt data, and monitor third-party usage
2020	**Slack**	Unauthorized access to corporate data	**Exposed API token** (credential management failure)	Rotate API keys and implement secrets management

What is the shared responsibility model?

In an on-premise data center, the company owns the complete tech stack; there's no question of who is responsible for security. But when the company transfers data to the cloud, some security responsibilities shift to cloud service providers (CSPs), and some rest with the users. The shared responsibility model dictates which security tasks the cloud provider handles and which tasks the company takes care of.



What is the shared responsibility model?

- Cloud storage vendors, who provide IaaS, take care of the underlying cloud infrastructure, both physical and virtualized. The users take care of data security and the protection of applications, virtualized networks, operating systems, and user access.
- In simple terms, the cloud storage providers are responsible for the security of the cloud, itself, and the users are responsible for the security of the content in the cloud.
- In line with this, cloud storage vendors have different security measures in place to safeguard their cloud infrastructure. Their dedicated cybersecurity team provides constant security updates, auto-patch, and built-in firewalls and backups. It also continuously audits, tests, and monitors for any vulnerability or cybersecurity risks.
- However, cloud users often misunderstand the shared responsibility model. They assume the CSP is responsible for the security of the cloud and the data stored there. As a result, users leave their data and assets unprotected on the cloud, making them vulnerable to attacks.

Attack Vectors for Cloud Computing

Sr No	Attack Vector	What is Targeted	Example
1	**Stolen Credentials**	User/admin accounts	Attacker obtains cloud username and password through phishing
2	**Weak IAM / Privilege Escalation**	Identity & Access Management	Normal user obtains administrator privileges
3	**Insecure APIs**	Cloud APIs	Manipulating API requests to access unauthorized resources
4	**Management Console**	Cloud control plane	Compromise of cloud administrator console
5	**Misconfiguration**	Storage, firewall, IAM, network	Publicly accessible cloud storage bucket
6	**Exposed Cloud Services**	EC2/VMs, databases, ports	Internet-exposed database attacked
7	**Vulnerable VMs**	Virtual machines	Exploitation of unpatched OS/software
8	**Hypervisor**	Virtualization layer	VM escape to underlying host
9	**Multi-Tenant Infrastructure**	Shared resources	One tenant attempts to access another tenant's data

Attack Vectors for Cloud Computing

Sr No.	Attack Vector	What is Targeted	Example
10	**Cloud Storage**	S3/object/file storage	Unauthorized access or deletion of stored data
11	**Network**	VPC/VNet, virtual networks	Network sniffing, spoofing, lateral movement
12	**Web Applications**	Cloud-hosted applications	SQL injection, XSS, authentication attacks
13	**Supply Chain**	Software/images/dependencies	Malicious library or compromised VM image
14	**Insider Threat**	CSP/customer environment	Authorized employee abuses privileges
15	**DDoS**	Cloud applications/network	Flooding service with huge traffic volume
16	**Malware/Ransomware**	VMs, storage, applications	Encrypting cloud-hosted files
17	**Secrets/API-Key Leakage**	Keys, tokens, credentials	API key accidentally published on GitHub
18	**Container/Kubernetes**	Containers/orchestration	Container escape or exposed Kubernetes API
19	**Serverless Functions**	Functions such as Lambda	Vulnerable function exploited
20	**Data Remanence**	Disks, snapshots, backups	Recovering residual information from improperly deleted resources

Cloud Vulnerabilities : Layer Wise

Cloud Layer	Major Vulnerabilities
User/Identity	Weak passwords, stolen credentials, no MFA, excessive privileges
Application	SQL injection, XSS, SSRF, RCE, broken access control
API	Weak authentication, authorization flaws, exposed keys
Management Plane	Console compromise, privilege escalation, configuration flaws
Network	Open ports, poor segmentation, weak firewall/security groups
VM	Unpatched OS, vulnerable applications, insecure images
Hypervisor	VM escape, isolation failure, hypervisor vulnerabilities
Container	Vulnerable images, privileged containers, container escape
Storage	Public buckets, weak ACLs, unencrypted data
Database	Public exposure, weak credentials, injection, excessive privileges
Physical Infrastructure	Hardware/firmware vulnerabilities, physical access
Operational	Poor monitoring, inadequate patching, weak incident response
Supply Chain	Compromised libraries, images, dependencies, CI/CD

What is cloud storage security?

- Cloud storage security includes a set of policies, technologies, tools, and practices to protect data stored in cloud storage systems from leaks, cyberattacks, unauthorized access, and other threats.
- Cloud storage security is fundamental for cloud data protection that includes data accessibility, data privacy, and data integrity.
- Protecting cloud storage involves fortifying cloud storage infrastructure, shielding networks, and endpoints, securing cloud data in rest and motion, and monitoring for vulnerabilities and risks.

Cloud storage vulnerabilities and risks

- **Lack of physical control :** Cloud storage providers offer security controls, but when you're operating in a zero-trust environment, users need to take additional precautions.
- **Multi-tenancy:** A flaw in a single shared application or a misconfiguration of a single user's system can lead to a chain of cyber attacks on multiple cloud users.
- **Insecure APIs and interfaces :** Cloud APIs are often the front door to cloud service, enabling applications and databases to work together seamlessly. However, when left unprotected, APIs are gateways for attacks. From misconfiguration to excessive data exposure, API loopholes create unwanted data exposure.
- **Poor identity and access management (IAM) :** Poor IAM can be slower in provisioning or de-provisioning privileged cloud access or undermanaged password, but these make it easier for malicious insiders and hackers to hijack your account.
- **DDoS attacks:** Distributed denial of service (DDoS) attacks are one of the most significant security concerns for businesses moving to the cloud. DDoS don't result in a data breach, but they can prevent users from accessing the cloud services, bringing operations to a halt.
- **Malicious insider:** The malicious actor is an insider who has access to and knows the ins and outs of a company's IT system.
- **Compliance and regulatory risks:** Many regions today have data protection, data sovereignty, and data localization laws, like General Data Protection Rules (GDPR). The rules stipulate where and how businesses can store, process, and use the personal data of their users. Businesses face heavy fines and penalties if they fail to comply with these rules.
- **Hackers, Organized Crime, and APTs:** Hackers, organized crime groups, and advanced persistent threats (APT) pose a significant risk to the cloud. While these attackers have different motives, they all aim to get into vulnerable systems and exploit them for their own gain.

Privacy & Security in Cloud Storages

- Is Google Drive secure?
- **In Transit**
  - To mitigate this threat, Google encrypts your data using TLS encryption before it's uploaded.
  - Google also encrypts your data whenever it is in transit within its internal network. This means that your data is always encrypted when it moves from one Google server to another, and during synchronization with your various devices.
- **At Rest**
  - Once your data arrives with Google, it's encrypted to keep it secure within its cloud servers – and Google uses 128-bit AES encryption for all data that is at rest. Although this isn't as strong as 256-bit encryption
  - Google encrypts the AES encryption keys used to encrypt your data with a rotating set of master keys.
  - Google holds the key to your files on your behalf, which means that the firm can take a look in your files if it really wants to.
- **Privacy policy**
  - Google asks for consent to access everything you upload, it can't claim HIPAA compliance.

- Is iCloud secure?
  - Apple states that all communication with iCloud servers is protected with TLS 1.2 encryption with Forward Secrecy.
  - For additional security, when you access iCloud services using native Apple apps such as Mail, Calendar, or Contacts, authentication is handled using a secure token.
  - Apple states that all data is stored on its servers using AES-128 encryption.
  - Apple's privacy policy makes it clear that iCloud user data may be accessed under some circumstances

Privacy & Security in Cloud Storages

- For data stored in the cloud (i.e., storage-as-a-service), referring to IaaS and not data associated with an application running in the cloud on PaaS or SaaS.
- **Confidentiality**:
  - AWS S3 does not encrypt a customer's data. Customers are able to encrypt their own data themselves prior to uploading, but S3 does not provide encryption.
  - If a CSP does encrypt a customer's data, the next consideration concerns what encryption algorithm it uses. : Symmetric encryption or Asymmetric encryption
  - what key length is used.
- **Integrity**
  - integrity also requires the use of message authentication codes (MACs).
  - The simplest way to use MACs on encrypted data is to use a block symmetric algorithm (as opposed to a streaming symmetric algorithm) in cipher block chaining (CBC) mode, and to include a one-way hash function.
- **Availability**
  - many cloud storage providers do not back up customer data, or do so only as an additional service for an additional cost. For example, "data stored in Amazon S3, Amazon SimpleDB, or Amazon Elastic Block Store is redundantly stored in multiple physical locations as a normal part of those services and at no additional charge." However, "data that is maintained within running instances on Amazon EC2, is all customer data and therefore AWS does not perform backups."

	Total downtime (HH:MM:SS)		
Availability	Per day	Per month	Per year
99.999%	00:00:00.4	00:00:26	00:05:15
99.99%	00:00:08	00:04:22	00:52:35
99.9%	00:01:26	00:43:49	08:45:56
99%	00:14:23	07:18:17	87:39:29

Privacy & Security in Cloud Storages

Feature	Google Drive	Microsoft OneDrive	Dropbox	Apple iCloud
Encryption in transit	Yes – HTTPS/TLS	Yes – TLS	Yes – SSL/TLS	Yes
Encryption at rest	Yes	Yes	AES-256	Yes
End-to-End Encryption (E2EE)	Limited; not default for normal personal Drive	Limited; not general E2EE for normal files	Not general E2EE	Strongest with Advanced Data Protection
MFA / 2FA	Yes	Yes	Yes	Yes; required for many E2EE features
Sensitive-data protection	DLP, IRM and sharing restrictions available, especially for organizations	Personal Vault provides additional authentication	Sharing/access controls	Advanced Data Protection
File sharing controls	Extensive sharing permissions and restrictions	Extensive permissions; password/expiry options for some links	File/folder/link sharing and access controls	Shared folders, files, photos etc.
Ransomware/File recovery	Version history and recovery features	Ransomware detection and recovery for supported Microsoft 365 subscriptions	File recovery/version history depending on plan	Versioning/recovery through Apple ecosystem
Zero-knowledge-style protection	No for ordinary Drive	No for ordinary OneDrive	No for ordinary Dropbox	Yes for many categories with Advanced Data Protection
Provider can potentially access ordinary stored content	Generally yes, subject to service/security architecture	Generally yes	Generally yes	Not for E2EE data protected by Advanced Data Protection
Main privacy strength	Strong account/security and enterprise controls	Strong enterprise security + Personal Vault	Strong encryption and sharing controls	User-controlled encryption keys for most data with ADP
Main limitation	Ordinary Drive is not E2EE	Ordinary OneDrive files are not E2EE	Ordinary files are not E2EE	E2EE increases recovery responsibility and some sharing/web features have limitations

Cloud Vulnerabilities : Multi Cloud Scenario

- **Inter-Cloud Lateral Movement**
- **Shared Secrets:** Compromised API keys stored in one cloud unlock access to another platform.
- **Transit Networks:** Attackers exploit unencrypted or poorly routed cross-cloud network connections to sniff traffic.
- **Over-Trusted Networks:** Vulnerable workloads in Cloud A use trusted peering to access private resources in Cloud B.
- **Storage Syncs:** Automated data pipelines replicate compromised or malicious files across multiple cloud storage systems.

- **Identity & Access Exploitation**
- **Cross-Cloud Trust Abuse:** Exploiting federated identity systems to assume high-privilege roles across different providers.
- **IAM Inconsistency:** Attackers target user accounts that lack Multi-Factor Authentication (MFA) on secondary, less-monitored cloud platforms.
- **Orphaned Accounts:** Discovered credentials for stale, forgotten test accounts give attackers a quiet entry point.
- **Privilege Escalation:** Exploiting platform-specific IAM quirks to elevate a standard user to a global administrator

Cloud Vulnerabilities : Multi Cloud Scenario

- **Management Plane & API Attacks**
- **Central Console Hijacking:** Gaining access to third-party multi-cloud management platforms to compromise all connected environments at once.
- **API Key Scraping:** Extracting cloud provider credentials from public repositories, unsecured codebases, or public deployment buckets.
- **Metadata Service Exploitation:** Using Server-Side Request Forgery (SSRF) to steal instance metadata tokens containing temporary security credentials.

- **DevOps & Supply Chain Vulnerabilities**
- **CI/CD Pipeline Poisoning:** Compromising a shared deployment pipeline to inject malicious code into workloads destined for multiple clouds.
- **Insecure Infrastructure as Code (IaC):** Exploiting misconfigured Terraform or Ansible scripts to deploy pre-compromised infrastructure.
- **Container Registry Poisoning:** Replacing legitimate application images with backdoored variants in a centralized, multi-cloud registry.

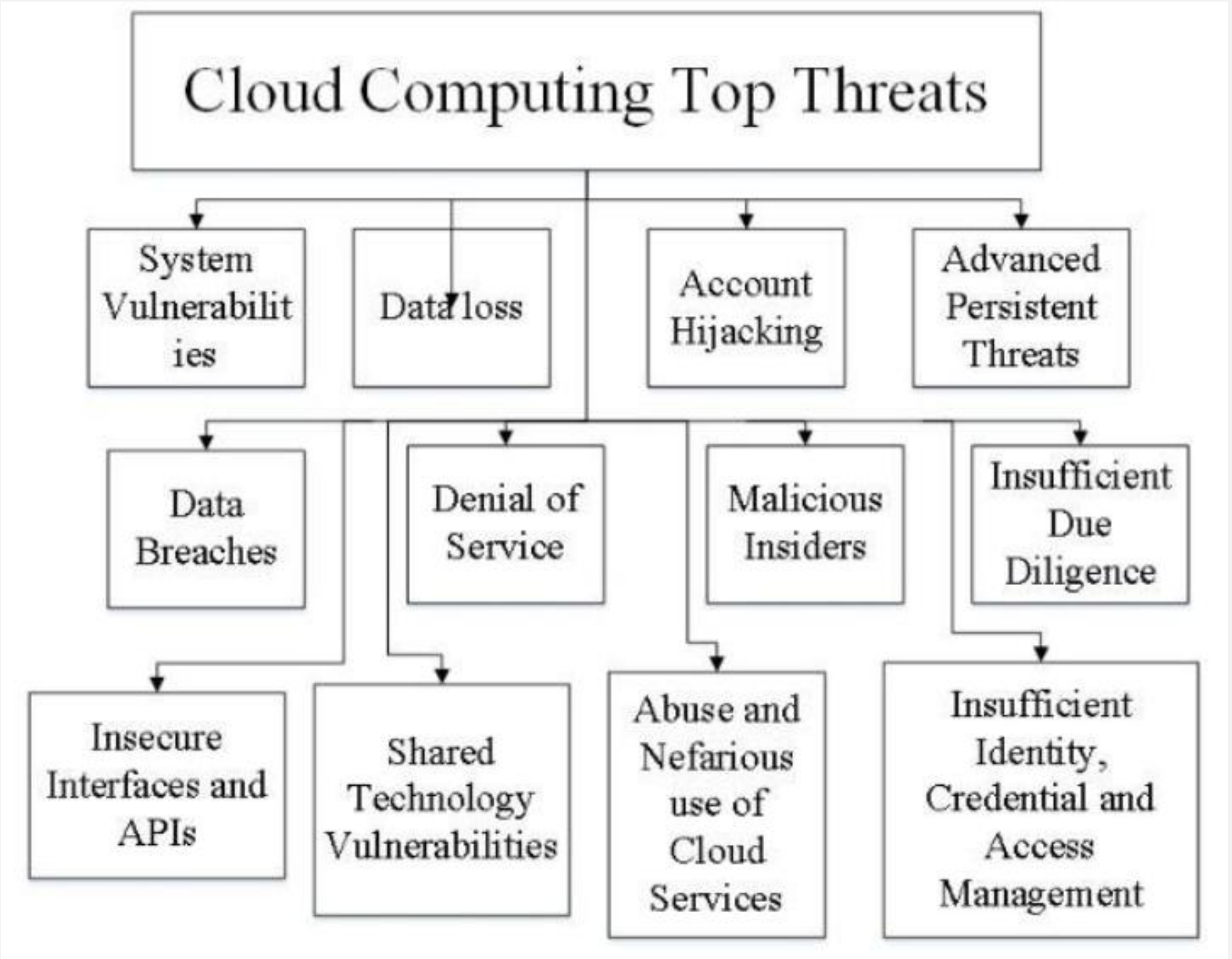
- **Visibility & Logging Gaps**
- **Log Siloing:** Exploiting delayed visibility across disconnected SIEM tools to execute actions before detection alerts trigger.
- **Alert Blindspots:** Launching attacks in secondary, less-monitored clouds where logging is disabled or improperly configured.

Service Provider Threats

Threat	Explanation
Malicious insider	CSP employee intentionally abuses privileges
Privileged administrator abuse	Administrator accesses customer resources without authorization
CSP account compromise	Provider's administrative account is compromised
Management-plane attack	Attacker compromises cloud management infrastructure
API compromise	Vulnerable provider APIs are exploited
Data leakage by provider	Customer information is accidentally or intentionally exposed
Weak tenant isolation	Provider fails to properly isolate customers
Supply-chain compromise	Third-party component used by CSP is compromised
Inadequate patching	Provider infrastructure contains exploitable vulnerabilities
Service outage	CSP infrastructure failure causes service unavailability
Data deletion/loss	Data is accidentally or maliciously deleted
Insufficient logging	Security events cannot be properly investigated
Privacy violation	Customer data is processed or disclosed improperly

Infrastructure and Host Threats

Threat	Explanation	Possible Impact
Physical attacks	Unauthorized access to cloud data centers	Hardware/data compromise
Hardware failure	Server, disk, network or power failure	Service disruption
Hypervisor attack	Exploitation of virtualization layer	Multiple VMs compromised
VM escape	Attacker escapes from VM to host	Host compromise
VM isolation failure	One tenant accesses another tenant	Cross-tenant data leakage
Side-channel attack	Information obtained through shared hardware resources	Confidentiality breach
Malicious VM/image	Compromised VM image introduced into cloud	Malware propagation
VM hopping	Compromised VM used to attack other VMs	Lateral movement
Resource exhaustion	Excessive CPU/RAM/storage consumption	Availability failure
Firmware attack	BIOS/UEFI/firmware compromised	Persistent host compromise
Storage attacks	Unauthorized access to disks/snapshots	Data leakage
Network attacks	Sniffing, spoofing, DDoS, etc.	Communication compromise

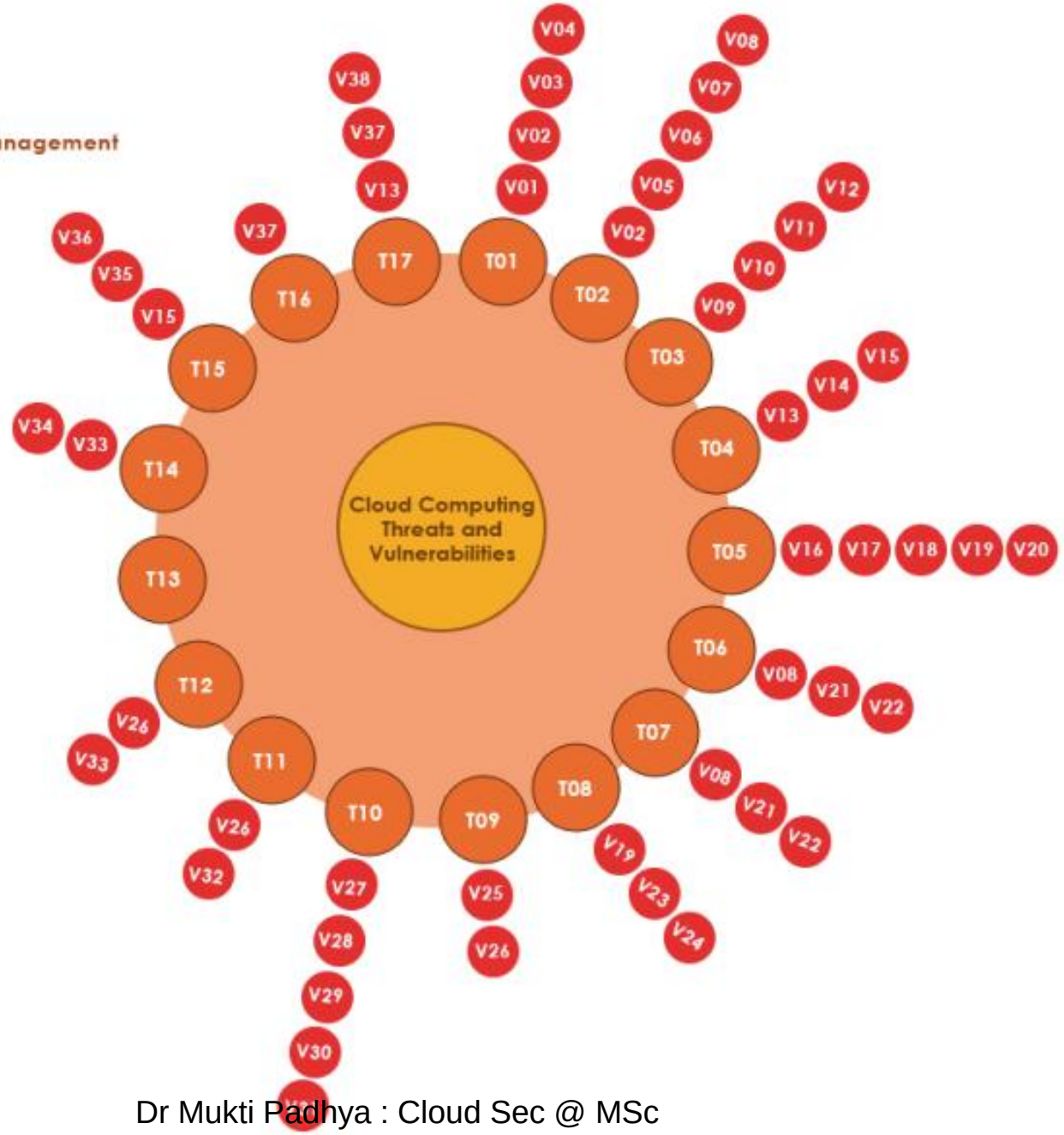

Cloud Computing Top Threats
System Vulnerabilities
Data loss
Account Hijacking
Advanced Persistent Threats
Data Breaches
Denial of Service
Malicious Insiders
Insufficient Due Diligence
Insecure Interfaces and APIs
Shared Technology Vulnerabilities
Abuse and Nefarious use of Cloud Services
Insufficient Identity, Credential and Access Management

THREATS

T01 - Data Breaches
T02 - Data Loss
T03 - Malicious Insiders
T04 - Denial of Service (DoS)
T05 - Vulnerable Systems and APIs
T06 - Weak Authentication and Identity Management
T07 - Account Hijacking
T08 - Shared Technology Vulnerabilities
T09 - Lacking Due Diligence
T10 - Advanced Persistent Threats (APT)
T11 - Abuse of Cloud Services
T12 - A Lack of Responsibility
T13 - Insufficient Security Tools
T14 - Human Error
T15 - Ransomware
T16 - Spectre and Meltdown
T17 - Unprotected IoT Devices

VULNERABILITIES

V01 - Targeted Attack
V02 - Simple Human Errors
V03 - Application Vulnerabilities
V04 - Poor Security Policies
V05 - Natural Disasters
V06 - Hard Drive Failures
V07 - Power Failures
V08 - Malware Infection
V09 - Former Employee
V10 - System Administrator
V11 - Third Party Contractor
V12 - Business Partner Weak
V13 - Weak Network Architecture
V14 - Insecure Network Protocol
V15 - Vulnerable Application
V16 - Weak API Credentials
V17 - Key Management
V18 - Operating System Bugs
V19 - Hypervisor Bugs
V20 - Unpatched Software
V21 - Social Engineering Attacks
V22 - Man-In-The-Middle (MITM) Attack
V23 - VM Vulnerabilities
V24 - Third-Party S/W Vulnerabilities
V25 - No Auditing
V26 - Service Level Agreement
V27 - Spear Phishing or Whaling
V28 - Direct Hacking
V29 - USB Malware
V30 - Network Penetration
V31 - Third-Party APIs
V32 - No Cloud Service Monitoring
V33 - Human Negligence
V34 - None or Insufficient Security Training
V35 - Infrastructure Vulnerabilities
V36 - Platform Vulnerabilities
V37 - Hardware Design Vulnerabilities
V38 - Weak Device Management



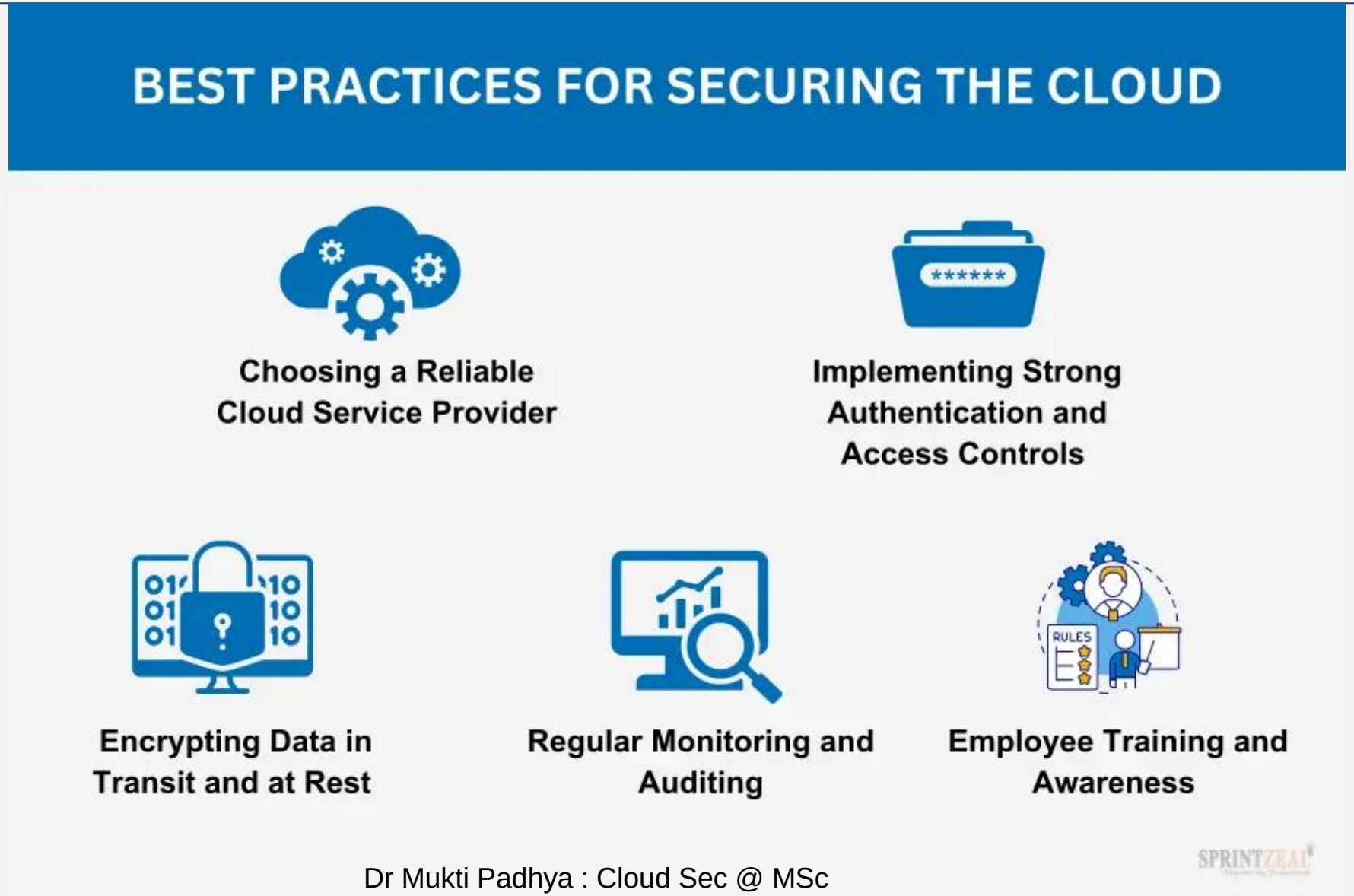
Classification of Cloud Threats:



Cloud Threats : Prevention Techniques

- Enhance security policies
- Use strong authentication
- Implement access management
- Protect data
  - at the source (on the user's side)
  - in transit (during its transfer from the user to the cloud server)
  - at rest (when stored in the cloud database)
- Detect intrusions
- Secure APIs and access

Cloud Security: Best Practices



Cloud Management Tools

AWS - Platform Specific Tools:

- **AWS Management Console** – Web-based interface to manage AWS resources.
- **AWS CloudFormation** – Infrastructure as Code (IaC) for automating resource provisioning.
- **AWS CloudWatch** – Monitoring and alerting service for cloud resources and applications.
- **AWS CloudTrail** – Logging and auditing of API activities for compliance and governance.
- **AWS Trusted Advisor** – Provides recommendations for cost optimization, performance, and security.
- **AWS Systems Manager (SSM)** – Unified interface to manage EC2, on-prem servers, and hybrid resources.

Cloud Management Tools

Multi-Cloud and Hybrid Cloud Management Platforms

These tools manage resources across **multiple cloud providers** and on-premise data centers.

VMware vRealize Suite / Aria Suite

- Automates provisioning, monitoring, and governance across private and public clouds.
- Integrates with AWS, Azure, and GCP.
- Supports self-service portals and policy-based automation.

IBM Cloud Pak for Multicloud Management

- Centralized dashboard for hybrid and multi-cloud operations.
- Offers monitoring, compliance, and cost governance.
- Kubernetes and container management support.

Red Hat CloudForms / Red Hat OpenShift

- Open-source hybrid cloud management and orchestration.
- Integrates with VMware, AWS, OpenStack, and Azure.
- Automates provisioning, lifecycle, and compliance.

Cloud Management Tools

Open Source Cloud Management Tools

OpenStack

- Popular open-source platform for building and managing private and hybrid clouds.
- Manages compute, storage, and networking resources.

Cloudify

- Open-source orchestration and automation platform.
- Supports multi-cloud environments using TOSCA blueprints.

Terraform

- Open-source Infrastructure as Code (IaC) tool.
- Supports AWS, Azure, GCP, and many other providers.

Cloud Management Tools

Cloud Cost and Performance Optimization Tools

CloudHealth (by VMware)
- Multi-cloud cost management, optimization, and governance.
- Tracks usage and provides cost-saving recommendations.

Spot.io (by NetApp)
- Automates cloud cost optimization using AI.
- Efficiently manages spot instances.

Flexera Cloud Management Platform
- Provides visibility, governance, and cost optimization across multiple clouds.

Turbonomic (by IBM)
- Application resource management using AI to optimize performance and cost.

Cloud Management Tools

Cloud Security and Compliance Management Tools

- Prisma Cloud (by Palo Alto Networks)
- Check Point CloudGuard
- Trend Micro Cloud One
- Qualys Cloud Platform

All provide cloud posture management (CSPM), threat detection, compliance reporting, and vulnerability management across multi-cloud environments.

Summary

- Cloud computing has become one of the most used technology components in modern day applications, which not only provides storage but also supports data analytics.
- Cloud services could be any one of the following:
  - **Software as a Service (SaaS):** Complete software application as a service is provided to the user.
  - **Platform as a Service (PaaS):** Development tools, APIs, libraries, etc. will be divided by the cloud service provider. User have to build, manage and maintain the applications.
  - **Infrastructure as a Service (IaaS):** User should be provided with virtual machine support, where the user does not need to know and worry about the infrastructure. Everything should be taken care by the service provider. User will manage the machines, select the OS and underlying applications.
- The three deployment models generally used for public, private and hybrid
- Private cloud deployment model can be opted wherever confidentiality matters the most.
- When its come to public cloud deployment model, the cloud service provider owns all the resource which include hardware/infrastructure and software. Cloud service provider will take care of all resource management.

Summary

- Hybrid development is a mix of both public and private deployment model. In this approach the resource offered and manage are both in-house and third party based.
- There are many challenges one could face while opting for cloud storage with IoT applications some of these are as follows:
  - Privacy and security
  - Bandwidth cost
  - Migration and portability
  - Reliability and availability
  - Costing
  - Data ownership
  - Expertise
- Selecting a CSP is not easy. Many parameters are to be considered before choosing the best option.